

MATH 452 INTRODUCTION TO FUNCTIONAL ANALYSIS

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ABSTRACT. These are the lecture notes I used for a 14-week introductory functional analysis class I taught at the Department of Mathematics of Middle East Technical University during Fall 2021. These lecture notes were mostly prepared by TeXing and reorganizing the hand-written sketch lecture notes of Zafer Nurlu, who has taught this course for many years at the department. I have added some additional course material, some slight modifications and the commentary to his lecture notes. Besides Prof. Nurlu's lecture notes, I also used the textbooks [Kre78], [Con90] and [RY08].

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First originated as the study of certain “function spaces”, functional analysis has evolved in time as any other mathematical discipline. Students who are interested in the history of the subject may take a look at [BK84].

While it is difficult to give an all-encompassing definition of functional analysis, we may quote Dieudonné [Die81] and define it as “the study of topological vector spaces and of mappings $u : \Omega \rightarrow F$ from a part Ω of a topological vector space E into a topological vector space F , these mappings being assumed to satisfy various algebraic and topological conditions.” Why am I emphasizing this?

By its nature, functional analysis requires anyone studying it to have a good understanding of linear algebra and topology of metric spaces. Indeed, it is reasonable to say to a certain extent that we are basically going to do infinite-dimensional linear algebra in a topological setting. For this reason, students are expected to have a working knowledge of MATH261 and MATH349.

Notational remarks. Throughout these lecture notes, $\mathbb{K}$ will always denote either of $\mathbb{R}$ or $\mathbb{C}$; and $\mathbf{0}$ will always denote the zero vector of the relevant vector space. Boldface variable names such as $\mathbf{x}, \mathbf{y}, \dots$ will usually be used for sequences.

1. NORMED LINEAR SPACES AND BANACH SPACES

In this section, we shall learn the definition, some important examples and some basic properties of the main object of our study: Normed spaces. A normed space is simply a vector space together with a function, its norm, which measures the “lengths” of vectors. Such functions are readily available when one studies finite-dimensional vector spaces such as $\mathbb{K}^n$; and consequently, one does not often pay specific attention to their properties and roles in various arguments. Let us now focus on how we can generalize the concept of “length” to abstract vector spaces.

1.1. Definition and some properties. Let V be a vector space over $\mathbb{K}$. A *norm* on V is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ such that

- (a) For all $x \in V$ with $x \neq \mathbf{0}$, $\|x\| > 0$.
- (b) For all $\alpha \in \mathbb{K}$ and $x \in V$, $\|\alpha x\| = |\alpha| \|x\|$.
- (c) For all $x, y \in V$, $\|x + y\| \leq \|x\| + \|y\|$.

Letting $\alpha = 0$ in Item (b), one obtains that $\|\mathbf{0}\| = 0$. It follows that for all $x \in V$ we have $\|x\| \geq 0$, and moreover, $\|x\| = 0$ iff $x = \mathbf{0}$. An inductive application of Items (b) and (c) gives that

$$\left\| \sum_{i=1}^n \alpha_i x_i \right\| \leq \sum_{i=1}^n |\alpha_i| \|x_i\|$$

for all $n \in \mathbb{N}^+$, $\alpha_1, \dots, \alpha_n \in \mathbb{K}$ and $x_1, \dots, x_n \in V$. The usual proof of the reverse triangle inequality also generalizes to norms and we have that for all $x, y \in V$

$$|\|x\| - \|y\|| \leq \|x - y\|$$

A *normed space* is simply a pair $(V, \|\cdot\|)$ where V is a vector space over $\mathbb{K}$ and $\|\cdot\|$ is a norm on V . Given a normed space $(V, \|\cdot\|)$, we can consider the map $d : V \times V \rightarrow [0, \infty)$ given by

$$d(x, y) = \|x - y\|$$

for all $x, y \in V$. It is not difficult to see that Items (a), (b) and (c) together imply that d is a metric on V . Consequently, we have a topological (indeed, metric) structure on V arising from the norm $\|\cdot\|$. Unless stated otherwise, a normed space will always be endowed with the topology induced by the metric associated to its norm defined as above.

Given a normed space $(V, \|\cdot\|)$, by the definition of its metric structure, we have that $x_n \rightarrow x$ iff $\|x_n - x\| \rightarrow 0$. Moreover, the map $x \rightarrow \|x\|$ from V to $\mathbb{R}$ is uniformly continuous. To see this, you can just take $\delta = \epsilon$ in the ϵ - δ definition of uniform continuity and use the reverse triangle inequality. As a result of this, we have that $x_n \rightarrow x$ implies $\|x_n\| \rightarrow \|x\|$ in a normed space. The converse implication is clearly false as exemplified by the sequence $(v, -v, v, -v, \dots)$ for any $v \neq \mathbf{0}$.

Before we proceed to provide examples, let us record two more basic but important points regarding the relationship between algebraic operations on V and its metric structure. First, for every $\alpha \in \mathbb{K}$, the scalar multiplication $x \mapsto \alpha x$ is a uniformly continuous map on V . To see this, in the ϵ - δ definition of uniform continuity, we can simply take $\delta = \epsilon/|\alpha|$ whenever $\alpha \neq 0$, and $\delta = 1$ whenever $\alpha = 0$. Indeed, the map $x \mapsto \alpha x$ is a self-homeomorphism of V for any $0 \neq \alpha \in K$. Second, the metric associated to a norm is translation-invariant in the sense that $d(x+w, y+w) = d(x, y)$ for all $x, y, w \in V$. As a result of this translation-invariance, the map $x \mapsto x + w$ is an isometry from V to V (and so, is a self-homeomorphism of V) for any $w \in V$, and hence the “local” topological structure around any two points are the same. This allows one to consider only the unit ball around $\mathbf{0}$ for topological purposes since every other ball can be dilated and translated to the unit ball around $\mathbf{0}$.

Before we proceed to see some examples, we would like to note that the arguments above show the continuity of addition and multiplication once one of the parameters is fixed¹. On the other hand, the vector addition $+: V \times V \rightarrow V$ and the scalar multiplication $\cdot: \mathbb{K} \times V \rightarrow V$ are also continuous maps when the domains are considered with product topology. Checking this fact is left to the reader as an exercise.

1.2. Some examples. In this subsection, we are going to see some examples of normed spaces.

1.2.1. Sequence spaces. Of particular importance among the examples that we are going to cover will be what are usually called as sequence spaces, which are (vectorial) subspaces of the vector space $\mathbb{K}^{\mathbb{N}}$. The following (vectorial) subspaces² of $\mathbb{K}^{\mathbb{N}}$ are going to be frequently used in the remaining of these notes.

$$\begin{aligned}\ell_p &= \left\{ \mathbf{x} \in \mathbb{K}^{\mathbb{N}} : \sum_{n=0}^{\infty} |x_n|^p < \infty \right\}, \quad 1 \leq p < \infty \\ \ell_{\infty} &= \left\{ \mathbf{x} \in \mathbb{K}^{\mathbb{N}} : \sup_{n \in \mathbb{N}} |x_n| < \infty \right\} \\ c_0 &= \left\{ \mathbf{x} \in \mathbb{K}^{\mathbb{N}} : \lim_{n \rightarrow \infty} x_n = 0 \right\}\end{aligned}$$

Observe that we have the following inclusions

$$\ell_p \subseteq \ell_q \subseteq c_0 \subseteq \ell_{\infty}, \quad 1 \leq p \leq q < \infty$$

To see the first inclusion, notice that if $\sum_{n=0}^{\infty} |x_n|^p < \infty$, then $\lim_{n \rightarrow \infty} x_n = 0$ and so $0 \leq |x_n| \leq 1$ for sufficiently large n , which subsequently implies that $|x_n|^q \leq |x_n|^p$

¹This is the difference between what people call *joint continuity* and *separate continuity*.

²That the first two subsets are indeed subspaces is going to follow from the inequalities that will be proven later.

for sufficiently large n and for all $1 \leq p \leq q$. The other inclusions follow from basic facts regarding convergent sequences.

Consider the following maps on each of these sequence spaces

$$\|\cdot\|_p : \ell_p \rightarrow \mathbb{R} \text{ given by } \|\mathbf{x}\|_p = \left(\sum_{n=0}^{\infty} |x_n|^p \right)^{1/p}, \quad 1 \leq p < \infty$$

$$\|\cdot\|_{\infty} : \ell_{\infty} \rightarrow \mathbb{R} \text{ given by } \|\mathbf{x}\|_{\infty} = \sup_{n \in \mathbb{N}} |x_n|$$

$$\|\cdot\|_{\infty} : c_0 \rightarrow \mathbb{R} \text{ given by } \|\mathbf{x}\|_{\infty} = \sup_{n \in \mathbb{N}} |x_n|$$

We shall prove that these maps are indeed norms on the corresponding vector spaces. In order to do this, we will need the following famous inequalities.

Theorem 1 (Hölder's inequality for sequences). *Let $p, q \in \mathbb{R}$ be such that $p, q > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$. Then, for every $\mathbf{x} \in \ell_p$ and $\mathbf{y} \in \ell_q$, we have*

$$\sum_{n=0}^N |x_n y_n| \leq \left(\sum_{n=0}^N |x_n|^p \right)^{1/p} \cdot \left(\sum_{n=0}^N |y_n|^q \right)^{1/q}$$

where $N \in \mathbb{N} \cup \{\infty\}$.

Proof. Note that the assumption $\frac{1}{p} + \frac{1}{q} = 1$ gives $(p-1)(q-1) = 1$ and so, $u = v^{q-1}$ implies $v = u^{p-1}$ for any $u, v \geq 0$. Now one can geometrically see that, for any $\alpha, \beta > 0$, the area of the region represented by the sum of the integrals

$$\int_0^{\alpha} u^{p-1} du + \int_0^{\beta} v^{q-1} dv$$

is greater than or equal to the area of rectangle $[0, \alpha] \times [0, \beta]$ in the uv -plane.

Therefore we have that

$$\alpha\beta \leq \int_0^{\alpha} u^{p-1} du + \int_0^{\beta} v^{q-1} dv = \frac{\alpha^p}{p} + \frac{\beta^q}{q}$$

for every $\alpha, \beta \geq 0$.³

We are now ready to prove Hölder's inequality. Let $\mathbf{x} \in \ell_p$ and $\mathbf{y} \in \ell_q$. We may assume without loss of generality that $\sum_{n=0}^N |x_n|^p \neq 0$ and $\sum_{n=0}^N |y_n|^q \neq 0$, since otherwise the inequality trivially holds.

Let $n \in \mathbb{N}$. Applying the inequality that we obtained earlier with

$$\alpha = \frac{|x_n|}{\left(\sum_{i=0}^N |x_i|^p \right)^{1/p}} \text{ and } \beta = \frac{|y_n|}{\left(\sum_{i=0}^N |y_i|^q \right)^{1/q}}$$

we have that

$$\frac{|x_n y_n|}{\left(\sum_{i=0}^N |x_i|^p \right)^{1/p} \left(\sum_{i=0}^N |y_i|^q \right)^{1/q}} \leq \frac{|x_n|^p}{p \left(\sum_{i=0}^N |x_i|^p \right)} + \frac{|y_n|^q}{q \left(\sum_{i=0}^N |y_i|^q \right)}$$

³This is called Young's inequality for products.

Since this is true for all $n \in \mathbb{N}$, summing over all n gives us

$$\sum_{n=0}^N \frac{|x_n y_n|}{\left(\sum_{i=0}^N |x_i|^p\right)^{1/p} \left(\sum_{i=0}^N |y_i|^q\right)^{1/q}} \leq \frac{1}{p} + \frac{1}{q} = 1$$

and hence we conclude that

$$\sum_{n=0}^N |x_n y_n| \leq \left(\sum_{i=0}^N |x_i|^p\right)^{1/p} \left(\sum_{i=0}^N |y_i|^q\right)^{1/q}$$

□

An immediate consequence of Hölder's inequality is Minkowski's inequality which is basically the triangle inequality for the $\|\cdot\|_p$ norm.

Theorem 2 (Minkowski's inequality for sequences). *Let $p \geq 1$. Then, for every $\mathbf{x}, \mathbf{y} \in \ell_p$, we have that*

$$\left(\sum_{n=0}^N |x_n + y_n|^p\right)^{1/p} \leq \left(\sum_{n=0}^N |x_n|^p\right)^{1/p} + \left(\sum_{n=0}^N |y_n|^p\right)^{1/p}$$

where $N \in \mathbb{N} \cup \{\infty\}$.

Proof. The case $p = 1$ immediately follows from the usual triangle inequality since

$$\sum_{n=0}^N |x_n + y_n| \leq \sum_{n=0}^N (|x_n| + |y_n|) = \sum_{n=0}^N |x_n| + \sum_{n=0}^N |y_n|$$

Suppose now that $p > 1$. Let $\mathbf{x}, \mathbf{y} \in \ell_p$. We may assume without loss of generality that $\left(\sum_{n=0}^N |x_n + y_n|^p\right)^{1/p} \neq 0$ since otherwise the inequality trivially holds. Set $q = \frac{p}{p-1}$ which means that $\frac{1}{p} + \frac{1}{q} = 1$. Let $K \in \mathbb{N}$. Applying the usual triangle inequality together with Hölder's inequality, we obtain

$$\begin{aligned} \sum_{n=0}^K |x_n + y_n|^p &= \sum_{n=0}^K |x_n + y_n| |x_n + y_n|^{p-1} \\ &\leq \sum_{n=0}^K (|x_n| + |y_n|) |x_n + y_n|^{p-1} \\ &\leq \sum_{n=0}^K |x_n| |x_n + y_n|^{p-1} + \sum_{n=0}^K |y_n| |x_n + y_n|^{p-1} \\ &\leq \left(\left(\sum_{n=0}^K |x_n|^p\right)^{1/p} + \left(\sum_{n=0}^K |y_n|^p\right)^{1/p} \right) \left(\sum_{n=0}^K |x_n + y_n|^{(p-1)q}\right)^{1/q} \\ &\leq \left(\left(\sum_{n=0}^K |x_n|^p\right)^{1/p} + \left(\sum_{n=0}^K |y_n|^p\right)^{1/p} \right) \left(\sum_{n=0}^K |x_n + y_n|^p\right)^{1-1/p} \end{aligned}$$

from which it follows that

$$\left(\sum_{n=0}^K |x_n + y_n|^p \right)^{1/p} \leq \left(\sum_{n=0}^K |x_n|^p \right)^{1/p} + \left(\sum_{n=0}^K |y_n|^p \right)^{1/p}$$

To finish the proof, we can now simply let $K = N$ if $N \in \mathbb{N}$ or $K \rightarrow \infty$ if $N = \infty$, in which case the convergence of both series on the right implies the convergence of the series on the left. $\square$

We would like to note that Hölder's and Minkowski's inequalities actually hold in a more general setting, namely, for measure spaces. The cases that we have considered here are simply these inequalities for the counting measure. As a result of Minkowski's inequality, we have the following.

Proposition 1. *Let $1 \leq p < \infty$. Then $(\ell_p, \|\cdot\|_p)$ is a normed space.*

Proof. That $\|\cdot\|_p$ satisfies Item (a) and (b) in the definition of a norm is clear. That ℓ_p is closed under addition and $\|\cdot\|_p$ satisfies the triangle inequality follows from Minkowski's inequality. That ℓ_p is closed under scalar multiplication is a triviality. $\square$

Perhaps at the risk of stating the obvious, we would like to say that Minkowski's inequality also implies that the map $\|\cdot\|_p$, when adjusted appropriately for finitely many coordinates, is a norm on $\mathbb{K}^n$ for each $n \in \mathbb{N}^+$.

Proposition 2. *$(\ell_\infty, \|\cdot\|_\infty)$ is a normed space*

Proof. That $\|\cdot\|_p$ satisfies Item (a) and (b) follow from the properties of supremum. Let $\mathbf{x}, \mathbf{y} \in \ell_\infty$. By the usual triangle inequality, we have

$$|x_n + y_n| \leq |x_n| + |y_n| \leq \sup_{i \in \mathbb{N}} |x_i| + \sup_{i \in \mathbb{N}} |y_i| = \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty$$

for all $n \in \mathbb{N}$ and hence, we have

$$\|\mathbf{x} + \mathbf{y}\|_\infty = \sup_{n \in \mathbb{N}} |x_n + y_n| \leq \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty$$

which shows both Item (c) in the definition of norm and that ℓ_∞ is closed under addition. That ℓ_∞ is closed under scalar multiplication is a triviality. $\square$

Before we conclude this subsection, we would like to point out the incompatibility between the topology of $\mathbb{K}^{\mathbb{N}}$ and the topology of these sequence spaces. Let us first recall the topological structure of $\mathbb{K}^{\mathbb{N}}$. Consider the map $d : \mathbb{K}^{\mathbb{N}} \rightarrow [0, \infty)$ given by

$$d(\mathbf{x}, \mathbf{y}) = \sum_{n=0}^{\infty} \frac{\min\{|x_n - y_n|, 1\}}{2^{n+1}}$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{\mathbb{N}}$. This is the *product metric* on $\mathbb{K}^{\mathbb{N}}$ obtained by using the standard bounded metric corresponding to the usual Euclidean metric in each component.⁴ As is the case with any product metric, it is compatible with the product topology on $\mathbb{K}^{\mathbb{N}}$ where the product topology is taken with Euclidean topology in each component.⁵

For each $i \in \mathbb{N}$, consider the vector $\mathbf{e}_i = (\delta_{in})_{n \in \mathbb{N}} = (0, \dots, 0, 1, 0, \dots) \in K^{\mathbb{N}}$ which has 1 at its i -th component and 0 everywhere else. It is clear that $\mathbf{e}_i \in \ell_p$ for all $1 \leq p \leq \infty$ and that $\|\mathbf{e}_i\|_{\infty} = \|\mathbf{e}_i\|_p = 1$. Therefore $\mathbf{e}_i \not\rightarrow \mathbf{0}$ in $(\ell_p, \|\cdot\|_p)$. However, in the product topology of $\mathbb{K}^{\mathbb{N}}$, we *do* have that $\mathbf{e}_i \rightarrow \mathbf{0}$. Therefore, the topologies on the vectorial subspaces ℓ_p and c_0 given by the corresponding norms are *not* the subspace topologies on these sets induced by the product topology on the ambient space $\mathbb{K}^{\mathbb{N}}$.

1.2.2. *Space of bounded and continuous maps.* Let X be a non-empty set. Then the set

$$B(X) = \{f : X \rightarrow \mathbb{K} \mid f \text{ is bounded}\}$$

can be turned into a vector space over $\mathbb{K}$ with the obvious operations, that is, $(f + g)(x) = f(x) + g(x)$ and $(\alpha \cdot f)(x) = \alpha f(x)$. It is not difficult to check that

$$\|f\|_{\infty} = \sup_{x \in X} |f(x)|$$

is a norm on $B(X)$. In the case that X is a compact topological space, we have that

$$C(X) = \{f : X \rightarrow \mathbb{K} \mid f \text{ is continuous}\}$$

is a subspace of $B(X)$ and is itself a norm space with the restriction of the supremum norm. We would like to note that the normed space $(\ell_{\infty}, \|\cdot\|_{\infty})$ is actually of this form, more specifically, it is simply $(B(\mathbb{N}), \|\cdot\|_{\infty})$.

If X has an additional measure space structure which allows integration with its usual properties, one may also define another norm on $C(X)$ via integration. For example, in the case of $X = [0, 1]$, consider

$$\|f\|_1 = \int_0^1 |f(x)| dx$$

Then $\|\cdot\|_1$ is a norm on $C([0, 1])$. Let us now show that this norm is not equivalent to the supremum norm $\|\cdot\|_{\infty}$. Consider that the maps

$$f_n(x) = \begin{cases} 0 & \text{if } 0 \leq x \leq 1 - \frac{1}{n} \\ nx - n + 1 & \text{if } 1 - \frac{1}{n} < x \leq 1 \end{cases}$$

⁴Recall that for any metric $\nu(\cdot, \cdot)$, the standard bounded metric $\bar{\nu}(\cdot, \cdot)$ corresponding to $\nu(\cdot, \cdot)$ is given by $\min\{1, \nu(\cdot, \cdot)\}$ and generates the same topology.

⁵While we will not use this fact, it is worth noting that $\mathbb{K}^{\mathbb{N}}$ is a Polish space and a locally convex topological vector space.

for each $n \in \mathbb{N}^+$. It is straightforward to check that $f_n \not\rightarrow \mathbf{0}$ in the $\|\cdot\|_\infty$ norm but $f_n \rightarrow \mathbf{0}$ in the $\|\cdot\|_1$ norm.

1.3. Equivalence of norms. In this subsection, we are going to learn when two norms on the same vector space are considered “the same” for our purposes.

Fix a vector space V over $\mathbb{K}$ and let $\|\cdot\|$ and $\|\cdot\|'$ be norms on V . The norm $\|\cdot\|$ is said to be *equivalent* to the norm $\|\cdot\|'$ if there exist constants $c, C > 0$ such that

$$c \|x\|' \leq \|x\| \leq C \|x\|'$$

for all $x \in V$.

Observe that the definition of a norm being equivalent to another one is not symmetric between these norms. However, a moment’s thought reveals that if $\|\cdot\|$ is equivalent to $\|\cdot\|'$, then $\|\cdot\|'$ is equivalent to $\|\cdot\|$ as well. Indeed, it is not difficult to check that being equivalent is an equivalence relation on the set of norms on V . For this reason, we shall say that two norms are equivalent without specifying an order.

Recall that, given two metrics d and d' on the same set X , if they are equivalent i.e. there are constants $c, C > 0$ such that for all $x, y \in X$ we have $c \cdot d(x, y) \leq d'(x, y) \leq C \cdot d(x, y)$, then they are topologically equivalent i.e. they induce the same topology on X . On the other hand, not all metrics that induce the same topology are equivalent. For example, it may be checked with some effort that the discrete metric on $\mathbb{N}$ is topologically equivalent but not equivalent to the metric $d(m, n) = \frac{|m-n|}{(m+1)(n+1)}$ on $\mathbb{N}$.

Suppose that d and d' are equivalent metrics on X . Then the sets of convergent sequences with respect to these metrics are the same because they induce the same topology. Moreover, it easily follows from the definition of equivalence of metrics that the set of Cauchy sequences with respect to these metrics are also the same. We would like to note that this last remark regarding Cauchy sequences may not hold in general for topologically equivalent metrics, as the last example shows.

Two norms are equivalent if and only if the associated metrics are equivalent. Consequently, equivalent norms give rise to the same topology, the same convergent sequences and the same Cauchy sequences. Contrary to the case of arbitrary metrics, it turns out that two normed spaces having the same topology *does* imply that the norms are equivalent i.e. the associated metrics are equivalent.

Proposition 3. *Let V be a vector space over $\mathbb{K}$ and $\|\cdot\|, \|\cdot\|'$ be norms on V . Then $\|\cdot\|$ and $\|\cdot\|'$ are equivalent if and only if the topologies induced by these norms are the same.*

Proof. The left-to-right direction is left to the reader as an exercise since it must have been already done in MATH349.

To prove the right-to-left direction, suppose that $\|\cdot\|$ and $\|\cdot\|'$ induce the same topology on V . Let d and d' be the associated metrics respectively. Consider the open unit ball

$$B_d(\mathbf{0}, 1) = \{x \in V : \|x\| < 1\}$$

By assumption, this set is also open in the metric space (V, d') and hence there exists $\delta > 0$ such that

$$B_{d'}(\mathbf{0}, \delta) \subseteq B_d(\mathbf{0}, 1)$$

However, again by assumption, the set $B_{d'}(\mathbf{0}, \delta)$ is open in the metric space (V, d) and hence there exists $\gamma > 0$ such that

$$B_d(\mathbf{0}, \gamma) \subseteq B_{d'}(\mathbf{0}, \delta) \subseteq B_d(\mathbf{0}, 1)$$

Set $c = \frac{\delta}{2}$ and $C = \frac{2\delta}{\gamma}$. Then $c, C > 0$. Let $x \in V$. Then we have that

$$\frac{\gamma x}{2\|x\|} \in B_d(\mathbf{0}, \gamma) \subseteq B_{d'}(\mathbf{0}, \delta)$$

and hence

$$\left\| \frac{\gamma x}{2\|x\|} \right\|' < \delta \quad \text{which implies} \quad \|x\|' < \frac{2\delta}{\gamma} \|x\| = C \|x\|$$

Similarly, we have that

$$\frac{\delta x}{2\|x\|'} \in B_{d'}(\mathbf{0}, \delta) \subseteq B_d(\mathbf{0}, 1)$$

and hence

$$\left\| \frac{\delta x}{2\|x\|'} \right\| < 1 \quad \text{which implies} \quad c\|x\| = \frac{\delta}{2} \|x\| < \|x\|'$$

Thus $c\|x\| \leq \|x\|' \leq C\|x\|$ for all $x \in V$ which means that $\|\cdot\|'$ and $\|\cdot\|$ are equivalent. $\square$

Let us next see some examples and non-examples of equivalent norms.

- We have already seen that $\|\cdot\|_1$ and the supremum norm $\|\cdot\|_\infty$ on $C([0, 1])$ are not equivalent since they have different convergent sequences.
- The norms $\|\cdot\|_p$ and $\|\cdot\|_\infty$ on ℓ_p are not equivalent. On the one hand, we have

$$\|\mathbf{x}\|_\infty = \sup_{n \in \mathbb{N}} |x_n| \leq \left(\sum_{n=0}^{\infty} |x_n|^p \right)^{1/p} = \|\mathbf{x}\|_p$$

for all $\mathbf{x} \in \ell_p$. On the other hand, considering $\mathbf{w}_n = \sum_{i=0}^n \mathbf{e}_i$ for each $n \in \mathbb{N}$ where $\mathbf{e}_i = (\delta_{in})_{n \in \mathbb{N}}$, we see that

$$\|\mathbf{w}_n\|_\infty = 1 \quad \text{but} \quad \|\mathbf{w}_n\|_p = n^{1/p}$$

which means that, for every $c > 0$, the inequality

$$c\|\mathbf{w}_n\|_p = cn^{1/p} \leq \|\mathbf{w}_n\|_\infty = 1$$

will fail for sufficiently large $n \in \mathbb{N}$. Therefore, $\|\cdot\|_p$ and $\|\cdot\|_\infty$ on ℓ_p are not equivalent.

- A similar argument shows that, for every $1 \leq p < q < \infty$, the norms $\|\cdot\|_p$ and $\|\cdot\|_q$ on ℓ_p are not equivalent.
- For every $n \in \mathbb{N}^+$, the norms $\|\cdot\|_p$ and $\|\cdot\|_\infty$ on $\mathbb{K}^n$ are equivalent. As

before, we already have $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_p$ for all $\mathbf{x} \in \mathbb{K}^n$. Set $c = \left(\frac{1}{n}\right)^{1/p}$.

Then for every $\mathbf{x} \in \mathbb{K}^n$ we have

$$c \|\mathbf{x}\|_p = \left(\frac{1}{n} \sum_{i=0}^{n-1} |x_i|^p \right)^{1/p} \leq \left(\max_{0 \leq i \leq n-1} |x_i|^p \right)^{1/p} = \max_{0 \leq i \leq n-1} |x_i| = \|\mathbf{x}\|_\infty$$

Therefore $\|\cdot\|_p$ and $\|\cdot\|_\infty$ on $\mathbb{K}^n$ are equivalent.

These norms, which are defined in the same way, being not equivalent on an infinite-dimensional space but being equivalent on a finite-dimensional space is no accident. Indeed, our next goal is to show that *all norms on a finite-dimensional normed space are equivalent*. Towards this goal, we will first learn about bounded linear maps between normed spaces.

1.4. Boundedness vs. Continuity. Throughout this subsection, fix normed spaces $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$. A linear map $T : E \rightarrow F$ is said to be *bounded* if there exists $M > 0$ such that

$$\|T(x)\|_F \leq M \|x\|_E$$

for all $x \in E$. It is not difficult to check that $T : E \rightarrow F$ is bounded if and only if T maps bounded sets to bounded sets.⁶ It turns out that a linear map being bounded is the same as it being continuous.

Theorem 3. *Let $T : E \rightarrow F$ be a linear map. Then the following are equivalent.*

- (a) *T is bounded.*
- (b) *T is uniformly continuous.*
- (c) *T is continuous.*
- (d) *T is continuous at $\hat{x}$ for some $\hat{x} \in E$.*

Proof. The implications (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d) are clear. We will only prove the implication (d) $\Rightarrow$ (a).

Suppose that T is continuous at $\hat{x}$ for some $\hat{x} \in E$. Using the definition of continuity at $\hat{x}$ with $\epsilon = 1$, we can find some $\delta > 0$ such that for every $x \in E$, if $\|x - \hat{x}\|_E < \delta$, then $\|T(x) - T(\hat{x})\|_F \leq 1$. Set $M = \frac{2}{\delta} > 0$. For every $x \in X$ with $x \neq \mathbf{0}$, we have

$$\left\| \hat{x} + \frac{\delta x}{2 \|x\|_E} - \hat{x} \right\|_E < \delta \quad \text{implying that} \quad \frac{\delta}{2 \|x\|_E} \|T(x)\|_F \leq 1$$

⁶Recall that a subset of a metric space is called bounded if it is contained in some ball.

which gives us $\|T(x)\|_F \leq M \|x\|_E$ for every $x \in E$ with $x \neq \mathbf{0}$. The inequality clearly holds for $x = \mathbf{0}$. Therefore T is bounded. $\square$

As a result of Theorem 3, while considering linear maps between normed spaces, we shall use the phrases “bounded linear maps” and “continuous linear maps” interchangeably. Let us now mention some basic but important consequences of Theorem 3.

Let $T : E \rightarrow F$ be a linear map. Set $G = T[E] = \{T(x) \in F : x \in E\}$. Then the inverse map $T^{-1} : G \rightarrow E$ exists and is continuous if and only if there exists $m > 0$ such that $m \|x\|_E \leq \|T(x)\|_F$ for all $x \in E$. To see the left-to-right direction, assume that $T^{-1} : G \rightarrow E$ exists and is continuous. Since T^{-1} is also linear, it follows from Theorem 3 that T^{-1} is bounded. So there exists $M > 0$ with $\|T^{-1}(y)\|_E \leq M \|y\|_F$ for all $y \in T[E]$. Setting $m = 1/M$, we have that $m \|x\|_E \leq \|T(x)\|_F$ for all $x \in E$. To see the right-to-left direction, suppose that there exists $m > 0$ such that $m \|x\|_E \leq \|T(x)\|_F$ for all $x \in E$. Then T has to be one-to-one since otherwise any non-zero element in the kernel would violate the inequality. Therefore $T^{-1} : G \rightarrow E$ exists. Setting $M = 1/m$, we see from the initial inequality that $\|T^{-1}(y)\|_E \leq M \|y\|_F$ for all $y \in G$. Therefore T^{-1} is bounded and hence is continuous by Theorem 3.

A map $T : E \rightarrow F$ is said to be an *isomorphism* if it is a linear map and a homeomorphism. This is one of the notions of “sameness” that we will be using for normed spaces.⁷ The other notion of “sameness” is the notion of isometric isomorphism. A map $T : E \rightarrow F$ is said to be an *isometric isomorphism* if T is a bijection and $\|T(x)\|_F = \|x\|_E$ for all $x \in E$.

An immediate corollary to our previous observation is that a surjective linear map $T : E \rightarrow F$ is an isomorphism if and only if there exist $m, M > 0$ such that for all $x \in E$ we have $m \|x\|_E \leq \|T(x)\|_F \leq M \|x\|_E$. Note that, in the case of $m = M = 1$, we have an isometric isomorphism.

Having defined isomorphism of normed spaces, it is now straightforward to check that if $\|\cdot\|$ and $\|\cdot\|'$ are equivalent norms on E , then $(E, \|\cdot\|)$ and $(E, \|\cdot\|')$ are isomorphic normed spaces via the identity map.

1.5. Finite-dimensional normed spaces. In this subsection, we shall show that all norms on a finite-dimensional normed space are equivalent to each other. In order to do this, instead of showing that any two norms are equivalent to each other, we shall show that every norm is equivalent to a fixed norm that we are going to define in a moment.

⁷One can also talk about mere isomorphisms of the underlying vector spaces which does not take the topological structure into the account and also talk about isometric isomorphisms which not only preserves the topological structure but also the metric structure.

Fix a finite-dimensional normed space $(E, \|\cdot\|)$. Since E is finite-dimensional, say, $\dim_{\mathbb{K}}(E) = n$, there exists a basis $B = \{v_1, \dots, v_n\} \subseteq E$. By scaling each basis element if necessary, we may assume without loss of generality that $\|v_i\| = 1$ for every $1 \leq i \leq n$. Consider the map $\|\cdot\|' : E \rightarrow \mathbb{R}$ given by

$$\|x\|' = \sum_{i=1}^n |\xi_i| \quad \text{where } x = \sum_{i=1}^n \xi_i v_i$$

Since B is a basis, every element of E can be uniquely represented as a linear combination of the elements of B and hence $\|\cdot\|'$ is indeed well-defined. We leave it to the reader to check that $\|\cdot\|'$ is indeed a norm on E . We are now ready to state and prove the main theorem of this subsection

Theorem 4. $\|\cdot\|$ and $\|\cdot\|'$ on E are equivalent.

Proof. Let $x \in E$ and suppose that $x = \sum_{i=1}^n \xi_i v_i$ is the unique representation of x as a linear combination of basis elements. Observe that, since $\|\cdot\|$ is a norm and the elements of B are unit vectors, we have that

$$\|x\| = \left\| \sum_{i=1}^n \xi_i v_i \right\| \leq \sum_{i=1}^n |\xi_i| \|v_i\| = \sum_{i=1}^n |\xi_i| = \|x\|'$$

It now remains to find a constant $c > 0$ such that $c\|\cdot\|' \leq \|\cdot\|$. Consider the map $T : \mathbb{K}^n \rightarrow E$ given by

$$T(\xi_1, \dots, \xi_n) = \sum_{i=1}^n \xi_i v_i$$

for every $(\xi_1, \dots, \xi_n) \in \mathbb{K}^n$. Clearly T is a linear map. Observe that B being linearly independent implies that T has trivial kernel and hence is one-to-one; and B spanning E implies that T is onto. Thus we have a vector space isomorphism from $\mathbb{K}^n$ to E . By construction, we also have

$$\|T(\alpha_1, \dots, \alpha_n)\|' = \|(\alpha_1, \dots, \alpha_n)\|_1$$

for every $(\alpha_1, \dots, \alpha_n) \in \mathbb{K}^n$ which implies that T is bounded. It now follows from Theorem 3 that T is continuous. Consider the set

$$S = \{x \in E : \|x\|' = 1\} = T \left(\left\{ (\xi_1, \dots, \xi_n) \in \mathbb{K}^n : \sum_{i=1}^n |\xi_i| = 1 \right\} \right)$$

Being the continuous image of a compact set, S is compact. Moreover, the map $\|\cdot\| : E \rightarrow \mathbb{R}$ is continuous with respect to the metric on E associated to $\|\cdot\|'$ because we have

$$|\|x\| - \|y\|| \leq \|x - y\| \leq \|x - y\|'$$

for all $x, y \in E$. Therefore, by the Extreme Value Theorem, $\|\cdot\|$ attains its minimum value on S at some $s_0 \in S$. In other words,

$$\|s_0\| = \min\{\|s\| : s \in S\}$$

Since $\|s_0\|' = 1$, we must have $s_0 \neq \mathbf{0}$ and so $c = \|s_0\| > 0$. Let $x \in E$. Then $x/\|x\|' \in S$ and so, by the choice of s_0 , we have that

$$c = \|s_0\| \leq \left\| \frac{x}{\|x\|'} \right\|$$

which implies that $c\|x\|' \leq \|x\|$. Therefore $\|\cdot\|$ is equivalent to $\|\cdot\|'$. $\square$

Corollary 5. *Let $(E, \|\cdot\|)$ be a normed space of dimension $n \in \mathbb{N}$. Then $(E, \|\cdot\|)$ is isomorphic to $(\mathbb{K}^n, \|\cdot\|_p)$.*

Corollary 6. *All finite-dimensional normed spaces are complete, separable and have the Heine-Borel property.*

Corollary 7. *Let $(E, \|\cdot\|)$ and $(F, \|\cdot\|')$ be normed spaces with $\dim_{\mathbb{K}}(E) < \infty$. Then every linear map $T : E \rightarrow F$ is continuous.*

Proof. Let $T : E \rightarrow F$ be a linear map. Consider the map $\|\cdot\|'' : E \rightarrow \mathbb{R}$ given by

$$\|x\|'' = \|x\| + \|T(x)\|'$$

for every $x \in E$. Then one can check that $\|\cdot\|''$ is indeed a norm. Since E is finite-dimensional, $\|\cdot\|''$ is equivalent to $\|\cdot\|$ by Theorem 4. Hence there exists $M > 0$ such that

$$\|T(x)\|' \leq \|x\| + \|T(x)\|' = \|x\|'' \leq M\|x\|$$

for all $x \in E$. It follows that T is bounded and so is continuous. $\square$

Let us now see examples which illustrate that Corollary 7 fails for infinite-dimensional spaces. Consider the subspace

$$C^1([0, 1]) = \{f : [0, 1] \rightarrow \mathbb{R} : f \text{ is continuously differentiable}\}$$

of the normed space $(C([0, 1]), \|\cdot\|_{\infty})$ with $\mathbb{K} = \mathbb{R}$, where $\|\cdot\|_{\infty}$ denotes the supremum norm given by

$$\|f\|_{\infty} = \sup_{x \in [0, 1]} |f(x)|$$

It is not difficult to check that the differentiation operator $T : C^1([0, 1]) \rightarrow C([0, 1])$ given by

$$T(f) = f'$$

is linear. Set $f_n(x) = \sin(2\pi nx)$ for each $n \in \mathbb{N}^+$. Then we have $\|f_n\|_{\infty} = 1$ for every $n \in \mathbb{N}^+$. However, we also have

$$\|T(f_n)\|_{\infty} = \|f_n'\|_{\infty} \geq |f_n'(0)| = 2\pi n$$

for each $n \in \mathbb{N}$. Hence T cannot be bounded and is not continuous. Another example can be given using the sequence spaces. Consider the subspace

$$c_{00} = \{\mathbf{x} \in c_0 : \exists n \in \mathbb{N} \forall m \in \mathbb{N} (m \geq n \rightarrow x_m = 0)\}$$

of the normed space $(c_0, \|\cdot\|_\infty)$. Then the map $T : c_{00} \rightarrow c_{00}$ given by

$$T(\mathbf{x}) = (nx_n)_{n \in \mathbb{N}}$$

for all $\mathbf{x} \in c_0$ is a linear map. Now consider $\mathbf{w}_n = \sum_{i=0}^n \mathbf{e}_i$ for each $n \in \mathbb{N}$ where $\mathbf{e}_i = (\delta_{in})_{n \in \mathbb{N}}$ as before. Then we have $\|\mathbf{w}_n\|_\infty = 1$ but $\|T(\mathbf{w}_n)\|_\infty = n$. This implies that T is not bounded and hence not continuous. Indeed, it is not difficult to mimic this construction to show that we can always find linear unbounded maps $T : E \rightarrow F$ whenever E is infinite-dimensional.

1.6. The closed unit ball. Analyzing the proof of Theorem 4, one can see that the closed unit ball

$$B_{\|\cdot\|}[\mathbf{0}, 1] = \{x \in E : \|x\| \leq 1\}$$

centered at $\mathbf{0}$ in a finite-dimensional normed space is compact. It turns out that the converse also holds. In order to prove this, we will need the following lemma which is also important on its own.

Lemma 1 (Riesz's lemma). *Let $(E, \|\cdot\|)$ be a normed space and $F \subsetneq E$ be a closed subspace. Then for every $0 < \theta < 1$ there exists $x_\theta \in E$ such that*

- $\|x_\theta\| = 1$ and
- $\|x - x_\theta\| \geq \theta$ for every $x \in F$.

Proof. Since $F \subsetneq E$, there exists $x' \in E - F$. Set $d = \inf_{x \in F} \|x - x'\|$. Then we have $d > 0$ as F is closed. Observe that we have $d < d/\theta$ and hence, by the definition of infimum, there exists $x'' \in F$ such that $d \leq \|x'' - x'\| \leq d/\theta$. Set

$$x_\theta = \frac{x' - x''}{\|x' - x''\|}$$

Clearly $\|x_\theta\| = 1$. Moreover, for every $x \in F$, we have that

$$\|x - x_\theta\| = \left\| x - \frac{x' - x''}{\|x' - x''\|} \right\| = \left\| \frac{\|x' - x''\| x + x'' - x'}{\|x' - x''\|} \right\|$$

Since F is a subspace, we clearly have $u = \|x'' - x'\| x + x'' \in F$. It now follows from the choice of d that

$$\|x - x_\theta\| = \frac{\|u - x'\|}{\|x' - x''\|} \geq \frac{d}{\|x' - x''\|} \geq \theta$$

which completes the proof. $\square$

Intuitively speaking, Riesz's lemma tells that that we can get away from points on a proper linear closed subspace by a distance close to 1 using points on the unit closed sphere. We are now ready to prove the main theorem of this subsection.

Theorem 8. *Let $(E, \|\cdot\|)$ be a normed space. Then*

$$S = B_{\|\cdot\|}[\mathbf{0}, 1] = \{x \in E : \|x\| \leq 1\}$$

is compact if and only if E is finite-dimensional.

Proof. The right-to-left direction follows from Corollary 6 as E has the Heine-Borel property.

To prove the left-to-right direction, assume that E is infinite-dimensional. We recursively define a sequence $(x_n)_{n \in \mathbb{N}}$ of points of S as follows.

- Pick any $s \in S$ and set $x_0 = s$.
- Let $n \in \mathbb{N}$ and suppose that $x_0, \dots, x_n \in S$ has been defined. Set

$$X_n = \langle x_0, \dots, x_n \rangle$$

Observe that X is closed subspace⁸. Moreover, since E is infinite-dimensional, $X_n \subsetneq E$ and hence, applying Riesz's lemma with $\theta = 1/2$, there exists $s \in S$ such that $\|s - x_i\| \geq 1/2$ for all $1 \leq i \leq n$. Set $x_{n+1} = s$.

By construction, the sequence $(x_n)_{n \in \mathbb{N}}$ does not have any Cauchy subsequence and hence not contains a convergent subsequence. This implies that S is not sequentially compact and hence is not compact. $\square$

Corollary 9. *In an infinite-dimensional normed space, the unit closed ball is not compact and so the Heine-Borel property does not hold.*

It is quite natural to ask whether the condition $\theta < 1$ is necessary in the statement of Riesz's lemma. Let us now provide a counterexample for the case $\theta = 1$. Set

$$X = \{f \in C([0, 1]) : f(0) = 0\}$$

with $\mathbb{K} = \mathbb{R}$. Then $(X, \|\cdot\|_\infty)$ is a normed space. Consider the subspace

$$Y = \{f \in X : \int_0^1 f(x)dx = 0\}$$

It is an exercise to the reader to check that Y is a proper closed subspace of X . If Riesz's lemma held for $\theta = 1$, then we would be able to find $f \in X$ such that $\|f\|_\infty = 1$ and $\|f - g\|_\infty \geq 1$ for all $g \in Y$. Observe that for all $h \in X - Y$ we have

$$f - \frac{\int_0^1 f(x)dx}{\int_0^1 h(x)dx} h \in Y$$

which implies

$$1 \leq \left\| f - \left(f - \frac{\int_0^1 f(x)dx}{\int_0^1 h(x)dx} h \right) \right\|_\infty = \left| \frac{\int_0^1 f(x)dx}{\int_0^1 h(x)dx} \right| \|h\|_\infty$$

and so

$$\left| \int_0^1 h(x)dx \right| \leq \left| \int_0^1 f(x)dx \right| \|h\|_\infty$$

⁸It is an exercise to the reader to check that a finite-dimensional subspace of an arbitrary normed space is closed.

For each $n \in \mathbb{N}^+$, set $g_n(x) = \sqrt[n]{x}$ in which case we have $\|g_n\|_\infty = 1$ and $g_n \in X - Y$. By the previous observation, it then follows that

$$1 = \sup_{n \in \mathbb{N}^+} \frac{n}{n+1} = \sup_{n \in \mathbb{N}^+} \int_0^1 g_n(x) dx \leq \left| \int_0^1 f(x) dx \right|$$

However, this cannot happen since f is continuous on $[0, 1]$, $f(0) = 0$ and $\|f\|_\infty = 1$. Thus Riesz's lemma is not true for $\theta = 1$ in general.

Back to the compactness of the closed unit ball... A variation of Theorem 8 can be proven without Riesz's lemma as follows.

Theorem 10. *Let $(E, \|\cdot\|)$ be a normed space. Then*

$$S = B_{\|\cdot\|}[\mathbf{0}, 1] = \{x \in E : \|x\| \leq 1\}$$

is totally bounded if and only if E is finite-dimensional.

Proof. As before, Corollary 6 immediately gives the -right-to-left direction. For the left-to-right direction, suppose that S is totally boundedness. Using $\epsilon = 1/2$ in the definition of total boundedness, there exist $s_1, \dots, s_n \in S$ such that

$$S \subseteq \bigcup_{i=1}^n B(s_i, 1/2)$$

Set $F = \langle s_1, \dots, s_n \rangle$. We claim that $S \subseteq F$. Let $x \in S$. We will inductively define a sequence $(x_n)_{n \in \mathbb{N}}$ of points of F satisfying $\|x - x_n\| < 1/2^n$ as follows.

- By construction, we can choose some $1 \leq i \leq n$ such that $\|x - s_i\| < 1/2$. Set $x_0 = s_i \in F$.
- Let $n \in \mathbb{N}$. Suppose that $x_n \in F$ has been defined so that $\|x - x_n\| < 1/2^n$. We wish to find $x_{n+1} \in F$ such that $\|x - x_{n+1}\| < 1/2^{n+1}$. Since

$$\frac{x - x_n}{\|x - x_n\|} \in S$$

there exists $1 \leq j \leq n$ such that

$$\left\| \frac{x - x_n}{\|x - x_n\|} - s_j \right\| < \frac{1}{2}$$

Therefore, by the inductive assumption, we have

$$\|x - x_n - \|x - x_n\| s_j\| < \frac{\|x - x_n\|}{2} < \frac{1}{2^{n+1}}$$

Set $x_{n+1} = x_n + \|x - x_n\| s_j \in F$. Then $x_{n+1} \in F$ because $x_n, s_j \in F$. Moreover, we have $\|x - x_{n+1}\| < 1/2^{n+1}$.

By construction, we have $\lim_{n \rightarrow \infty} x_n = x$. Since F is closed, we have $x \in F$. Therefore $S \subseteq F$. This immediately implies that $E = F$ since every vector can be scaled to fall into S . Hence E is finite-dimensional. $\square$

1.7. Banach spaces. In this subsection, we will learn the definition, some basic properties and some examples Banach spaces which will be central to the remaining of the course.

A normed space $(E, \|\cdot\|)$ is said to be a *Banach space* if the associated metric $d(x, y) = \|x - y\|$ is complete. It turns out that most of the spaces that we have dealt with up to now are indeed Banach.

Theorem 11. *The following normed spaces are Banach.*

- Every finite-dimensional normed space.
- $(\ell_p, \|\cdot\|_p)$ where $1 \leq p \leq \infty$.
- $(c_0, \|\cdot\|_\infty)$.
- $(B(X), \|\cdot\|_\infty)$ for any non-empty set X .
- $(C(X), \|\cdot\|_\infty)$ for any compact topological space X .

Proof. We have already proven that finite-dimensional normed spaces are Banach. All the remaining examples must have been shown to be complete metric spaces in MATH349 while covering complete metrics. In case these examples were not covered, we will just show that $(\ell_\infty, \|\cdot\|_\infty)$ is Banach as an example.

Let $(\mathbf{x}_n)_{n \in \mathbb{N}}$ be a Cauchy sequence in $(\ell_\infty, \|\cdot\|_\infty)$. Observe that each $\mathbf{x}_n$ is itself a bounded sequence of elements of $\mathbb{K}$. To avoid confusion due to using multiple indices, we will denote the entries of such a sequence $\mathbf{x}_n$ as follow.

$$\mathbf{x}_n = (x_n(0), x_n(1), \dots) = (x_n(i))_{i \in \mathbb{N}}$$

Given $\epsilon \in \mathbb{R}^+$, by Cauchyness of $(\mathbf{x}_n)_{n \in \mathbb{N}}$, we can find some $N \in \mathbb{N}$ such that for all integers $n, m \geq N$ we have that

$$\sup_{i \in \mathbb{N}} |x_n(i) - x_m(i)| = \|\mathbf{x}_n - \mathbf{x}_m\|_\infty < \epsilon$$

This immediately implies that, for every $i \in \mathbb{N}$, the sequence $(x_n(i))_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{K}$. Since $\mathbb{K}$ is complete, there exists $\alpha_i \in \mathbb{K}$ such that

$$\lim_{n \rightarrow \infty} x_n(i) = \alpha_i$$

Consider the sequence $\alpha = (\alpha_i)_{i \in \mathbb{N}}$. We will first show that $\alpha \in \ell_\infty$.

Recall that any norm is a uniformly continuous function. In particular, the map $\cdot \mapsto \|\cdot\|_\infty$ is uniformly continuous with Lipschitz constant 1. It follows $(\|\mathbf{x}_n\|_\infty)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{K}$ which necessarily is bounded. Choose $K \in \mathbb{K}$ such that

$$\sup_{n \in \mathbb{N}} \|\mathbf{x}_n\|_\infty \leq K$$

Let $i \in \mathbb{N}$. Using $\epsilon = 1$ in the definition of convergence, we see that there exists $M \in \mathbb{N}$ such that for all integers $m \geq M$ we have

$$|\alpha_i - x_m(i)| < 1$$

It follows that

$$|\alpha_i| \leq |\alpha_i - x_M(i)| + |x_M(i)| \leq 1 + \sup_{j \in \mathbb{N}} |x_M(j)| = 1 + \|\mathbf{x}_M\| \leq 1 + K$$

Since this inequality holds for all $i \in \mathbb{N}$, we have that $\alpha \in \ell_\infty$. It now remains to show that $\mathbf{x}_n \rightarrow \alpha$ in $(\ell_\infty, \|\cdot\|_\infty)$.

Given $\epsilon \in \mathbb{R}^+$, using the definition of the Cauchy-ness of the sequence $(\mathbf{x}_n)_{n \in \mathbb{N}}$ with $\epsilon/2$, we can get $N \in \mathbb{N}$ such that for all integers $n, m \geq N$ we have

$$\sup_{i \in \mathbb{N}} |x_n(i) - x_m(i)| = \|\mathbf{x}_n - \mathbf{x}_m\|_\infty < \epsilon/2$$

In particular, for all $i \in \mathbb{N}$ and for all integers $n, m \geq N$, we have

$$|x_n(i) - x_m(i)| < \epsilon/2$$

Taking the limit as $m \rightarrow \infty$, we obtain

$$|x_n(i) - \alpha_i| \leq \epsilon/2$$

It follows that

$$\sup_{i \in \mathbb{N}} |x_n(i) - \alpha_i| = \|\mathbf{x}_n - \alpha\| \leq \epsilon/2 < \epsilon$$

Therefore $\lim_{n \rightarrow \infty} \mathbf{x}_n = \alpha$ in $(\ell_\infty, \|\cdot\|_\infty)$ which completes the proof that $(\ell_\infty, \|\cdot\|_\infty)$ is a Banach space. $\square$

Given a complete metric space (X, d) and a subset $Y \subseteq X$, recall from MATH349 that, the metric space (Y, d_Y) is complete if and only if Y is closed in X . Here d_Y denotes the restriction of d to the subset Y . Consequently, we have the following proposition.

Proposition 4. *Let $(E, \|\cdot\|)$ be a Banach space and $F \subseteq E$ be a subspace. Then $(F, \|\cdot\|)$ is Banach if and only if F is closed in E .*

Using this proposition, one can deduce that $(c_0, \|\cdot\|_\infty)$ is Banach since c_0 is closed in ℓ_∞ . On the other hand, $(c_{00}, \|\cdot\|_\infty)$ is not Banach since the subspace c_{00} of ℓ_∞ is not closed. To see the latter claim, for each $i \in \mathbb{N}$, consider the sequence $\mathbf{x}_i = (x_i(0), x_i(1), \dots)$ where $x_i(n)$ is defined as follows.

$$x_i(n) = \begin{cases} 0 & \text{if } n > i \\ \frac{1}{n+1} & \text{if } n \leq i \end{cases}$$

Then $\mathbf{x}_i \in c_{00}$ for all $i \in \mathbb{N}$. Setting $\mathbf{y} = \left(\frac{1}{n+1} \right)_{n \in \mathbb{N}}$, we see that

$$\|\mathbf{y} - \mathbf{x}_i\|_\infty = \frac{1}{i+2} \xrightarrow{i \rightarrow \infty} 0$$

which means that $\lim_{i \rightarrow \infty} \mathbf{x}_i = \mathbf{y}$. However $\mathbf{y} \notin c_{00}$ and hence c_{00} is not closed in ℓ_∞ .

We will next see that any normed space can be embedded into a Banach space as a dense subset.

Theorem 12. *Let $(E, \|\cdot\|_E)$ be a normed space. Then there exists a Banach space $(F, \|\cdot\|_F)$ so that there is a linear isometry $\Phi : E \rightarrow F$ with the property that $\Phi(E)$ is dense in F .*

Proof sketch. Set $K = \{(x_n)_{n \in \mathbb{N}} \in E^{\mathbb{N}} : (x_n)_{n \in \mathbb{N}} \text{ is Cauchy}\}$ and consider the equivalence relation $\sim$ on K given by

$$(x_n)_{n \in \mathbb{N}} \sim (y_n)_{n \in \mathbb{N}} \text{ if and only if } \lim_{n \rightarrow \infty} (x_n - y_n) = \mathbf{0}$$

Consider the operations $+$ and $\cdot$ on the quotient set $K/\sim$ given by

$$\begin{aligned} [(x_n)_{n \in \mathbb{N}}] + [(y_n)_{n \in \mathbb{N}}] &= [(x_n + y_n)_{n \in \mathbb{N}}] \\ \alpha \cdot [(x_n)_{n \in \mathbb{N}}] &= [(\alpha x_n)_{n \in \mathbb{N}}] \end{aligned}$$

for all $(x_n)_{n \in \mathbb{N}}, (y_n)_{n \in \mathbb{N}} \in K$ and $\alpha \in K$. A straightforward but tedious argument shows that these operations are indeed well-defined. Moreover, one can check that the quotient set $K/\sim$ together with these operations is a $\mathbb{K}$ -vector space. Now consider the map $\|\cdot\|' : K \rightarrow \mathbb{R}$ given by

$$\|[(x_n)_{n \in \mathbb{N}}]\|' = \lim_{n \rightarrow \infty} \|x_n\|$$

for all $(x_n)_{n \in \mathbb{N}} \in K$. Then the map $\|\cdot\|'$ is well-defined and is a norm.

Following the same arguments done in MATH349 for the Cauchy-completion construction of a metric space, one sees that $(K/\sim, \|\cdot\|')$ is a Banach space and that the map $\Phi : E \rightarrow K/\sim$ given by

$$\Phi(e) = [(e, e, e, \dots)]$$

for all $e \in E$ is an isometry with dense image. The linearity of Φ trivially follows from the definition of $+$ and $\cdot$ on $K/\sim$. $\square$

Before we conclude this subsection, let us record the following proposition that will be of use in later topics.

Proposition 5. *Let $(E, \|\cdot\|_E)$ be a Banach space and let $(x_n)_{n \in \mathbb{N}}$ be a sequence of points of E . If the series $\sum_{i=0}^{\infty} \|x_i\|$ converges, then $\sum_{i=0}^{\infty} x_i$ converges in $(E, \|\cdot\|_E)$.*

Proof. Assume $\sum_{i=0}^{\infty} \|x_i\|$. For each $n \in \mathbb{N}$, set $s_n = \sum_{i=0}^n x_i \in E$. We will show that $(s_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in E . Let $\epsilon \in \mathbb{R}^+$.

Recall that the tail of convergent series of real numbers goes to 0 and hence there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have that

$$\sum_{i=n}^{\infty} \|x_i\| < \epsilon$$

The triangle inequality now implies that

$$\|s_m - s_n\| \leq \sum_{i=n+1}^m \|x_i\| \leq \sum_{i=n}^{\infty} \|x_i\| < \epsilon$$

for all $m > n \geq N$. Therefore $(s_n)_{n \in \mathbb{N}}$ is a Cauchy sequence. Since $(E, \|\cdot\|_E)$ is a Banach space, $\sum_{i=0}^{\infty} x_i$ converges. $\square$

We would like to note that this proposition fails if one drops the Banachness condition. For example, consider the sequence $(\mathbf{x}_n)_{n \in \mathbb{N}}$ of points of $(c_{00}, \|\cdot\|_{\infty})$ where

$$\mathbf{x}_n = (x_n(i))_{i \in \mathbb{N}} = \left(\frac{\delta_{ni}}{(i+1)^2} \right)_{i \in \mathbb{N}}$$

Then we have

$$\sum_{n=0}^{\infty} \|\mathbf{x}_n\|_{\infty} = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

but $\sum_{n=0}^{\infty} \mathbf{x}_n$ does not converge to any point in c_{00} . On the other hand, it *does* converge to the sequence $\left(\frac{1}{(i+1)^2} \right)_{i \in \mathbb{N}}$ in the Banach space $(\ell_{\infty}, \|\cdot\|_{\infty})$.

1.8. Quotient spaces and quotient norm. In this section, we will learn when and how a quotient (vector) space can be given a norm space structure. Let $(E, \|\cdot\|_E)$ be a normed space and $F \subseteq E$ be a subspace. We know that the quotient set E/F is a vector space with the operations given by

$$(x + F) + (y + F) = (x + y) + F$$

$$\alpha(x + F) = \alpha x + F$$

for all $\alpha \in \mathbb{K}$ and all $x, y \in E$.

What is a reasonable norm on the quotient space E/F ? How should we define the norm of a coset, possibly using its representatives? One natural approach would be to measure the distance of vector x to the set F by setting

$$\|x + F\| = \inf_{w \in F} \|x - w\|_E$$

Clearly we should first check that this definition is independent of the coset representative x that we used. Suppose that $x, x' \in E$ are such that $x + F = x' + F$. Then, by definition, we have $x - x' \in F$. Now, for every $y \in F$, we have that $(x - x') + y \in F$ as F is a subspace and so

$$\|x + F\| = \inf_{w \in F} \|x - w\|_E \leq \|x - ((x - x') - y)\|_E = \|x' - y\|_E$$

Taking the infimum of right-hand side as $y \in F$ now gives

$$\|x + F\| = \inf_{w \in F} \|x - w\|_E \leq \inf_{y \in F} \|x' - y\|_E = \|x' + F\|$$

Applying the same argument symmetrically, we obtain that $\|x + F\| = \|x' + F\|$. Hence the definition of $\|\cdot\|$ indeed makes sense.

Let us now check whether $\|\cdot\| : E/F \rightarrow \mathbb{R}$ is a norm. The reader may verify using the properties of infimum that $\|\cdot\|$ is actually a *seminorm*, i.e. it satisfies the latter two properties of being a norm, namely, that

- $\|\alpha(x + F)\| = |\alpha| \|x + F\|$ for all $\alpha \in \mathbb{K}$ and for all $x \in E$, and
- $\|(x + F) + (y + F)\| \leq \|x + F\| + \|y + F\|$ for all $x, y \in E$.

On the other hand, there is no reason for $\|x + F\| = 0$ to imply $x + F = F = \mathbf{0}_{E/F}$ which simply means $x \in F$. It is possible that the distance of x to F is zero, i.e. x is a limit point of F but x is not in F . Of course, this would mean that F is not closed. It turns out that this is the only obstruction for $\|\cdot\|$ to be not a norm.

Proposition 6. $\|\cdot\|$ is a norm if and only if F is closed.

Proof. Suppose that $\|\cdot\|$ is a norm. We wish to show that F is closed. Let $x \in E$ be a limit point of F . Then there exists a sequence $(x_n)_{n \in \mathbb{N}}$ of elements of F such that $\lim_{n \rightarrow \infty} x_n = x$ which means that

$$0 \leq \|x + F\| = \inf_{y \in F} \|x - y\|_E = \inf_{n \in \mathbb{N}} \|x - x_n\|_E = 0$$

Since $\|\cdot\|$ is a norm, we obtain that $x + F = \mathbf{0}_{E/F} = F$, so $x \in F$.

Now suppose F is closed. We already know that $\|\cdot\|$ is a seminorm. So it suffices to show that, for all $S \in E/F$ with $S \neq \mathbf{0}_{E/F}$, we have $\|S\| > 0$. Assume towards a contradiction that there is $S \in E/F$ such that $S \neq \mathbf{0}_{E/F} = F$ and $\|S\| > 0$. Let $x \in E$ be a representative for the coset S . Since

$$\|S\| = \|x + F\| = \inf_{y \in F} \|x - y\|_E = 0$$

for each $n \in \mathbb{N}$, we can choose $x_n \in F$ such that $\|x - x_n\| < \frac{1}{n+1}$. But then, we have $\lim_{n \rightarrow \infty} x_n = x$ and hence $x \in F$, by the closedness of F . This contradicts that $x + F = S \neq \mathbf{0}_{E/F} = F$. $\square$

From now on, we will refer to $\|\cdot\|$ as the quotient norm, whenever the subspace F is closed. It is an exercise to the reader to check that the canonical surjection $\pi : E \rightarrow E/F$ given by $\pi(x) = x + F$ is a continuous map whenever F is closed and E/F is endowed with the quotient norm structure.

Having determined when $\|\cdot\|$ is a norm on the quotient set, the next obvious question is to ask which properties are inherited by the quotient normed space from the ambient space. It turns out that quotient normed spaces are Banach provided that the ambient space is Banach.

Theorem 13. Let $(E, \|\cdot\|_E)$ be a Banach space and F be a closed subspace and let $\|\cdot\|$ be the quotient norm on E/F . Then $(E/F, \|\cdot\|)$ is Banach.

Proof. Let $(x_n + F)_{n \in \mathbb{N}}$ be a Cauchy sequence in E/F . We wish to show that this sequence has a limit in E/F . Observe that, if a *subsequence* of this sequence has

a limit in E/F , then the sequence itself will have the same limit because it is a Cauchy sequence. Thus, for the purposes of finding a limit, we may pass to any subsequence. So we may assume without loss of generality that

$$\|(x_{n+1} + F) - (x_n + F)\| < \frac{1}{2^n}$$

for all $n \in \mathbb{N}$, by passing to a suitable subsequence if necessary. We will next construct a sequence $(z_n)_{n \in \mathbb{N}}$ of points of E such that

- $z_n + F = x_n + F$,
- $\|z_n - z_{n+1}\|_E < \frac{1}{2^n}$

for all $n \in \mathbb{N}$ by recursion as follows.

- For $n = 0$, we set $z_0 = x_0$,
- Let $n \in \mathbb{N}$ and suppose that z_n is chosen so that the required properties are satisfied. We will show that we can choose a point z_{n+1} that satisfies the required properties. Note that, since $z_n + F = x_n + F$, we have that $x_n - z_n = f$ for some $f \in F$. Since

$$\begin{aligned} \frac{1}{2^n} &> \|(x_{n+1} + F) - (x_n + F)\| = \|(x_{n+1} - x_n) + F\| \\ &= \inf_{y \in F} \|x_{n+1} - x_n - y\|_E \\ &= \inf_{y \in F} \|x_{n+1} - (f + z_n) - y\|_E \\ &= \inf_{y \in F} \|(x_{n+1} - f - y) - z_n\|_E \end{aligned}$$

Therefore, for some $y' \in F$, we must have

$$\frac{1}{2^n} > \|(x_{n+1} - f - y') - z_n\|_E$$

Choose $z_{n+1} = x_{n+1} - f - y'$. Then $z_{n+1} + F = x_{n+1} + F$ as $-f - y' \in F$. Moreover, by construction itself, the inequality

$$\|z_n - z_{n+1}\|_E < \frac{1}{2^n}$$

is satisfied.

Thus such a sequence $(z_n)_{n \in \mathbb{N}}$ of points of E can be constructed by induction. We claim that $(z_n)_{n \in \mathbb{N}}$ is Cauchy. Given $\epsilon > 0$, as the tail of a convergent sequence goes to zero, there exists $N \in \mathbb{N}$ such that $\sum_{i=N}^{\infty} 2^{-i} < \epsilon$. But then, for all integers $m > n \geq N$, we have

$$\|z_n - z_m\|_E = \left\| \sum_{i=n}^{m-1} (z_i - z_{i+1}) \right\|_E \leq \sum_{i=n}^{m-1} \|z_i - z_{i+1}\|_E \leq \sum_{i=n}^{\infty} \frac{1}{2^i} < \epsilon$$

by the triangle inequality. Since $(z_n)_{n \in \mathbb{N}}$ is Cauchy and E is Banach, there exists a point $z \in E$ such that $\lim_{n \rightarrow \infty} z_n = z$. The projection map $\pi : E \rightarrow E/F$ being

continuous now gives us that

$$\lim_{n \rightarrow \infty} x_n + F = \lim_{n \rightarrow \infty} z_n + F = \lim_{n \rightarrow \infty} \pi(z_n) = \pi(z)$$

which is the desired limit.

□

2. LINEAR MAPS AND THE DUAL SPACE

Having learned the basic facts regarding normed spaces, in this section, we are going to focus on “morphisms” between normed spaces.

2.1. The normed space of continuous linear maps. Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed spaces. Consider the set

$$L(E, F) = \{T : E \rightarrow F \mid T \text{ is continuous and linear}\}$$

Recall by Theorem 3 that the set $L(E, F)$ is precisely the set of bounded linear maps from E to F .

Together with addition defined by $(T_1 + T_2)(x) = T_1(x) + T_2(x)$ and scalar multiplication defined by $(\alpha T)(x) = \alpha T(x)$, the set $L(E, F)$ turns into a $\mathbb{K}$ -vector space. It is also not difficult to check that the map $\|\cdot\| : L(E, F) \rightarrow \mathbb{R}$ given by

$$\|T\| = \sup_{\|x\|_E \leq 1} \|T(x)\|_F$$

for all $T \in L(E, F)$ is a norm. Thus $(L(E, F), \|\cdot\|)$ itself is a normed space. While it is almost trivial to observe, let us note the following proposition since it is going to be used over and over again in the course of our study.

Proposition 7. $\|T(x)\|_F \leq \|T\| \|x\|_E$ for all $x \in E$.

Proof. The claim clearly holds for $x = \mathbf{0}$. Let $x \in E$ be such that $x \neq \mathbf{0}$. Since $\left\| \frac{x}{\|x\|_E} \right\|_E \leq 1$, by the definition of $\|T\|$, we have

$$\left\| T \left(\frac{x}{\|x\|_E} \right) \right\|_F \leq \|T\|$$

from which the result follows by linearity of T . □

Let us next see some other ways to compute $\|T\|$ aside from its definition.

Proposition 8. *We have that*

- $\|T\| = \inf\{M > 0 : \|T(x)\|_F \leq M \|x\|_E \text{ for all } x \in E\}$
- $\|T\| = \sup \left\{ \frac{\|T(x)\|_F}{\|x\|_E} : x \neq \mathbf{0} \right\}$
- $\|T\| = \sup\{\|T(x)\|_F : \|x\|_E = 1\}$

Proof. We will prove a chain of inequalities which at the end is going to show that all these quantities are the same.

Let $M > 0$ be such that $\|T(x)\|_F \leq M \|x\|_E$ for all $x \in E$. We wish to show $\|T\| \leq M$. But this is almost immediate since, for every $x \in E$ with $\|x\|_E \leq 1$, we have

$$\|T(x)\|_F \leq M \|x\|_E \leq M$$

by our assumption. This means that

$$\|T\| = \sup_{\|x\|_E \leq 1} \|T(x)\|_F \leq M$$

Since this inequality is true for all such M , taking the infimum over these, we obtain

$$\|T\| \leq \inf\{M > 0 : \|T(x)\|_F \leq M \|x\|_E \text{ for all } x \in E\}$$

Now set $K = \sup\left\{\frac{\|T(x)\|_F}{\|x\|_E} : x \neq \mathbf{0}\right\}$. Then, by definition, for every $x \in E$ with $x \neq \mathbf{0}$ we have

$$\frac{\|T(x)\|_F}{\|x\|_E} \leq K$$

which means $\|T(x)\|_F \leq K \|x\|_E$ for all $x \in E$. Hence we must have

$$\inf\{M > 0 : \|T(x)\|_F \leq M \|x\|_E \text{ for all } x \in E\} \leq \sup\left\{\frac{\|T(x)\|_F}{\|x\|_E} : x \neq \mathbf{0}\right\}$$

Observe that, by the linearity of T , we have that

$$\sup\left\{\frac{\|T(x)\|_F}{\|x\|_E} : x \neq \mathbf{0}\right\} = \sup\left\{T\left(\frac{x}{\|x\|_E}\right) : x \neq \mathbf{0}\right\} \leq \sup\{\|T(x)\|_F : \|x\|_E = 1\}$$

Finally, using Proposition 7, we see that

$$\sup\{\|T(x)\|_F : \|x\|_E = 1\} \leq \sup\{\|T\| \|x\|_E : \|x\|_E = 1\} = \|T\|$$

which completes the proof. $\square$

Theorem 14. $(L(E, F), \|\cdot\|)$ is a Banach space whenever $(F, \|\cdot\|_F)$ is Banach.

Proof. Let $(T_n)_{n \in \mathbb{N}}$ be a Cauchy sequence in $(L(E, F), \|\cdot\|)$. Observe that, for each $x \in E$, we have

$$\|T_n(x) - T_m(x)\|_F = \|(T_n - T_m)(x)\|_F \leq \|T_n - T_m\| \|x\|_E$$

Hence, the Cauchyness of $(T_n)_{n \in \mathbb{N}}$ implies that, for every $x \in E$, the sequence $(T_n(x))_{n \in \mathbb{N}}$ is a Cauchy sequence in F . Since $(F, \|\cdot\|_F)$ is Banach, there exists $y \in F$ with

$$\lim_{n \rightarrow \infty} T_n(x) = y$$

Define the map $T : E \rightarrow F$ by

$$T(x) = \lim_{n \rightarrow \infty} T_n(x)$$

for all $x \in E$. This map is defined at every $x \in E$ by the preceding argument. The linearity of the limit operation, more precisely, the continuity of addition and scalar multiplication of a normed space implies that T is a linear map.

We next check that T is continuous. Since every Cauchy sequence is bounded, there exists $M > 0$ such that $\|T_n\| \leq M$ for all $n \in \mathbb{N}$. It now follows from Proposition 7 that

$$\|T(x)\|_F \leq \|T\| \|x\|_E \leq M \|x\|_E$$

for all $x \in E$. This means that T is bounded and hence is continuous by Theorem 3. Therefore we have $T \in L(E, F)$. It remains to check that the sequence $(T_n)_{n \in \mathbb{N}}$ indeed converges to T in $(L(E, F), \|\cdot\|)$.

Let $\epsilon \in \mathbb{R}^+$. Using the Cauchy-ness of $(T_n)_{n \in \mathbb{N}}$, we can find $N \in \mathbb{N}$ such that for all $n, m \geq N$, we have

$$\|T_n - T_m\| < \frac{\epsilon}{4}$$

Let $x \in E$. Since $T(x) = \lim_{n \rightarrow \infty} T_n(x)$, there exists $N_x \in \mathbb{N}$ such that for all $n \geq N_x$, we have

$$\|T(x) - T_n(x)\|_F \leq \frac{\epsilon}{4} \|x\|_E$$

While it is not going to create a problem for our argument, we would like to warn the reader that N_x indeed depends on the point $x \in E$. Now, for all $n \geq N$ and $m \geq \max\{N, N_x\}$, we get

$$\begin{aligned} \|T(x) - T_n(x)\|_F &\leq \|T(x) - T_m(x)\|_F + \|T_m(x) - T_n(x)\|_F \\ &\leq \frac{\epsilon}{4} \|x\|_E + \|T_m - T_n\| \|x\|_E \\ &\leq \frac{\epsilon}{4} \|x\|_E + \frac{\epsilon}{4} \|x\|_E = \frac{\epsilon}{2} \|x\|_E \end{aligned}$$

It follows that

$$\|T - T_n\| = \sup_{\|x\|_E \leq 1} \|(T - T_n)(x)\|_F \leq \frac{\epsilon}{2} < \epsilon$$

for all $n \geq N$. This completes the proof that $\lim_{n \rightarrow \infty} T_n = T$. Hence $(L(E, F), \|\cdot\|)$ is a Banach space. $\square$

2.2. Dual space. Consider the normed space $(L(E, \mathbb{K}), \|\cdot\|)$ of continuous linear functionals on E . This normed space is called the *dual space* of E and is denoted by E' . An immediate corollary of Theorem 14 is the following.

Theorem 15. *E' is a Banach space for every normed space E .*

The following is a useful criterion that can be used to check whether a linear functional is in the dual space.

Proposition 9. *Let $(E, \|\cdot\|)$ be a normed space and $f : E \rightarrow \mathbb{K}$ be a linear functional. Then $f \in E'$ if and only if $\ker(f)$ is closed.*

Proof. For the left-to-right direction, suppose that $f \in E'$. Since f is continuous and $\{0\} \subseteq \mathbb{K}$ is closed, $f^{-1}(0) = \ker(f)$ is closed.

For the right-to-left direction, suppose that $\ker(f)$ is closed. If $f = \mathbf{0}$, then the statement clearly holds. So assume that $f \neq \mathbf{0}$. Fix $x_0 \in E$ with $f(x_0) \neq 0$. Set $e = x_0/f(x_0)$. Then, by linearity, $f(e) = 1$. Since $\ker(f)$ is closed and $e \notin \ker(f)$, we have that

$$\inf\{\|e - x\| : x \in \ker(f)\} = r > 0$$

Let $x \in E$. We claim that $|f(x)| \leq \frac{1}{r} \|x\|$. We split into two cases.

- If $x \in \ker(f)$, then clearly $0 = |f(x)| \leq \frac{1}{r} \|x\|$.
- If $x \notin \ker(f)$, then

$$\|x\| = \|f(x)e - (f(x)e - x)\| = |f(x)| \cdot \left\| e - \frac{f(x)e - x}{f(x)} \right\| \geq r|f(x)|$$

$$\text{since } \frac{f(x)e - x}{f(x)} \in \ker(f).$$

Therefore f is bounded, that is, $f \in E'$. $\square$

Before we proceed to compute some examples, we need to learn an important notion.

2.3. Schauder basis. In MATH261, you learned about bases of vector spaces. Recall that a basis of a vector space is a linearly independent spanning subset, whose cardinality is defined as the dimension of the vector space. Such bases are actually called *Hamel bases*.

It is a consequence of the Baire category theorem that the dimension of a Banach space, i.e. the cardinality of a Hamel basis of a Banach space, is either finite or uncountable.¹ Hence Hamel bases are not “practical” to use for infinite-dimensional Banach spaces, unless you are comfortable with transfinite methods.

In a normed space, we have an obvious notion of convergence coming from the metric structure which we did not have for abstract vector spaces. So why not incorporating this into a new notion of basis? This brings us to the notion of a Schauder basis.

Let $(E, \|\cdot\|)$ be an infinite-dimensional normed space. A sequence $(b_n)_{n \in \mathbb{N}}$ of elements of E is said to be a *Schauder basis* for $(E, \|\cdot\|)$ if for every $e \in E$ there exists a *unique* sequence $(\alpha_n)_{n \in \mathbb{N}}$ of elements of $\mathbb{K}$ such that

$$e = \sum_{n=0}^{\infty} \alpha_n \cdot b_n$$

We would like to point out that the infinite series

$$\sum_{n=0}^{\infty} \alpha_n \cdot b_n = \lim_{m \rightarrow \infty} \sum_{n=0}^m \alpha_n \cdot b_n$$

may be conditionally converging in the sense that we may be obtaining different limits when the entries of the sequence $(b_n)_{n \in \mathbb{N}}$ are permuted. For this reason, a Schauder basis is an actually *ordered* basis, which is why we defined it not as a set but as a sequence.

The motivation behind this definition comes from the basic property of Hamel bases that every vector being able to be written as a unique finite linear combination

¹The proof of this fact is postponed to Section 3.

of basis elements. We simply replace *finite* by *countable* using the convergence of the underlying space.

In this definition, we required the dimension of the underlying normed space to be infinite as a part of the definition. However, by allowing finite sequences $(b_n)_{n \leq k}$ instead of sequence of countably infinite length, we may indeed consider usual Hamel basis of finite-dimensional normed spaces as Schauder basis. For this reason, we will simply say *basis* for finite-dimensional normed spaces without specifying adjective, while we must specify the type of the basis for infinite-dimensional normed spaces.

An example of a Schauder basis is the sequence

$$(\mathbf{e}_n)_{n \in \mathbb{N}} \quad \text{where} \quad \mathbf{e}_n = (\delta_{in})_{i \in \mathbb{N}}$$

It is not difficult to check that $(\mathbf{e}_n)_{n \in \mathbb{N}}$ is a Schauder basis for the Banach spaces $(\ell_p, \|\cdot\|_p)$ with $1 \leq p < \infty$ and $(c_0, \|\cdot\|_\infty)$.

What about non-examples? We know that every vector space has a Hamel basis. Are there any normed spaces without Schauder basis? Let us first explore some basic properties of Schauder basis that follow from the definition.

Proposition 10. *Let $(E, \|\cdot\|)$ be a normed space with Schauder basis $(b_n)_{n \in \mathbb{N}}$. Then $\{b_n : n \in \mathbb{N}\}$ is linearly independent.*

Proof. This immediately follows from the uniqueness assumption on coefficients. $\square$

Lemma 2. *Let $(E, \|\cdot\|)$ be a normed space over $\mathbb{R}$ with Schauder basis $(b_n)_{n \in \mathbb{N}}$. Then*

$$S = \left\{ \sum_{n=0}^N \alpha_n \cdot b_n : N \in \mathbb{N}, \alpha_n \in \mathbb{Q} \right\}$$

is dense in E .

Proof. Let $e \in E$ and $\epsilon > 0$. We wish to show that $B_{\|\cdot\|}(e, \epsilon)$ contains an element of S which will prove that S is dense in E . Since $(b_n)_{n \in \mathbb{N}}$ is a Schauder basis, there exists $(\beta_n)_{n \in \mathbb{N}}$ of elements of $\mathbb{R}$ such that

$$e = \sum_{n=0}^{\infty} \beta_n \cdot b_n$$

Therefore, we can find some $N \in \mathbb{N}$ such that

$$\left\| e - \sum_{n=0}^N \beta_n \cdot b_n \right\| < \frac{\epsilon}{2}$$

Set $M = \max\{\|b_i\| : 0 \leq i \leq N\}$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, we can find scalars $\{\alpha_n : 0 \leq n \leq N\} \subseteq \mathbb{Q}$ such that

$$|\beta_i - \alpha_n| < \frac{\epsilon}{(2N+3) \cdot M}$$

for all $1 \leq n \leq N$. It now follows that

$$\begin{aligned} \left\| e - \sum_{n=0}^N \alpha_n \cdot b_n \right\| &\leq \left\| e - \sum_{n=0}^N \beta_n \cdot b_n \right\| + \left\| \sum_{n=0}^N \beta_n \cdot b_n - \sum_{n=0}^N \alpha_n \cdot b_n \right\| \\ &\leq \frac{\epsilon}{2} + \sum_{n=0}^N |\beta_n - \alpha_n| \|b_n\| \\ &\leq \frac{\epsilon}{2} + (N+1) \cdot \frac{\epsilon}{(2N+3) \cdot M} \cdot M < \epsilon \end{aligned}$$

Therefore S is dense in E . $\square$

We would like to note that the exact same argument works for normed spaces over $\mathbb{C}$ if one replaces $\mathbb{Q}$ with $\mathbb{Q}(i)$ which still is a countable field.

Corollary 16. *Every normed space with a Schauder basis is separable.*

Proof. This follows from the fact that the dense subset constructed in the proof of Lemma 2 is countable. $\square$

In particular, every Banach space with a Schauder basis is separable. Banach asked whether the converse statement is true: Does every separable Banach space have a Schauder basis?

This famous problem, known as the basis problem, was answered negatively by Per Enflo in 1973. The details of this result, as expected, is beyond the scope of this course. That said, using Corollary 16, we can easily give examples of Banach spaces with no Schauder basis.

Consider the Banach space $(\ell_\infty, \|\cdot\|_\infty)$. For every sequence $\alpha \in \{0, 1\}^{\mathbb{N}}$, consider the open ball

$$O_\alpha = B_{\|\cdot\|}(\alpha, 1/2)$$

It is straightforward to verify that, for all $\alpha, \beta \in \{0, 1\}^{\mathbb{N}}$ with $\alpha \neq \beta$, we have that $O_\alpha \cap O_\beta = \emptyset$. Since $\{0, 1\}^{\mathbb{N}}$ is uncountable, there are uncountably many disjoint non-empty open sets. This means that ℓ_∞ cannot be separable. Consequently, $(\ell_\infty, \|\cdot\|_\infty)$ does not have a Schauder basis.

To see another example, consider the Banach space $(B([0, 1]), \|\cdot\|_\infty)$. What is the cardinality of this space? An easy cardinality argument shows that $B([0, 1])$ has cardinality

$$\mathfrak{c}^{\mathfrak{c}} > \mathfrak{c}$$

where $\mathfrak{c}$ denotes continuum, i.e. the cardinality of $\mathbb{R}$. Recall from MATH349 that every separable metric space is of cardinality at most $\mathfrak{c}$. Consequently, $(B([0, 1]), \|\cdot\|_\infty)$ is not separable and hence does not have a Schauder basis. Of course, in this last example, the issue simply stems from the fact that the space is too big to be separable and have a Schauder basis. Indeed, it is not difficult to see that the cardinality of a normed space with a Schauder basis must be exactly $\mathfrak{c}$.

2.4. Some examples of dual spaces.

2.4.1. *Finite-dimensional normed spaces.* Let $(E, \|\cdot\|)$ be a finite-dimensional normed space. Using Corollary 5, we may assume without loss of generality that $(E, \|\cdot\|)$ is indeed $(\mathbb{K}^n, \|\cdot\|_1)$ for some $n \in \mathbb{N}$.

Let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ denote its standard basis. For each $1 \leq i \leq n$, consider the functional $f_i : E \rightarrow \mathbb{K}$ given by

$$f_i(e) = \alpha_i \quad \text{where} \quad e = \sum_{k=1}^n \alpha_k \mathbf{e}_k$$

Recall from MATH261 that $\{f_i : 1 \leq i \leq n\}$ form a basis for the vector space of all linear functionals on $\mathbb{K}^n$. Since every linear functional on a finite-dimensional normed space is continuous by Corollary 7, we have that $\{f_i : 1 \leq i \leq n\}$ form a basis for E' . Consider the map $T : E' \rightarrow E$ given by

$$T\left(\sum_{k=1}^n \alpha_k f_k\right) = \sum_{k=1}^n \alpha_k \mathbf{e}_k$$

One can easily check that T is a vector space isomorphism. Observe that for every $f \in E'$ with $f = \sum_{k=1}^n \alpha_k f_k$, we have

$$\begin{aligned} \|f\| &= \left\| \sum_{k=1}^n \alpha_k f_k \right\| \leq \sum_{k=1}^n |\alpha_k| \|f_k\| = \sum_{k=1}^n |\alpha_k| \sup_{\|\mathbf{x}\|_1 \leq 1} \|f_k(\mathbf{x})\|_1 \\ &= \sum_{k=1}^n |\alpha_k| \\ &= \left\| T\left(\sum_{k=1}^n \alpha_k f_k\right) \right\|_1 = \|T(f)\|_1 \end{aligned}$$

Therefore T^{-1} is continuous. Since $\dim_{\mathbb{K}}(E') < \infty$, the map T itself is continuous by Corollary 7. Hence T is not only a vector space isomorphism but also a normed space isomorphism.

2.4.2. *Sequence spaces.* Let $1 < p < q < \infty$ be such that $\frac{1}{p} + \frac{1}{q} = 1$. We claim that $(\ell_p, \|\cdot\|_p)'$ is isomorphic to $(\ell_q, \|\cdot\|_q)$. Consider the map $T : \ell_q \rightarrow \ell_p'$ given by

$$T(\mathbf{a}) = f_{\mathbf{a}} \quad \text{where} \quad \mathbf{a} = (a_n)_{n \in \mathbb{N}} \quad \text{and} \quad f_{\mathbf{a}}((x_n)_{n \in \mathbb{N}}) = \sum_{n=0}^{\infty} a_n x_n$$

for all $\mathbf{a} \in \ell_q$. Note that $f_{\mathbf{a}}$ is indeed a linear functional for each $(a_n)_{n \in \mathbb{N}} \in \ell_q$ because $\sum_{n=0}^{\infty} a_n x_n$ is convergent for all $\mathbf{x} \in \ell_p$ by Hölder's inequality.

It is easily seen that T is a linear map. We claim that T is an isometry. Let $\mathbf{a} \in \ell_q$ be such that $\mathbf{a} \neq \mathbf{0}$. First observe that

$$\begin{aligned} \|T(\mathbf{a})\| &= \sup_{\|\mathbf{x}\|_p \leq 1} |f_{\mathbf{a}}(\mathbf{x})| = \sup_{\|\mathbf{x}\|_p \leq 1} \left| \sum_{n=0}^{\infty} a_n x_n \right| \\ &\leq \sup_{\|\mathbf{x}\|_p \leq 1} \sum_{n=0}^{\infty} |a_n x_n| \leq \sup_{\|\mathbf{x}\|_p \leq 1} \|\mathbf{a}\|_q \|\mathbf{x}\|_p = \|\mathbf{a}\|_q \end{aligned}$$

It remains to prove the reverse inequality. Let $I = \{n \in \mathbb{N} : a_n \neq 0\}$. Consider the sequence $\mathbf{b}$ given by

$$b_n = \begin{cases} \frac{|a_n|^q}{\|\mathbf{a}\|_q^{q-1}} a_n & \text{if } n \in I \\ 0 & \text{if } n \notin I \end{cases}$$

Then we have

$$\begin{aligned} \sum_{n=0}^{\infty} |b_n|^p &= \sum_{n \in I} |b_n|^p = \sum_{n \in I} \left| \frac{|a_n|^q}{\|\mathbf{a}\|_q^{q-1}} a_n \right|^p \\ &= \sum_{n \in I} \frac{|a_n|^{qp}}{\|\mathbf{a}\|_q^{(q-1)p} |a_n|^p} \\ &= \sum_{n \in I} \frac{|a_n|^{p+q}}{\|\mathbf{a}\|_q^{(q-1)p} |a_n|^p} = \sum_{n \in I} \frac{|a_n|^q}{\|\mathbf{a}\|_q^q} = \sum_{n=0}^{\infty} \frac{|a_n|^q}{\|\mathbf{a}\|_q^q} = 1 \end{aligned}$$

Hence $\mathbf{b} \in \ell_p$. It follows that

$$\|T(\mathbf{a})\| = \sup_{\|\mathbf{x}\|_p \leq 1} |f_{\mathbf{a}}(\mathbf{x})| \geq |f_{\mathbf{a}}(\mathbf{b})| = \left| \sum_{n=0}^{\infty} a_n b_n \right| = \sum_{n=0}^{\infty} \frac{|a_n|^q}{\|\mathbf{a}\|_q^{q-1}} = \|\mathbf{a}\|_q$$

It follows that $\|T(\mathbf{a})\| = \|\mathbf{a}\|_q$ for all $\mathbf{a} \in \ell_q$. Thus T is an isometry. Being an isometry, T is one-to-one.

We will now show that T is onto which will complete the proof that it is an isometric isomorphism. Let $f \in \ell'_p$. Construct the sequence

$$\mathbf{a} = (a_n)_{n \in \mathbb{N}} = (f(\mathbf{e}_n))_{n \in \mathbb{N}}$$

We leave it to the reader to verify² that $\mathbf{a} \in \ell_q$. For any $\mathbf{x} \in \ell_p$, we see that

$$f_{\mathbf{a}}(\mathbf{x}) = \sum_{n=0}^{\infty} a_n x_n = \sum_{n=0}^{\infty} f(\mathbf{e}_n) x_n = f\left(\sum_{n=0}^{\infty} x_n \mathbf{e}_n\right) = f(\mathbf{x})$$

Observe that for the third equality above, we use that f is continuous and $\{\mathbf{e}_n\}_{n \in \mathbb{N}}$ is a Schauder basis for ℓ_p . Finally, we see that $T(\mathbf{a}) = f_{\mathbf{a}} = f$ which means that T is onto. Hence T is an isometric isomorphism from ℓ_q to ℓ'_p .

²This is not a straightforward exercise. You have to imitate the argument above where we obtained $f_{\mathbf{a}}(\mathbf{b}) = \|\mathbf{a}\|_q$ using truncations of $\mathbf{a}$ to finitely many terms (so that the relevant sequence is in ℓ_q) and taking limits. In order to take care of limits, try to use Schauder basis argument that follows this footnote.

Note that the argument above works for $1 < p < q < \infty$. It is an exercise to the reader to prove using similar ideas that

- the dual of $(\ell_1, \|\cdot\|_1)$ is isomorphic to $(\ell_\infty, \|\cdot\|_\infty)$.
- the dual of $(c_0, \|\cdot\|_\infty)$ is isomorphic to $(\ell_1, \|\cdot\|_1)$.

2.5. The Hahn-Banach theorem. In this subsection, we will prove the Hahn-Banach theorem, which unarguably is one of the most important theorems that you will learn in this course.

Let us first introduce some terminology. Given a vector space V over $\mathbb{K}$, a subspace $W \subseteq V$ and a linear function $f : W \rightarrow \mathbb{K}$, we say that a map $F : V \rightarrow \mathbb{K}$ is a *linear extension of f to V* if F is a linear map and $F(x) = f(x)$ for all $x \in W$.

An obvious question to ask is the following: Given a linear functional with some “nice” properties defined on a subspace of a vector space, when can we find a linear extension to the whole space while preserving its “nice” properties?

Observe that, without requiring preservation of “nice” properties, this can always be done in the following way: Given a functional $f : W \rightarrow \mathbb{K}$, choose a Hamel basis B for W and a Hamel basis $\hat{B} \supseteq B$ of V . Then the function

$$F(x) = \begin{cases} f(x) & \text{if } x \in B \\ 0 & \text{if } x \in \hat{B} \setminus B \end{cases}$$

defined on the basis $\hat{B}$ can be linearly extended to all of V . On the other hand, this procedure will produce extensions with “bad” properties even if the original functional has “nice” properties.

For example, set $V = c_{00}$ and $W = \langle \mathbf{e}_0 \rangle$. The linear functional $f : W \rightarrow \mathbb{K}$ given by

$$f(\alpha \mathbf{e}_0) = \alpha$$

for all $\alpha \in \mathbb{K}$ is clearly continuous with respect to the topology arising from $\|\cdot\|_\infty$. Consider the vectors $\{u_i\}_{i \in \mathbb{N}}$ recursively defined as follows.

$$\begin{aligned} u_0 &= \mathbf{e}_0 \\ u_{n+1} &= \mathbf{e}_{n+1} - (n+1) \cdot \mathbf{e}_n \quad \text{for all } n \in \mathbb{N} \end{aligned}$$

It is not difficult to check that $\{u_i\}_{i \in \mathbb{N}}$ is a Hamel basis for V . Suppose that we apply the procedure described above with $B = \{\mathbf{e}_0\}$ and $\hat{B} = \{u_i\}_{i \in \mathbb{N}}$. Then we get a linear functional $F : V \rightarrow \mathbb{K}$ such that $F(u_0) = 1$ and $F(u_i) = 0$ for all $i \in \mathbb{N}^+$. One can easily check that $F(\mathbf{e}_n) = n$ for all $n \in \mathbb{N}^+$. It follows that $F : V \rightarrow \mathbb{K}$ is not continuous even though $f : W \rightarrow \mathbb{K}$ is continuous.

The Hahn-Banach theorem tells us that one can always extend linear functionals that satisfy a certain boundedness property to the whole space while preserving this boundedness property.

In order to state the Hahn-Banach theorem, we will need to recall the notion of a seminorm. Let V be a vector space over $\mathbb{K}$. A function $p : V \rightarrow [0, \infty)$ is called a *seminorm* on V if

- $p(\alpha x) = |\alpha|p(x)$ for all $\alpha \in \mathbb{K}$ and for all $x \in V$, and
- $p(x + y) \leq p(x) + p(y)$ for all $x, y \in V$.

In other words, a seminorm is simply a norm without the property of taking the value zero only at the zero vector. We can now state the Hahn-Banach theorem.

Theorem 17 (The Hahn-Banach theorem). *Let V be a vector space over $\mathbb{K}$ and $p : V \rightarrow [0, \infty)$ be a seminorm on V . Let $W \subseteq V$ be a subspace and $f : W \rightarrow \mathbb{K}$ be a linear functional such that*

$$|f(x)| \leq p(x)$$

*for all $x \in W$. Then there exists a **linear extension** $F : V \rightarrow \mathbb{K}$ of the functional f to V such that*

$$|F(x)| \leq p(x)$$

for all $x \in V$.

In order to prove the Hahn-Banach theorem, we will first prove the statement for the case $\mathbb{K} = \mathbb{R}$ and obtain the statement for the case $\mathbb{K} = \mathbb{C}$ as a corollary.

Let us say a couple words about our plan of attack. How can such a result be proven at all? The idea is to first show that such a linear functional f can always be extended to a larger subspace of dimension one more than that of its domain and then to iterate this procedure “transfinitely” until there is no more room to extend at which point the domain will be all of V .

Such constructions often employ basic set-theoretic tools such as the axiom of choice and transfinite recursion, both of which we think most students are not familiar with. For this reason, we will use their famous proxy, Zorn’s lemma, to take care of this transfinite procedure all at once. We will not include background material on Zorn’s lemma in these lecture notes due to this class being not a set theory class. However, we will go over Zorn’s lemma and its basic uses in class before we start proving the Hahn-Banach theorem.

Let us now state the following lemma which corresponds to the first step in our plan of attack and which, in a sense, is the heart of the matter.

Lemma 3. *Let V be a vector space over $\mathbb{R}$ and $p : V \rightarrow [0, \infty)$ be a seminorm on V . Let $W \subseteq V$ be a proper subspace, $e \in V - W$ and $f : W \rightarrow \mathbb{R}$ be a linear functional such that*

$$f(x) \leq p(x)$$

for all $x \in W$. Then there exists a **linear extension** $F : \langle W, e \rangle \rightarrow \mathbb{R}$ of the functional f to $\langle W, e \rangle$ such that

$$F(x) \leq p(x)$$

for all $x \in \langle W, e \rangle$.

Before we start the actual proof of Lemma 3, let us try to *reverse engineer* what we are supposed to do. Suppose that there is such a linear extension F . Then we must have $F(w + \alpha e) = f(w) + \alpha F(e)$ for all $w \in W$. Hence the real problem is to determine what $F(e)$ must be. By the boundedness condition given to us, we also know that we must have $f(w) + \alpha F(e) \leq p(w + \alpha e)$ for all $w \in W$ and $\alpha \in \mathbb{R}$. In particular, for $\alpha = \pm 1$, we obtain

$$f(w) - F(e) \leq p(w - e) \text{ and } f(w) + F(e) \leq p(w + e)$$

and hence

$$f(w) - p(w - e) \leq F(e) \leq p(w + e) - f(w)$$

for all $w \in W$. This inequality tell us to make sure to choose $F(e)$ in a specific interval. Having obtained enough insight, we can now start the actual proof.

Proof of Lemma 3. Observe that, by assumption, we have

$$f(x) + f(y) = f(x + y) \leq p(x + y) = p(x + e + y - e) \leq p(x + e) + p(y - e)$$

for all $x, y \in W$, from which it follows that

$$f(y) - p(y - e) \leq p(x + e) - f(x)$$

for all $x, y \in W$. Hence

$$\sup_{w \in W} \{f(w) - p(w - e)\} \leq \inf_{w \in W} \{p(w + e) - f(w)\}$$

Fix $\lambda \in \mathbb{R}$ such that

$$\sup_{w \in W} \{f(w) - p(w - e)\} \leq \lambda \leq \inf_{w \in W} \{p(w + e) - f(w)\}$$

Consider the map $F : \langle W, e \rangle \rightarrow \mathbb{R}$ given by

$$F(w + \alpha e) = f(w) + \alpha \lambda$$

for all $w \in E$ and $\alpha \in \mathbb{R}$. Observe that, every element in $\langle W, e \rangle$ can be uniquely written as $w + \alpha e$ with $w \in W$ and $\alpha \in \mathbb{R}$, which is why this definition makes sense at all. It is easily checked that F is a linear extension of f to $\langle W, e \rangle$.

It remains to check the boundedness condition. Let $w \in W$ and $\alpha \in \mathbb{R}$. We split into three cases.

- Suppose that $\alpha = 0$. Then $F(w + \alpha e) = f(w) \leq p(w) = p(w + \alpha e)$.

- Suppose $\alpha > 0$. Then

$$\begin{aligned} F(w + \alpha e) &= f(w) + \alpha \lambda = \alpha \left(f\left(\frac{w}{\alpha}\right) + \lambda \right) \\ &\leq \alpha \cdot p\left(\frac{w}{\alpha} + e\right) = p(w + \alpha e) \end{aligned}$$

- Suppose that $\alpha < 0$. Then

$$\begin{aligned} F(w + \alpha e) &= f(w) - (-\alpha)\lambda = -\alpha \left(f\left(\frac{w}{-\alpha}\right) - \lambda \right) \\ &\leq (-\alpha) \cdot p\left(\frac{w}{-\alpha} - e\right) = p(w + \alpha e) \end{aligned}$$

This completes the proof that $F(x) \leq p(x)$ for all $x \in \langle W, e \rangle$. $\square$

We are now ready to prove the Hahn-Banach theorem for $\mathbb{R}$ -vector spaces.

Theorem 18 (The Hahn-Banach theorem for $\mathbb{R}$ -vector spaces). *Let V be a vector space over $\mathbb{R}$ and $p : V \rightarrow [0, \infty)$ be a seminorm on V . Let $W \subseteq V$ be a subspace and $f : W \rightarrow \mathbb{R}$ be a linear functional such that*

$$f(x) \leq p(x)$$

*for all $x \in W$. Then there exists a **linear extension** $F : V \rightarrow \mathbb{R}$ of the functional f to V such that*

$$F(x) \leq p(x)$$

for all $x \in V$.

Proof. Set $\mathbb{P}$ to be the set of all pairs of the form (G, g) where $W \subseteq G \subseteq V$ is a subspace and $g : G \rightarrow \mathbb{R}$ is a linear extension of f to G such that $g(x) \leq p(x)$ for all $x \in G$. Consider the partial order relation $\preceq$ on $\mathbb{P}$ given by

$$(G, g) \preceq (G', g') \text{ if and only if } G \subseteq G' \text{ and } g'(x) = g(x) \text{ for all } x \in G$$

Observe that $\mathbb{P} \neq \emptyset$ as $(W, f) \in \mathbb{P}$. Moreover, the partially ordered set $(\mathbb{P}, \preceq)$ satisfies the hypothesis of Zorn's lemma: Given a chain $\{(G_\alpha, g_\alpha)_{\alpha \in I}\}$, the pair

$$\left(\bigcup_{\alpha \in I} G_\alpha, \bigcup_{\alpha \in I} g_\alpha \right)$$

is an element of $\mathbb{P}$ which is an upper bound for this chain.³ Thus, by Zorn's lemma, there exists a maximal element $(H, h) \in \mathbb{P}$.

If it were the case that $H \neq V$, then we would pick $e \in V - H$ and obtain an extension $h' : \langle H, e \rangle \rightarrow \mathbb{R}$ satisfying $h'(x) \leq p(x)$ by Lemma 3. This would lead to a contradiction as $(\langle H, e \rangle, h')$ would be a strictly bigger element than (H, h) in the partially ordered set $(\mathbb{P}, \preceq)$. $\square$

³Being a set theorist, of course I treat functions as sets of ordered pairs, so the union $\bigcup_{\alpha \in I} g_\alpha$ is the function $g : G \rightarrow \mathbb{R}$ such that $g(x) = g_\alpha(x)$ for any $\alpha \in I$ and $x \in G_\alpha$, which is well-defined since g_α 's are compatible with each other on their common domain.

Having proven it for $\mathbb{R}$ -vector spaces, we can now prove Theorem 17 in its full generality.

Proof of the Hahn-Banach theorem. We split into two cases.

- Suppose that $\mathbb{K} = \mathbb{R}$. Then, by Theorem 18, we can find an extension F of f to V such that $F(x) \leq p(x)$ for all $x \in V$. But then, since F is linear and p is a seminorm, we obtain that

$$-F(x) = F(-x) \leq p(-x) = |-1| \cdot p(x) = p(x)$$

from which it follows that $|F(x)| \leq p(x)$ for all $x \in V$.

- Now suppose that $\mathbb{K} = \mathbb{C}$. Let us write $f(x) = \operatorname{Re}(f(x)) + i \cdot \operatorname{Im}(f(x))$ for the real and imaginary parts of f .⁴ Observe that the map $g : W \rightarrow \mathbb{R}$ given by

$$g(x) = \operatorname{Re}(f(x))$$

is an $\mathbb{R}$ -linear map with the property that

$$|g(x)| = |\operatorname{Re}(f(x))| \leq |f(x)| \leq p(x)$$

By previous case $\mathbb{K} = \mathbb{R}$, we know that we can find an $\mathbb{R}$ -linear extension $G : V \rightarrow \mathbb{R}$ of g to V such that $|G(x)| \leq p(x)$ for all $x \in V$. We now define $F : V \rightarrow \mathbb{C}$ by

$$F(x) = G(x) - i \cdot G(ix)$$

for all $x \in V$. It is not difficult to see that F is $\mathbb{C}$ -linear. Moreover, using the footnote below, we obtain that

$$F(x) = G(x) - i \cdot G(ix) = \operatorname{Re}(f(x)) - i \cdot \operatorname{Re}(f(ix)) = f(x)$$

for all $x \in W$. Let $x \in V$. If $F(x) = 0$, then clearly $0 = |F(x)| \leq p(x)$ is satisfied. So suppose $F(x) \neq 0$. Then we have

$$|F(x)| = F\left(\frac{|F(x)|}{F(x)}x\right) = G\left(\frac{|F(x)|}{F(x)}x\right) \leq p\left(\frac{|F(x)|}{F(x)}x\right) = p(x)$$

Thus F is the extension we are looking for.

This completes the proof. □

Next will be covered some important consequences of the Hahn-Banach theorem.

Corollary 19. *Let $(E, \|\cdot\|)$ be a normed space, let $W \subseteq E$ be a subspace and $f : W \rightarrow \mathbb{K}$ be a continuous linear functional. Then there exists a continuous linear extension $F : E \rightarrow \mathbb{K}$ of f to E such that $\|F\| = \|f\|$.*

⁴We would like to add a side comment here. Since $f : W \rightarrow \mathbb{C}$ is $\mathbb{C}$ -linear, we have that

$$\operatorname{Re}(f(ix)) + i \cdot \operatorname{Im}(f(ix)) = f(ix) = if(x) = i \cdot \operatorname{Re}(f(x)) - \operatorname{Im}(f(x))$$

Thus $\operatorname{Im}(f(x)) = -\operatorname{Re}(f(ix))$. In other words, a complex linear functional is indeed determined by its real part.

Proof. Consider the map $p : E \rightarrow [0, \infty)$ given by

$$p(x) = \|f\| \|x\|$$

for all $x \in E$. It is easily seen that p is a seminorm and indeed, is a norm whenever $f \neq \mathbf{0}$. Moreover, by Proposition 7, we have $|f(x)| \leq \|f\| \|x\| = p(x)$ for all $x \in E$. It now follows from the Hahn-Banach theorem that there exists a linear extension $F : E \rightarrow \mathbb{K}$ of f to E such that

$$|F(x)| \leq p(x) = \|f\| \|x\|$$

for all $x \in E$. Consequently, we have

$$\|F\| = \sup_{\substack{\|x\| \leq 1 \\ x \in E}} |F(x)| \leq \|f\|$$

On the other hand, we also have

$$\|F\| = \sup_{\substack{\|x\| \leq 1 \\ x \in E}} |F(x)| \geq \sup_{\substack{\|x\| \leq 1 \\ x \in W}} |F(x)| = \sup_{\substack{\|x\| \leq 1 \\ x \in W}} |f(x)| = \|f\|$$

Hence $\|F\| = \|f\|$. □

Corollary 20. *Let $(E, \|\cdot\|)$ be a normed space and $e \in E$ be such that $e \neq \mathbf{0}$. Then there exists a continuous linear functional $F : E \rightarrow \mathbb{K}$ such that $F(e) = \|e\|$ and $\|F\| = 1$.*

Proof. Set $W = \langle e \rangle$. Then the linear functional $f : W \rightarrow \mathbb{K}$ given by $f(\alpha e) = \alpha \|e\|$ is continuous. Hence, by Corollary 19, there exists a continuous linear extension $F : E \rightarrow \mathbb{K}$ of f to E such that

$$\|F\| = \|f\| = \sup_{\substack{\|x\| \leq 1 \\ x \in W}} |f(x)| = 1$$

Moreover, $F(e) = f(e) = \|e\|$. □

Corollary 21. *Let $(E, \|\cdot\|)$ be a normed space. Then for all $x \in E$, we have*

$$\|x\| = \sup\{|f(x)| : f \in E', \|f\| = 1\}$$

Proof. Recall that, by Proposition 7, we already have $|f(x)| \leq \|f\| \|x\|$ for all $x \in E$ and $f \in E'$. Thus

$$\|x\| \geq \sup\{|f(x)| : f \in E', \|f\| = 1\}$$

for all $x \in E$. The reverse inequality follows from Corollary 20 since, given $x \in E$, we can find $f_x \in E'$ with $\|f_x\| = 1$ and $f_x(x) = \|x\|$. □

Before we state the next corollary of the Hahn-Banach theorem, we need to introduce the canonical injection map from a normed space to its double dual. Let $(E, \|\cdot\|)$ be a normed space. Consider the map $i : E \rightarrow E''$ given by

$$i(e) = \varphi_e \text{ where } \varphi_e : E' \rightarrow \mathbb{K} \text{ is given by } \varphi_e(f) = f(e) \text{ for all } f \in E'$$

for all $e \in E$. A straightforward computation shows that i is a linear map.

Corollary 22. *The map i is an isometry.*

Proof. For all $e \in E$, we have that

$$\|i(e)\| = \|\varphi_e\| = \sup_{\substack{\|f\| \leq 1 \\ f \in E'}} |\varphi_e(f)| = \sup_{\substack{\|f\| \leq 1 \\ f \in E'}} |f(e)| \leq \sup_{\substack{\|f\| \leq 1 \\ f \in E'}} \|f\| \|e\| = \|e\|$$

Applying Corollary 21, we obtain that

$$\|i(e)\| = \|\varphi_e\| = \sup_{\substack{\|f\| \leq 1 \\ f \in E'}} |\varphi_e(f)| = \sup_{\substack{\|f\| \leq 1 \\ f \in E'}} |f(e)| \geq \sup_{\substack{\|f\|=1 \\ f \in E'}} |f(e)| = \|e\|$$

□

Using this corollary, one can give another proof of Theorem 12 by choosing F to be the closure of $i(E)$ in E'' and $\Phi = i$. In other words, the completion of a normed space is simply the closure of its canonical image in its double dual.

The Hahn-Banach theorem has many other important consequences that we will skip due to time constraints. You can find these consequences in many functional analysis books. So far we have only seen some important “abstract” consequences that hold for general normed spaces. In the next subsection, we will learn one “concrete” application of the Hahn-Banach theorem.

2.6. Banach limits. A continuous linear functional $\mathbf{Lim} : \ell_\infty \rightarrow \mathbb{K}$ such that

- For all $\mathbf{a} \in \ell_\infty$ such that $a_n \geq 0$ for every $n \in \mathbb{N}$, we have

$$\mathbf{Lim} \mathbf{a} \geq 0$$

- For all $\mathbf{a} \in \ell_\infty$

$$\mathbf{Lim} \mathbf{a} = \mathbf{Lim} S(\mathbf{a})$$

where $S : \ell_\infty \rightarrow \ell_\infty$ is the left-shift operator given by

$$S((a_n)_{n \in \mathbb{N}}) = (a_{n+1})_{n \in \mathbb{N}}$$

- For all convergent sequences $\mathbf{a} \in \ell_\infty$

$$\mathbf{Lim} \mathbf{a} = \lim_{n \rightarrow \infty} a_n$$

is called a *Banach limit*. A moment's thought reveals that a Banach limit is a generalization of the (usual) notion of the limit of a sequence to the space of *all* bounded sequences. In other words, a Banach limit allows us to take the “limit” of *any* bounded sequence in a way that has the expected properties and that coincides with usual limits.

Theorem 23. *There exists a Banach limit $\mathbf{Lim} : \ell_\infty \rightarrow \mathbb{K}$.*

Proof. Consider the subspace $W = \{\mathbf{x} - S(\mathbf{x}) : \mathbf{x} \in \ell_\infty\}$ and $\mathbf{1} = (1, 1, \dots)$. We will first show that

$$\inf\{\|\mathbf{1} - \mathbf{w}\| : \mathbf{w} \in W\} > 0$$

Let $\mathbf{w} \in W$, say, $\mathbf{w} = \mathbf{x} - S(\mathbf{x})$ for some $\mathbf{x} \in \ell_\infty$. Then, for all $K \in \mathbb{N}$, we have

$$\begin{aligned} \|\mathbf{1} - \mathbf{w}\| &\geq \frac{1}{K} \sum_{i=0}^{K-1} |1 - w_i| \\ &\geq \frac{1}{K} \sum_{i=0}^{K-1} |1 - x_i + x_{i+1}| \\ &\geq \left| \frac{1}{K} \sum_{i=0}^{K-1} 1 - x_i + x_{i+1} \right| = \left| \frac{1}{K} K + \frac{1}{K} (-x_0 + x_K) \right| \end{aligned}$$

Taking the limit of both sides as $K \rightarrow \infty$, we see that $\|\mathbf{1} - \mathbf{w}\| \geq 1$. Thus

$$d = \inf\{\|\mathbf{1} - \mathbf{w}\| : \mathbf{w} \in W\} \geq 1$$

It follows that $\mathbf{1} \notin W$ and so, every element in $\langle W, \mathbf{1} \rangle$ can be written as a unique linear combination $\mathbf{w} + \beta \mathbf{1}$ of an element of W and a multiple of $\mathbf{1}$. Consider the map $f : \langle W, \mathbf{1} \rangle \rightarrow \mathbb{K}$ given by

$$f(\mathbf{w} + \beta \mathbf{1}) = \beta$$

This map is linear due to the uniqueness of representation that was mentioned above. We next argue that f is continuous. For all $\beta \in \mathbb{K}$ with $\beta \neq 0$ and $\mathbf{w} \in W$, we have that

$$|f(\mathbf{w} + \beta \mathbf{1})| = |\beta| \leq \frac{|\beta|}{d} \left\| \mathbf{1} - \left(\frac{-\mathbf{w}}{\beta} \right) \right\| = \frac{1}{d} \|\mathbf{w} + \beta \mathbf{1}\| \leq \|\mathbf{w} + \beta \mathbf{1}\|$$

For $\beta = 0$, the inequality $|f(\mathbf{w} + \beta \mathbf{1})| = 0 \leq \|\mathbf{w} + \beta \mathbf{1}\|$ clearly holds. Hence f is bounded and so, is continuous. It is not difficult to see that $\|f\| = 1$. Applying Corollary 19, we obtain a continuous linear extension $\mathbf{Lim} : \ell_\infty \rightarrow \mathbb{K}$ such that $\|\mathbf{Lim}\| = \|f\| = 1$. We claim that $\mathbf{Lim}$ is as desired. Let us check each property.

- Let $\mathbf{x} \in \ell_\infty$ be such that $x_n \geq 0$ for all $n \in \mathbb{N}$. Set $M = \|\mathbf{x}\|$. Recall that $\mathbf{Lim} \mathbf{1} = f(\mathbf{1}) = 1$. So we have

$$M - \mathbf{Lim} \mathbf{x} = \mathbf{Lim} (M \cdot \mathbf{1} - \mathbf{x}) \leq \|\mathbf{Lim}\| \|M \cdot \mathbf{1} - \mathbf{x}\| \leq M$$

Consequently, we have that $\mathbf{Lim} \mathbf{x} \geq 0$. This verifies the first property of being a Banach limit.

- Note that $f(\mathbf{w}) = 0$ for all $\mathbf{w} \in W$. It follows that

$$\mathbf{Lim} \mathbf{x} - \mathbf{Lim} S(\mathbf{x}) = \mathbf{Lim}(\mathbf{x} - S(\mathbf{x})) = f(\mathbf{x} - S(\mathbf{x})) = 0$$

for all $\mathbf{x} \in \ell_\infty$. This verifies the second property of being a Banach limit. Clearly we can also prove $\mathbf{Lim} S^n(\mathbf{x}) = \mathbf{Lim} \mathbf{x}$ for all $n \in \mathbb{N}^+$ and $\mathbf{x} \in \ell_\infty$ by induction.

- Let $\mathbf{x} \in \ell_\infty$ be a convergent sequence, say, $\lim_{n \rightarrow \infty} x_n = L$. Given $\epsilon > 0$, we can find $N \in \mathbb{N}$ such that for all integers $n \geq N$, we have $L - \epsilon < x_n < L + \epsilon$. In other words, the sequences

$$\mathbf{y} = (L + \epsilon) \cdot \mathbf{1} - S^N(\mathbf{x})$$

$$\mathbf{z} = S^N(\mathbf{x}) - (L - \epsilon) \cdot \mathbf{1}$$

are elements of ℓ_∞ such that $y_n, z_n \geq 0$ for all $n \in \mathbb{N}$. But then, applying the first and second properties of being a Banach limit, we have that

$$0 \leq \mathbf{Lim} \mathbf{y} = \mathbf{Lim} (L + \epsilon) \cdot \mathbf{1} - \mathbf{Lim} S^N(\mathbf{x}) = L + \epsilon - \mathbf{Lim} \mathbf{x}$$

$$0 \leq \mathbf{Lim} \mathbf{z} = \mathbf{Lim} S^N(\mathbf{x}) - \mathbf{Lim} (L - \epsilon) \cdot \mathbf{1} = \mathbf{Lim} \mathbf{x} - L + \epsilon$$

Combining these we get,

$$L - \epsilon \leq \mathbf{Lim} \mathbf{x} \leq L + \epsilon$$

Since $\epsilon > 0$ was arbitrary, this means that $\mathbf{Lim} \mathbf{x} = L = \lim_{n \rightarrow \infty} x_n$. This verifies the third property of being a Banach limit.

□

Let us now see this concept in action. Fix some Banach limit $\mathbf{Lim}$. Since we have

$$\mathbf{1} = (1, 1, 1, 1, \dots) = (1, 0, 1, 0, \dots) + S(1, 0, 1, 0, \dots)$$

it follows that

$$1 = \mathbf{Lim} \mathbf{1} = \mathbf{Lim} (1, 0, 1, 0, \dots) + \mathbf{Lim} S(1, 0, 1, 0, \dots)$$

$$1 = \mathbf{Lim} \mathbf{1} = \mathbf{Lim} (1, 0, 1, 0, \dots) + \mathbf{Lim} (1, 0, 1, 0, \dots)$$

and hence $\mathbf{Lim} (1, 0, 1, 0, \dots) = \frac{1}{2}$. So the sequence $(1, 0, 1, 0, \dots)$ which does not converge in the traditional sense converges to $\frac{1}{2}$ in this generalized setting. If you think about this for a moment, you will realize that *if* there is to be a limit to this sequence at all, it must be $\frac{1}{2}$; this sequence is trying to converge to $\frac{1}{2}$ but it keeps missing it at every stage!

One may wonder whether it is possible to get an even broader notion of limit defined on all of $\mathbb{K}^{\mathbb{N}}$. Of course, in this case, we have to drop the continuity requirement since $\mathbb{K}^{\mathbb{N}}$ has no normed space structure. It is an exercise to the reader to prove that there is no linear functional $\Psi : \mathbb{K}^{\mathbb{N}} \rightarrow \mathbb{K}$ which even has the latter two properties of a Banach limit.⁵

⁵Show that if there were such a shift-invariant linear functional Ψ , we would have that $\Psi(\mathbf{1}) = 0$.

2.7. Convexity and hyperplane separation. Throughout this subsection only, we shall be working with vector and normed spaces over $\mathbb{R}$.

Let V be a vector space over $\mathbb{R}$. A subset $A \subseteq V$ is said to be *convex* if

$$\alpha x + (1 - \alpha)y \in A$$

for all $x, y \in A$ and for all real numbers $0 \leq \alpha \leq 1$. In other words, A is convex if the “line segment” joining any two points of A is completely contained in A .

Our goal in this subsection is to show that certain disjoint convex subsets of a normed space over $\mathbb{R}$ can be “separated by affine hyperplanes”. Before we proceed, let us now explain what we mean by “separating two subsets by affine hyperplanes”.

Given a vector space V over $\mathbb{R}$ and $H \subseteq V$, we say that H is a *hyperplane*⁶ of V if $H = \ker(f)$ for some non-zero linear functional $f : V \rightarrow \mathbb{R}$. An *affine hyperplane* is simply a hyperplane translated by some vector, that is, a subset of the form

$$v + H = \{v + h \in V : h \in H\}$$

for some $v \in V$. Observe that if T is an affine hyperplane, say, $T = v + H$ where $H = \ker(f)$, then $T = f^{-1}(f(v))$. Therefore affine hyperplanes are inverse images of singletons under non-zero functionals. We would like to note that V has a normed space structure and $f : V \rightarrow \mathbb{R}$ is continuous, then the affine hyperplane $f^{-1}(c)$ is closed for every $c \in \mathbb{R}$ since singletons are closed.

An affine hyperplane $f^{-1}(c)$ splits the space into two convex “half spaces”, namely, $\{x \in V : f(x) < c\}$ and $\{x \in V : f(x) > c\}$. Our goal in this subsection is to show that, in a normed space over $\mathbb{R}$, a compact convex subset can be separated from a disjoint closed convex subset via a closed affine hyperplane in the sense that they reside on the opposite “sides” of this affine hyperplane. Towards this goal, we will first introduce the gauge of a convex subset which is an important notion on its own.

Fix a normed space $(E, \|\cdot\|)$ over $\mathbb{R}$ and a convex subset $A \subseteq E$ such that $B_{\|\cdot\|}(\mathbf{0}, \delta) \subseteq A$ for some $\delta \in \mathbb{R}^+$. We define the *gauge* (or, *Minkowski functional*) of A to be the function $q_A : E \rightarrow [0, \infty)$ given by

$$q_A(x) = \inf\{\lambda > 0 : x \in \lambda A\}$$

for all $x \in E$. Observe that the set on the right-hand side is non-empty because the assumption $B_{\|\cdot\|}(\mathbf{0}, \delta) \subseteq A$ implies that $x \in \frac{2\|x\|}{\delta}A$ for all $x \in E$ and hence the

⁶Classically, for a finite-dimensional vector space V of dimension $n \in \mathbb{N}^+$, a hyperplane is defined to be a subspace $H \subseteq V$ of dimension $n - 1$. This definition does not work for infinite dimensional spaces. A moment’s thought reveals that, for finite-dimensional spaces, being a hyperplane is equivalent to being the kernel of a non-zero linear functional. Clearly the latter statement makes sense for infinite-dimensional spaces as well. This is exactly why we are going to use this condition as the definition of a hyperplane.

infimum is indeed defined.⁷ Moreover, we have $q_A(x) \leq 1$ for all $x \in A$ as well. Let us now investigate some basic properties of the gauge function.

Proposition 11. *Let $r \in \mathbb{R}^+$ and $x \in E$. Then $x \in (q_A(x) + r)A$.*

Proof. Since $q_A(x) + r > q_A(x) = \inf\{\lambda > 0 : x \in \lambda A\}$, we know by the definition of infimum that there exists $q_A(x) \leq \theta < q_A(x) + r$ such that $x \in \theta A$, say, $x = \theta a$ for some $a \in A$. Since $\mathbf{0}, a \in A$ and A is convex, we now obtain

$$\left(1 - \frac{\theta}{q_A(x) + r}\right) \mathbf{0} + \frac{\theta}{q_A(x) + r} a \in A$$

which gives $x = \theta a \in (q_A(x) + r)A$. $\square$

Proposition 12.

- $q_A(\alpha x) = \alpha q_A(x)$ for all $\alpha \geq 0$ and all $x \in E$.
- $q_A(x + y) \leq q_A(x) + q_A(y)$ for all $x, y \in E$.
- There exists $M \in \mathbb{R}^+$ such that $q_A(x) \leq M \|x\|$ for all $x \in E$.
- If A is open, then $A = \{x \in E : q_A(x) < 1\}$.

Proof. To prove the first item, let $\alpha \geq 0$ and $x \in E$. If $\alpha = 0$, then the equality is clearly satisfied. So suppose $\alpha > 0$. Then we have

$$\begin{aligned} q_A(\alpha x) &= \inf\{\lambda > 0 : \alpha x \in \lambda A\} \\ &= \inf\left\{\lambda > 0 : x \in \frac{\lambda}{\alpha} A\right\} \\ &= \inf\{\eta \alpha > 0 : x \in \eta A\} = \alpha \inf\{\eta > 0 : x \in \eta A\} = \alpha q_A(x) \end{aligned}$$

To prove the second item, let $x, y \in E$. We wish to show that

$$q_A(x + y) = \inf\{\lambda > 0 : x + y \in \lambda A\} \leq q_A(x) + q_A(y)$$

Let $\epsilon \in \mathbb{R}^+$. Set $\lambda_1 = q_A(x) + \epsilon/2$ and $\lambda_2 = q_A(y) + \epsilon/2$. By Proposition 11, we have that $x \in \lambda_1 A$ and $y \in \lambda_2 A$, say, $x = \lambda_1 a$ and $y = \lambda_2 b$ for some $a, b \in A$. Observe that

$$x + y = \lambda_1 a + \lambda_2 b = (\lambda_1 + \lambda_2) \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} a + \frac{\lambda_2}{\lambda_1 + \lambda_2} b \right)$$

Since $a, b \in A$ and A is convex, we obtain that

$$x + y \in (\lambda_1 + \lambda_2)A = (q_A(x) + q_A(y) + \epsilon)A$$

which subsequently implies $q_A(x + y) \leq q_A(x) + q_A(y) + \epsilon$. As this inequality is true for arbitrary $\epsilon \in \mathbb{R}^+$, we must have that $q_A(x + y) \leq q_A(x) + q_A(y)$.

⁷Let us remark that, in the definition of gauge, we could have considered scalars $\lambda \geq 0$ instead of scalars $\lambda > 0$. However, this would not change the gauge function. This is because $0A = \{\mathbf{0}\}$ and hence the only vector for which the right-hand side sets are different for these two definitions is $\mathbf{0}$ but we already have $q_A(\mathbf{0}) = 0$ in both definitions.

To prove the third item, first recall that $B_{\|\cdot\|}(\mathbf{0}, \delta) \subseteq A$ for some $\delta \in \mathbb{R}^+$ by assumption. Set $M = \frac{2}{\delta} > 0$. We claim that $q_A(x) \leq M \|x\|$ for all $x \in E$. Let $x \in E$. The inequality clearly holds for $x = \mathbf{0}$. So suppose that $x \neq \mathbf{0}$. Then we have

$$\frac{\delta}{2\|x\|}x \in B_{\|\cdot\|}(\mathbf{0}, \delta) \subseteq A$$

which means that

$$q_A\left(\frac{\delta}{2\|x\|}x\right) \leq 1$$

Applying the first item, we now obtain $q_A(x) \leq M \|x\|$.

To prove the fourth item, let us first prove the left-to-right inclusion. Let $x \in A$. If $x = \mathbf{0}$, then clearly $q_A(x) = 0 < 1$. So suppose that $x \neq \mathbf{0}$. Since A is open, there exists $\epsilon \in \mathbb{R}^+$ such that $B_{\|\cdot\|}(x, \epsilon) \subseteq A$. Choose some

$$1 < \alpha < 1 + \frac{\epsilon}{\|x\|}$$

Then we have

$$\|\alpha x - x\| = |\alpha - 1| \|x\| < \frac{\epsilon}{\|x\|} \|x\| = \epsilon$$

Thus $\alpha x \in B_{\|\cdot\|}(x, \epsilon) \subseteq A$ and consequently we have $x \in \frac{1}{\alpha}A$. It now follows that

$$q_A(x) \leq \frac{1}{\alpha} < 1$$

This proves the left-to-right inclusion. To prove the converse inclusion, let $x \in E$ be such that $q_A(x) < 1$. Then, by Proposition 11, we have

$$x \in (q_A(x) + (1 - q_A(x)))A = A$$

□

We are now ready to prove that a convex open subset of a normed space over $\mathbb{R}$ containing $\mathbf{0}$ can be separated from a point not in this subset by a closed affine hyperplane so that this set is contained in one side of this hyperplane and the point is contained either on the other side or on the hyperplane itself. More precisely, we have the following.

Theorem 24. *Let $(E, \|\cdot\|)$ be a normed space over $\mathbb{R}$. Let $\hat{x} \in E$ and $A \subseteq E$ be such that*

- *A is convex open,*
- *$\mathbf{0} \in A$, and*
- *$\hat{x} \notin A$.*

Then there exists $f \in E'$ such that

$$f(x) < 1 \leq f(\hat{x})$$

for all $x \in A$.

Proof. Observe that A being open and $\mathbf{0} \in A$ together imply that $B_{\|\cdot\|}(\mathbf{0}, \delta) \subseteq A$ for some $\delta \in \mathbb{R}^+$. Since A is also convex, we have the gauge function q_A which satisfies the previous properties.

Set $F = \langle \hat{x} \rangle$ and consider the map $g : F \rightarrow \mathbb{R}$ given by

$$g(\alpha \hat{x}) = \alpha q_A(\hat{x})$$

for all $\alpha \in \mathbb{R}$. It is easily seen that g is a linear functional.

We claim that $g(x) \leq q_A(x)$ for all $x \in F$. Let $x \in F$. Then $x = \alpha \hat{x}$ for some $\alpha \in \mathbb{R}$. If $\alpha \geq 0$, then we have $g(\alpha \hat{x}) = \alpha q_A(\hat{x}) = q_A(\alpha \hat{x})$ by the first item of Proposition 12. If $\alpha < 0$, then we have $g(x) = \alpha q_A(\hat{x}) \leq 0 < q_A(\alpha \hat{x})$.

We wish to apply the Hahn-Banach theorem for $\mathbb{R}$ -vector spaces. However, we cannot apply the statement that we proved as is, since we do not have that q_A is a seminorm. It turns out that the Hahn-Banach theorem for $\mathbb{R}$ -vector spaces actually holds for not just seminorms but *sublinear functions*, i.e. functions that satisfy the first two items of Proposition 12. We refer the reader to [Kre78, 4.2-1] for a precise statement and proof of this variation. The reader is also welcome to check that our proof actually goes through for this variation.

Applying the Hahn-Banach theorem, we obtain a linear extension $f : E \rightarrow \mathbb{R}$ of g to E such that $f(x) \leq q_A(x)$ for all $x \in E$. Now, by the third item of Proposition 12, we know that there exists $M \in \mathbb{R}^+$ such that $q_A(x) \leq M \|x\|$ for all $x \in E$. Combining these inequalities, we see that f is bounded and hence is continuous. Having constructed a candidate $f \in E'$, we finally check whether it has the desired properties. Since A is open, by the fourth item of Proposition 12, we have

$$f(x) \leq q_A(x) < 1$$

for all $x \in A$. Since $\hat{x} \notin A$, we must have $q_A(\hat{x}) \geq 1$; otherwise we would have $\hat{x} \in (q_A(\hat{x}) + (1 - q_A(\hat{x})))A = A$ by Proposition 11. Hence $f(\hat{x}) = q_A(\hat{x}) \geq 1$, which finishes the proof. $\square$

We would like to point out that the non-strict inequality $1 \leq f(\hat{x})$ cannot be strengthened to $1 < f(\hat{x})$. To see this, choose A to be the open unit disc in $\mathbb{R}^2$ and $\hat{x}$ to be any point on its boundary. Then an affine hyperplane, a line in $\mathbb{R}^2$ in this case, which “separates” the open unit disc and this point has to contain this point.

Next will be proven a generalization of this result where we separate a convex open subset from an arbitrary disjoint convex set.

Theorem 25. *Let $(E, \|\cdot\|)$ be a normed space over $\mathbb{R}$. Let $A, B \subseteq E$ be such that*

- *A is convex open,*
- *B is convex, and*
- *A and B are disjoint.*

Then there exists $f \in E'$ and $\gamma \in \mathbb{R}$ such that

$$f(x) < \gamma \leq f(y)$$

for all $x \in A$ and $y \in B$.

Proof. If $A = \emptyset$ or $B = \emptyset$, then the conclusion vacuously holds. Suppose that $A \neq \emptyset$ and $B \neq \emptyset$. Choose $a \in A$ and $b \in B$. Set $\hat{x} = b - a$ and consider the set

$$D = \hat{x} + A - B$$

We claim that D is convex open. To see that D is open, observe that

$$D = \bigcup_{y \in B} \hat{x} + A - y$$

The map $\cdot \mapsto \hat{x} + A - y$ is a homeomorphism for all $y \in B$ and hence D is a union of open sets which means that it is open. To see that D is convex, let $\hat{x} + x - y$ and $\hat{x} + x' - y'$ be in D where $x, x' \in A$ and $y, y' \in B$. Let $0 \leq \alpha \leq 1$ and observe that

$$\alpha(\hat{x} + x - y) + (1 - \alpha)(\hat{x} + x' - y') = \hat{x} + (\alpha x + (1 - \alpha)x') - (\alpha y + (1 - \alpha)y')$$

Since A and B are convex, we must have $(\alpha x + (1 - \alpha)x') \in A$ and $(\alpha y + (1 - \alpha)y') \in B$ and hence

$$\alpha(\hat{x} + x - y) + (1 - \alpha)(\hat{x} + x' - y') \in \hat{x} + A - B = D$$

Thus D is convex. If it were that $\hat{x} \in D = \hat{x} + A - B$, then we would have that A and B have non-empty intersection which contradicts the given assumption. Hence $\hat{x} \notin D$.

Applying Theorem 24, we see that there exists $f \in E'$ such that

$$f(z) < 1 \leq f(\hat{x})$$

for all $z \in D$. It follows that, for all $x \in A$ and $y \in B$, we have

$$1 > f(\hat{x} + x - y) = f(\hat{x}) + f(x) - f(y) \geq 1 + f(x) - f(y)$$

which implies that $f(x) < f(y)$.

We have just shown that for each $y \in B$, we have $f(x) < f(y)$ for all $x \in A$. Hence, setting $\gamma = \sup\{f(x) : x \in A\}$, we obtain from the definition of supremum that $\gamma \leq f(y)$ for all $y \in B$. This proves half of what we desire. In order to get the other inequality, we will use a lemma that will be proven after this proof, namely, that a non-zero linear functional is an open map. By Lemma 4, we have that $f(A)$ is open as A is open. Thus $f(x) < \gamma$ for all $x \in A$. It follows that

$$f(x) < \gamma \leq f(y)$$

for all $x \in A$ and $y \in B$. □

Now we have to fulfill our promise and prove that linear functionals are open maps.

Lemma 4. *Let $(E, \|\cdot\|)$ be a normed space over $\mathbb{R}$ and let $f : E \rightarrow \mathbb{R}$ be a non-zero linear functional. Then f is an open map, i.e. it takes open sets to open sets.*

Proof. Since f is non-zero, it must be surjective and hence there exists $\hat{x} \in E$ such that $f(\hat{x}) = 1$. Let $O \subseteq E$ be an open set. We wish to show that $f(O)$ is open. Let $y \in f(O)$. We wish to produce some $\epsilon > 0$ such that $B_{|\cdot|}(y, \epsilon) = (y - \epsilon, y + \epsilon) \subseteq f(O)$. Observe that, by definition, $y = f(x)$ for some $x \in O$. Since O is open, there exists $\delta > 0$ such that $B_{\|\cdot\|}(x, \delta) \subseteq O$. We claim that $\epsilon = \frac{\delta}{\|\hat{x}\|}$ is as desired.

Let $z \in B_{|\cdot|}(y, \epsilon) = (y - \epsilon, y + \epsilon)$. Then $z = y + \eta$ for some $|\eta| < \epsilon$. Set $y_\eta = \eta\hat{x} + x$. Then clearly $\|y_\eta - x\| = |\eta|\|\hat{x}\| < \epsilon\|\hat{x}\| = \delta$. This means that $y_\eta \in B_{\|\cdot\|}(x, \delta) \subseteq O$ which subsequently implies that $f(y_\eta) \in f(O)$. But we also have

$$f(y_\eta) = f(\eta\hat{x} + x) = \eta f(\hat{x}) + f(x) = f(x) + \eta = y + \eta = z$$

which means that $z \in f(O)$. Thus $B_{|\cdot|}(y, \epsilon) \subseteq f(O)$ which completes the proof that $f(O)$ is open. $\square$

We are now ready to prove the main theorem of this subsection which states that a convex compact subset of a normed space over $\mathbb{R}$ can be separated from a disjoint closed subset via an affine hyperplane in the sense that these sets reside on different sides of this hyperplane. More precisely, we have the following.

Theorem 26. *Let $(E, \|\cdot\|)$ be a normed space over $\mathbb{R}$. Let $A, B \subseteq E$ be such that*

- *A is convex compact,*
- *B is convex closed, and*
- *A and B are disjoint.*

Then there exists $f \in E'$ and $\gamma_1, \gamma_2 \in \mathbb{R}$ such that

$$f(x) < \gamma_1 < \gamma_2 < f(y)$$

for all $x \in A$ and $y \in B$.

Proof. As before, in the case that A or B is empty, there is nothing to prove. So suppose that A and B are both non-empty. For each $r \in \mathbb{R}^+$, set

$$A_r = \{z \in E : \inf_{a \in A} \|z - a\| < r\}$$

In other words, A_r is defined as the set of points of E whose distance to the set A is less than r . A straightforward verification, which we leave to the reader, shows that A_r is convex open for all $r \in \mathbb{R}^+$.

We claim that there exists $n \in \mathbb{N}^+$ such that $A_{1/n} \cap B = \emptyset$. Assume towards a contradiction that for each $n \in \mathbb{N}^+$ there exists $b_n \in A_{1/n} \cap B$. Since $b_n \in$

$A_{1/n}$, there exists $a_n \in A$ such that $\|a_n - b_n\| < 1/n$. Since A is compact, it is sequentially compact and so there exists a convergent subsequence $(a_{n_k})_{k \in \mathbb{N}}$ with $\lim_{k \rightarrow \infty} a_{n_k} = a \in A$. But it is easily check that $\lim_{n \rightarrow \infty} b_n = a$. As B is closed, we obtain $a \in A \cap B$ which contradicts the disjointness assumption.

Fix some $n \in \mathbb{N}^+$ such that $A_{1/n} \cap B = \emptyset$. Applying Theorem 25 to the open convex set $A_{1/n}$ and the set B , we see that there exists $f \in E'$ and $\gamma \in \mathbb{R}$ such that

$$f(x) < \gamma \leq f(y)$$

for all $x \in A_{1/n}$ and $y \in B$.

We now claim that there exists $w \in E$ such that $\|w\| < 1$ and $f(w) > \frac{2}{3} \|f\|$. Suppose to the contrary that for all $w \in E$ with $\|w\| < 1$, we have $f(w) \leq \frac{2}{3} \|f\|$. Let $x \in E$. Then we have

$$\left| f\left(\frac{3}{4\|x\|}x\right) \right| \leq \frac{2}{3} \|f\|$$

and hence

$$\left| f\left(\frac{x}{\|x\|}\right) \right| \leq \frac{8}{9} \|f\|$$

which means that

$$|f(x)| \leq \frac{8}{9} \|f\| \|x\|$$

Since f is non-zero, we have $\|f\| > 0$ and consequently, the previous inequality leads to the contradiction that

$$\|f\| = \sup_{\|x\|_E \leq 1} |f(x)| \leq \frac{8}{9} \|f\|$$

Now fix $w \in E$ with $\|w\| < 1$ and $f(w) > \frac{2}{3} \|f\|$. Let $a \in A$. Then $a + \frac{1}{n}w \in A_{1/n}$ which implies that

$$\gamma > f\left(a + \frac{1}{n}w\right) = f(a) + \frac{1}{n}f(w) > f(a) + \frac{1}{n} \frac{2}{3} \|f\|$$

Hence

$$f(a) < \gamma - \frac{1}{n} \frac{2}{3} \|f\| < \gamma \leq f(b)$$

for all $a \in A$ and $b \in B$. Finally choose $\gamma_1, \gamma_2 \in \mathbb{R}$ such that

$$\gamma - \frac{1}{n} \frac{2}{3} \|f\| < \gamma_1 < \gamma_2 < \gamma$$

Then we get the desired inequality. $\square$

3. BAIRE CATEGORY THEOREM AND ITS CONSEQUENCES

In this section, we will see some important applications of the Baire category theorem in functional analysis. Although we have recalled the related notions and proved the Baire category theorem in class, this background material will not be included in these lecture notes, since this material is actually a part of MATH349. The reader who wishes to learn this background material may consult to Lecture 14 of MATH535 lecture notes of the author [from this link](#).

In this section, we shall prove three fundamental results: The uniform boundedness principle, the open mapping theorem and the closed graph theorem. These results themselves have important consequences (for example, the first isomorphism theorem for Banach spaces) which we shall also learn.

3.1. The dimension of a Banach space. Before we proceed to prove these results, let us first fulfill our earlier promise and prove that there is no countably infinite-dimensional Banach space. We first need the following basic but important lemma.

Lemma 5. *Let $(E, \|\cdot\|)$ be a normed space and $F \subseteq E$ be a proper subspace. Then $\text{Int}(F) = \emptyset$.*

Proof. Assume to the contrary that $\text{Int}(F) \neq \emptyset$. Then there exists $\hat{x} \in E$ and $\epsilon \in \mathbb{R}^+$ such that $B_{\|\cdot\|}(\hat{x}, \epsilon) \subseteq F$. Since F is a subspace, we clearly have $\mathbf{0} \in F$. Now, for any $x \in E$ with $x \neq \mathbf{0}$, we also have that

$$\hat{x} + \frac{\epsilon}{2\|x\|}x \in B_{\|\cdot\|}(\hat{x}, \epsilon) \subseteq F$$

which implies $x \in F$, since F is a subspace and $\hat{x} \in F$. Thus $F = E$ which contradicts the assumption that F is proper. $\square$

The following result is an immediate consequence of this lemma.

Corollary 27. *Let $(E, \|\cdot\|)$ be a normed space and $F \subseteq E$ be a proper closed subspace. Then F is nowhere dense.*

Proof. It follows from Lemma 5 that $\text{Int}(\overline{F}) = \text{Int}(F) = \emptyset$. $\square$

We are now ready to prove that every infinite-dimensional Banach space is of uncountable dimension.

Theorem 28. *Let $(E, \|\cdot\|)$ be a Banach space over $\mathbb{K}$. Then we have*

$$\dim_{\mathbb{K}}(E) < \aleph_0 \quad \text{or} \quad \aleph_0 < \dim_{\mathbb{K}}(E)$$

Proof. Assume to the contrary that $\dim_{\mathbb{K}}(E) = \aleph_0$, say, $H = \{h_n\}_{n \in \mathbb{N}}$ is a Hamel basis for E . For each $n \in \mathbb{N}$, consider $W_n = \langle h_i : 0 \leq i \leq n \rangle$. Observe that W_n is closed since it is a finite-dimensional subspace. Moreover, since $\dim_{\mathbb{K}}(E) = \aleph_0$, we

have that W_n is proper. Thus, by Lemma 27, W_n is nowhere dense for all $n \in \mathbb{N}$. However, H being a Hamel basis, we must have

$$E = \bigcup_{n \in \mathbb{N}} W_n$$

This implies that E is meager. This is a contradiction since the Baire category theorem implies that every non-empty open subset of a completely metrizable topological space, in particular, of E is non-meager. $\square$

It turns out that we actually have the stronger result that $\dim_{\mathbb{K}}(E) < \aleph_0$ or $\mathfrak{c} \leq \dim_{\mathbb{K}}(E)$ for every Banach space $(E, \|\cdot\|)$. The reader is referred to an elegant half-page proof of this fact in [Lac73].

3.2. The uniform boundedness principle and its consequences. Next will be proven the first important result of Section 3, namely, the uniform boundedness principle which is often called *the Banach-Steinhaus theorem*.

Theorem 29 (The Uniform Boundedness Principle). *Let $(E, \|\cdot\|_E)$ be a Banach space and let $(F, \|\cdot\|_F)$ be a normed space. Let $\mathcal{T} \subseteq L(E, F)$ be a set of bounded linear operators such that $\sup_{T \in \mathcal{T}} \|T(x)\|_F < \infty$ for every $x \in E$. Then we have*

$$\sup_{T \in \mathcal{T}} \|T\| < \infty$$

Proof. For each $n \in \mathbb{N}$, consider the set

$$\begin{aligned} A_n &= \{x \in E : \forall T \in \mathcal{T} \quad \|T(x)\|_F \leq n\} \\ &= \bigcap_{T \in \mathcal{T}} \{x \in E : \|T(x)\|_F \leq n\} \end{aligned}$$

Observe that, for each $T \in \mathcal{T}$, the set $\{x \in E : \|T(x)\|_F \leq n\}$ is closed as it is the inverse image of the closed set $(-\infty, n]$ under the continuous map $\cdot \mapsto \|T(\cdot)\|_F$. Hence A_n is closed for all $n \in \mathbb{N}$. By the given assumption, we know that

$$E = \bigcup_{n \in \mathbb{N}} A_n$$

Since E is Banach, it cannot be meager by the Baire category theorem and hence A_k is not nowhere dense for some $k \in \mathbb{N}$. Thus there exists $\hat{x} \in A_k$ and $\epsilon > 0$ such that $B_{\|\cdot\|_E}(\hat{x}, \epsilon) \subseteq \text{Int}(A_k) \subseteq A_k$.

We will now find a uniform bound for the norms of maps in $\mathcal{T}$ using this ϵ . Let $T \in \mathcal{T}$. Let $u \in E$ be such that $\|u\|_E = 1$. Then we have that

$$\begin{aligned} \|T(u)\|_F &= \left\| T \left(\frac{2}{\epsilon} \left(\hat{x} + \frac{\epsilon}{2} u - \hat{x} \right) \right) \right\|_F \\ &= \frac{2}{\epsilon} \left\| T \left(\hat{x} + \frac{\epsilon}{2} u \right) - T(\hat{x}) \right\|_F \end{aligned}$$

Since $\hat{x}, \hat{x} + \frac{\epsilon}{2}u \in B_{\|\cdot\|_E}(\hat{x}, \epsilon) \subseteq A_k$, we have that

$$\|T(u)\|_F \leq \frac{2}{\epsilon} \left(\left\| T\left(\hat{x} + \frac{\epsilon}{2}u\right) \right\|_F + \|T(\hat{x})\|_F \right) \leq \frac{2}{\epsilon}(k + k) = \frac{4k}{\epsilon}$$

Thus $\|T\| = \sup_{\|u\|_E=1} \|T(u)\|_F \leq 4k/\epsilon$ for all $T \in \mathcal{T}$, which immediately gives

$$\sup_{T \in \mathcal{T}} \|T\| \leq \frac{4k}{\epsilon} < \infty$$

□

We are now going to see some consequences of the uniform boundedness principle. The first consequence is that any sequence of continuous linear maps between Banach spaces which “defines a Cauchy sequence pointwise” gives rise to a continuous linear functional whose norm is bounded by the supremum of the norms of the maps in the sequence. More precisely, we have the following.

Theorem 30. *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be Banach spaces. Let $(T_n)_{n \in \mathbb{N}}$ be a sequence of elements of $L(E, F)$ such that $(T_n(x))_{n \in \mathbb{N}}$ is Cauchy, for all $x \in E$. Then the map $T : E \rightarrow F$ given by*

$$T(x) = \lim_{n \rightarrow \infty} T_n(x)$$

for all $x \in E$ is in $L(E, F)$ and moreover, $\|T\| \leq \sup_{n \in \mathbb{N}} \|T_n\| < \infty$.

Proof. First observe that the map T is indeed defined for all $x \in E$ because the Cauchyness of $(T_n(x))_{n \in \mathbb{N}}$ and that F is Banach together imply that $\lim_{n \rightarrow \infty} T_n(x)$ exists. The linearity of the map T follows from the continuity of addition and scalar multiplication of a normed space, since $\lim$ commutes with (these) continuous functions. The non-trivial steps are to prove that T is indeed bounded and to derive the given bound on its norm.

For all $x \in E$, the sequence $(T_n(x))_{n \in \mathbb{N}}$ is bounded as it is Cauchy. Hence we have $\sup_{n \in \mathbb{N}} \|T_n(x)\|_F < \infty$ for all $x \in E$. By the uniform boundedness principle, we now have that

$$\sup_{n \in \mathbb{N}} \|T_n\| = M < \infty$$

We claim that $\|T(x)\|_F \leq M \|x\|_E$ for all $x \in E$. To see this, let $\epsilon > 0$ and $x \in E$. By definition of T , we can find some $N \in \mathbb{N}$, which possibly depends on x , such that

$$\|T(x) - T_N(x)\|_F < \epsilon$$

It then follows that

$$\begin{aligned} \|T(x)\|_F &= \|T_N(x) - (T_N(x) - T(x))\|_F \leq \|T_N(x)\|_F + \|T(x) - T_N(x)\|_F \\ &\leq \|T_N\| \|x\|_E + \epsilon \\ &\leq M \|x\|_E + \epsilon \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, we obtain that $\|T(x)\|_F \leq M \|x\|_E$ for all $x \in E$. This shows that T is bounded and hence is continuous. Moreover, using this bound for each $x \in E$, we also obtain

$$\|T\| = \sup_{\|x\|_E \leq 1} \|T(x)\|_F \leq \sup_{\|x\|_E \leq 1} M \|x\|_E = M = \sup_{n \in \mathbb{N}} \|T_n\| = M$$

□

The second consequence of the uniform boundedness principle that we shall learn is the equivalence of the notions of boundedness and weak boundedness.

Let $(E, \|\cdot\|)$ be a normed space and $A \subseteq E$. The set A is said to be *weakly bounded* if $f(A) \subseteq \mathbb{K}$ is bounded for all $f \in E'$. We would like to remark that if A is bounded, then it is weakly bounded as continuous linear maps take bounded sets to bounded sets. There is no a priori reason that a weakly bounded set is bounded. It turns out that this is the case.

Theorem 31. *Let $(E, \|\cdot\|)$ be a normed space and let $A \subseteq E$ be weakly bounded. Then A is bounded.*

Proof. By assumption, we know that

$$\sup_{x \in A} |f(x)| < \infty$$

for all $f \in E'$. We shall now rewrite this statement using the isometric linear map $i : E \rightarrow E''$ into the double dual which was introduced at the end of Section 2.5. We have

$$\sup_{i(x) \in i(A)} |(i(x))(f)| < \infty$$

for all $f \in E'$. Note that this statement is precisely the hypothesis of the uniform boundedness principle applied between spaces E' and $\mathbb{K}$ where the set of bounded linear maps $\mathcal{T}$ is to be taken $i(A) \subseteq E'' = L(E', \mathbb{K})$. Hence, by the uniform boundedness principle, we have that

$$\sup_{i(x) \in i(A)} \|i(x)\|_{E''} = M < \infty$$

Since i is an isometry, this implies that

$$\sup_{x \in A} \|x\|_E = M < \infty$$

Hence A is bounded. □

The last consequence of the uniform boundedness principle that we shall see is the discontinuity of “most” coefficient functionals associated to a Hamel basis of an infinite-dimensional Banach space.

Let $(E, \|\cdot\|)$ be a Banach space and let $H \subseteq E$ be a Hamel basis, say,

$$H = \{h_\alpha : \alpha \in I\}$$

for some index set I . For each $\alpha \in I$, consider the map $T_\alpha : E \rightarrow \mathbb{K}$ given by

$$T_\alpha(x) = k_\alpha \quad \text{where} \quad x = \sum_{\alpha \in I} k_\alpha h_\alpha$$

for all $x \in E$. Note that, since every element of E can be written as a unique finite linear combination of elements of H , we know that the terms in the sum $\sum_{\alpha \in I} k_\alpha h_\alpha$ are all zero except for finitely many $\alpha \in I$. It is not difficult to check that T_α is a linear functional which we will refer to as the α -th coefficient functional. We will prove that only finitely many of these functionals are continuous.

Theorem 32. *The set $J = \{\alpha \in I : T_\alpha \text{ is continuous}\}$ is finite.*

Proof. Assume to the contrary that J is infinite. Then there exists a countably infinite subset $\{\alpha_n\}_{n \in \mathbb{N}^+} \subseteq J$. Consider the sequence $(\lambda_\alpha)_{\alpha \in I}$ given by

$$\lambda_\alpha = \begin{cases} 0 & \text{if } \alpha \neq \alpha_n \text{ for every } n \in \mathbb{N}^+ \\ \frac{n}{\|T_{\alpha_n}\|} & \text{if } \alpha = \alpha_n \text{ for some } n \in \mathbb{N}^+ \end{cases}$$

It is immediately seen that if T_α is continuous, then $\lambda_\alpha T_\alpha$ is continuous. Therefore we have

$$\{\lambda_\alpha T_\alpha : \alpha \in J\} \subseteq L(E, \mathbb{K})$$

This is the collection of bounded linear maps to which we shall apply the uniform boundedness principle. A moment's thought reveals that, for every $x \in E$, the set

$$\{\lambda_\alpha T_\alpha(x) : \alpha \in I\}$$

is finite due to our previous observation that, for every $x \in E$, we have $T_\alpha(x) = 0$ for all but finitely many $\alpha \in I$. Thus we have

$$\sup_{\alpha \in J \subseteq I} |\lambda_\alpha T_\alpha(x)| < \infty$$

for all $x \in E$. It now follows from the uniform boundedness principle that

$$\sup_{\alpha \in J} \|\lambda_\alpha T_\alpha\| = \sup_{\alpha \in J} |\lambda_\alpha| \|T_\alpha\| < \infty$$

But this is a contradiction as we must have

$$\infty = \sup_{n \in \mathbb{N}^+} \frac{n}{\|T_{\alpha_n}\|} \|T_{\alpha_n}\| = \sup_{n \in \mathbb{N}^+} |\lambda_{\alpha_n}| \|T_{\alpha_n}\| \leq \sup_{\alpha \in J} |\lambda_\alpha| \|T_\alpha\| < \infty$$

□

3.3. The open mapping theorem and its consequences. In this subsection, we shall prove the open mapping theorem which itself has many consequences. Recall that an *open map* between two topological spaces is map that takes open sets to open sets.

Theorem 33 (The Open Mapping Theorem). *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be Banach spaces. Let $T : E \rightarrow F$ be a continuous surjective linear map. Then T is an open map.*

Proof. Let $O \subseteq E$ be an open set. We wish to prove that $T(O) \subseteq F$ is open, that is, for every $y \in T(O)$ there exists $\hat{r} > 0$ such that $B_{\|\cdot\|_F}(y, \hat{r}) \subseteq T(O)$.

Let $y \in T(O)$. By definition, there exists $x \in O$ such that $T(x) = y$. Since O is open and $x \in O$, there exists $\epsilon > 0$ such that $B_{\|\cdot\|_E}(x, \epsilon) \subseteq O$. Suppose for the moment that

$$(\dagger) \text{ There exists } r > 0 \text{ such that } B_{\|\cdot\|_F}(\mathbf{0}, r) \subseteq T(B_{\|\cdot\|_E}(\mathbf{0}, 1))$$

Set $\hat{r} = r\epsilon$. Then the statement $(\dagger)$, the linearity of T and this footnote¹ together imply that

$$\begin{aligned} B_{\|\cdot\|_F}(y, \hat{r}) &= B_{\|\cdot\|_F}(T(x), r\epsilon) = T(x) + B_{\|\cdot\|_F}(\mathbf{0}, r\epsilon) \\ &= T(x) + \epsilon B_{\|\cdot\|_F}(\mathbf{0}, r) \\ &\subseteq T(x) + \epsilon T(B_{\|\cdot\|_E}(\mathbf{0}, 1)) \\ &\subseteq T(x) + T(\epsilon B_{\|\cdot\|_E}(\mathbf{0}, 1)) \\ &\subseteq T(x) + T(B_{\|\cdot\|_E}(\mathbf{0}, \epsilon)) \\ &\subseteq T(x + B_{\|\cdot\|_E}(\mathbf{0}, \epsilon)) \\ &\subseteq T(B_{\|\cdot\|_E}(x, \epsilon)) \subseteq T(O) \end{aligned}$$

This completes the proof that $T(O)$ is open, modulo the assumption $(\dagger)$. It now remains to prove $(\dagger)$. Suppose for the moment that

$$(\dagger\dagger) \text{ There exists } r > 0 \text{ such that for all } n \in \mathbb{N} \text{ we have}$$

$$B_{\|\cdot\|_F}\left(\mathbf{0}, \frac{r}{2^n}\right) \subseteq \overline{T\left(B_{\|\cdot\|_E}\left(\mathbf{0}, \frac{1}{2^{n+1}}\right)\right)}$$

We will show that $(\dagger)$ holds for the same $r > 0$ value. Let $z \in B_{\|\cdot\|_F}(\mathbf{0}, r)$. Next will be constructed a sequence $(x_n)_{n \in \mathbb{N}}$ of points in E such that, for all $n \in \mathbb{N}$,

$$\bullet \ x_n \in B_{\|\cdot\|_E}\left(\mathbf{0}, \frac{1}{2^{n+1}}\right) \text{ and}$$

¹The reader should check the following easy facts: In any normed space $(X, \|\cdot\|_X)$, we have $B_{\|\cdot\|_X}(a + b, r) = a + B_{\|\cdot\|_X}(b, r)$ and $B_{\|\cdot\|_X}(a, \delta r) = \delta B_{\|\cdot\|_X}(a/\delta, r)$ for every $a, b \in X$ and $\delta > 0$ and $r \geq 0$.

- $\left\| z - \sum_{i=0}^n T(x_i) \right\|_F < \frac{r}{2^{n+1}}$

inductively as follows.

- For $n = 0$, applying $(\dagger\dagger)$ with $n = 0$, we know that

$$z \in B_{\|\cdot\|_F}(\mathbf{0}, r) \subseteq \overline{T\left(B_{\|\cdot\|_E}\left(\mathbf{0}, \frac{1}{2}\right)\right)}$$

Since z is in the closure of the image of $B_{\|\cdot\|_E}(\mathbf{0}, \frac{1}{2})$ under T , we can get arbitrarily close to z with points of the form $T(x')$ with $x' \in B_{\|\cdot\|_E}(\mathbf{0}, \frac{1}{2})$. In particular, we can choose a point $x_0 \in B_{\|\cdot\|_E}(\mathbf{0}, \frac{1}{2})$ such that

$$\|z - T(x_0)\|_F < \frac{r}{2}$$

- Let $n \in \mathbb{N}$ and suppose that such points $x_0, x_1, \dots, x_n$ have been already constructed. We wish to find a point $x_{n+1} \in B_{\|\cdot\|_E}(\mathbf{0}, \frac{1}{2^{n+2}})$ that satisfies the given properties. Since we have

$$z - \sum_{i=0}^n T(x_i) \in B_{\|\cdot\|_F}\left(\mathbf{0}, \frac{r}{2^{n+1}}\right)$$

as an inductive assumption, we obtain by applying $(\dagger\dagger)$ for $n + 1$ that

$$z - \sum_{i=0}^n T(x_i) \in \overline{T\left(B_{\|\cdot\|_E}\left(\mathbf{0}, \frac{1}{2^{n+2}}\right)\right)}$$

Therefore, we can choose a point $x_{n+1} \in B_{\|\cdot\|_E}(\mathbf{0}, \frac{1}{2^{n+2}})$ such that

$$\left\| z - \sum_{i=0}^n T(x_i) - T(x_{n+1}) \right\|_F < \frac{r}{2^{n+2}}$$

Thus we can construct such a sequence $(x_n)_{n \in \mathbb{N}}$ by induction. Observe that

$$\sum_{n=0}^{\infty} \|x_n\|_E < \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} = 1 < \infty$$

It now follows from Proposition 5 that $\sum_{n=0}^{\infty} x_n$ converges, say, $x = \sum_{n=0}^{\infty} x_n$. Moreover, by the exact same reasoning, we have

$$\|x\|_E = \left\| \sum_{n=0}^{\infty} x_n \right\|_E \leq \sum_{n=0}^{\infty} \|x_n\|_E < \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} = 1$$

which means $x \in B_{\|\cdot\|_E}(\mathbf{0}, 1)$. Finally, the continuity and linearity of T implies that

$$T(x) = T\left(\sum_{n=0}^{\infty} x_n\right) = T\left(\lim_{N \rightarrow \infty} \sum_{n=0}^N x_n\right) = \lim_{N \rightarrow \infty} \sum_{n=0}^N T(x_n) = z$$

Thus $z \in T(B_{\|\cdot\|_E}(\mathbf{0}, 1))$ which completes the proof of $(\dagger)$ assuming $(\dagger\dagger)$. It now remains to prove $(\dagger\dagger)$ from the hypothesis of the theorem. Since T is surjective and linear, we have

$$F = T(E) = T\left(\bigcup_{n \in \mathbb{N}^+} B_{\|\cdot\|_E}(\mathbf{0}, n)\right) = \bigcup_{n \in \mathbb{N}^+} T(B_{\|\cdot\|_E}(\mathbf{0}, n))$$

But then we also have $F = \bigcup_{n \in \mathbb{N}^+} \overline{T(B_{\|\cdot\|_E}(\mathbf{0}, n))}$ and hence, by the Baire category theorem, we must have that

$$\text{Int}\left(\overline{T(B_{\|\cdot\|_E}(\mathbf{0}, k))}\right) \neq \emptyset$$

for some $k \in \mathbb{N}^+$, otherwise, E would be meager. Therefore there exists some ball $B_{\|\cdot\|_E}(w, \delta) \subseteq \overline{T(B_{\|\cdot\|_E}(\mathbf{0}, k))}$. Set

$$r = \frac{\delta}{2k}$$

We claim that $(\dagger\dagger)$ is satisfied. Let $n \in \mathbb{N}$. Let $z \in B_{\|\cdot\|_F}(\mathbf{0}, \frac{r}{2^n})$. Note that $\|2kz\|_F = 2k\|z\|_F < \frac{2kr}{2^n} = \frac{\delta}{2^n}$. Thus we have

$$\frac{w}{2^n}, \frac{w}{2^n} + 2kz \in B_{\|\cdot\|_E}\left(\frac{w}{2^n}, \frac{\delta}{2^n}\right) = \frac{1}{2^n} B_{\|\cdot\|_E}(w, \delta) \subseteq \frac{1}{2^n} \overline{T(B_{\|\cdot\|_E}(\mathbf{0}, k))}$$

Since multiplying by a fixed scalar is a continuous function from F to F , we also have that

$$\frac{w}{2^n}, \frac{w}{2^n} + 2kz \in \frac{1}{2^n} \overline{T(B_{\|\cdot\|_E}(\mathbf{0}, k))} \subseteq \overline{\frac{1}{2^n} T(B_{\|\cdot\|_E}(\mathbf{0}, k))} = \overline{T\left(B_{\|\cdot\|_E}\left(\mathbf{0}, \frac{k}{2^n}\right)\right)}$$

Consequently, there are sequences $(x_i)_{i \in \mathbb{N}}$ and $(x'_i)_{i \in \mathbb{N}}$ of elements of $B_{\|\cdot\|_E}(\mathbf{0}, \frac{k}{2^n})$ such that

$$\lim_{i \rightarrow \infty} T(x_i) = \frac{w}{2^n} \quad \text{and} \quad \lim_{i \rightarrow \infty} T(x'_i) = \frac{w}{2^n} + 2kz$$

It follows that

$$z = \frac{w/2^n + 2kz - w/2^n}{2k} = \frac{\lim_{i \rightarrow \infty} T(x'_i) - \lim_{i \rightarrow \infty} T(x_i)}{2k} = \lim_{i \rightarrow \infty} T\left(\frac{x'_i}{2k} - \frac{x_i}{2k}\right)$$

But this means that

$$z \in \overline{T\left(B_{\|\cdot\|_E}\left(\mathbf{0}, \frac{1}{2^{n+1}}\right)\right)}$$

because have $\frac{x_i}{2k}, \frac{x'_i}{2k} \in B_{\|\cdot\|_E}(\mathbf{0}, \frac{1}{2^{n+1}})$ for all $i \in \mathbb{N}$. This completes the proof of $(\dagger\dagger)$ and the main theorem. $\square$

The open mapping theorem has many important consequences. The first consequence that we will cover is the Banach isomorphism theorem which says that continuous vector space isomorphisms between Banach spaces are indeed Banach space isomorphisms.

Theorem 34 (The Banach Isomorphism Theorem). *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be Banach spaces. Let $T : E \rightarrow F$ be a continuous bijective linear map. Then T is an isomorphism.*

Proof. For all open $O \subseteq E$, we have that $(T^{-1})^{-1}(O) = T(O)$ is open by the open mapping theorem. Thus T^{-1} is continuous which means that T is a linear homeomorphism. $\square$

The second consequence that we will cover is that, given a continuous linear surjection, there exists a right inverse (not necessarily linear) map which is bounded. More precisely, we have the following.

Corollary 35. *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be Banach spaces. Let $T : E \rightarrow F$ be a continuous surjective linear map. Then there exists $M > 0$ such for all $y \in Y$ there exists $x \in X$ with*

$$T(x) = y \quad \text{and} \quad \|x\|_E \leq M \|y\|_F$$

Proof. By the open mapping theorem, T is open and hence, $T(B_{\|\cdot\|_E}(\mathbf{0}, 1))$ is open. Then there exists $\epsilon > 0$ such that $B_{\|\cdot\|_F}(\mathbf{0}, \epsilon) \subseteq T(B_{\|\cdot\|_E}(\mathbf{0}, 1))$. Set $M = 2/\epsilon$. We claim that this M works.

Let $y \in F$. If $y = \mathbf{0}$, then we can choose $x = \mathbf{0}$ and the inequality clearly holds. If $y \neq \mathbf{0}$, then we have

$$\frac{\epsilon}{2 \|y\|_F} y \in B_{\|\cdot\|_F}(\mathbf{0}, \epsilon) \subseteq T(B_{\|\cdot\|_E}(\mathbf{0}, 1))$$

which means that there exists $\hat{x} \in B_{\|\cdot\|_E}(\mathbf{0}, 1)$ such that

$$T(\hat{x}) = \frac{\epsilon}{2 \|y\|_F} y$$

Set $x = (2 \|y\|_F / \epsilon) \hat{x}$. Then clearly $T(x) = y$ and moreover

$$\|x\|_E = \left\| \frac{2 \|y\|_F}{\epsilon} \hat{x} \right\|_E = \frac{2 \|y\|_F}{\epsilon} \|\hat{x}\|_E \leq M \|y\|_F$$

$\square$

The third consequence of the open mapping theorem that we will learn is that if two norms that give Banach space structures on the same vector space are comparable, then they are actually equivalent.

Corollary 36. *Let $(E, \|\cdot\|_1)$ and $(E, \|\cdot\|_2)$ be Banach spaces. Suppose that there exists $M > 0$ such that $\|x\|_2 \leq M \|x\|_1$ for all $x \in E$. Then $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent norms.*

Proof. Consider the identity map $T : (E, \|\cdot\|_1) \rightarrow (E, \|\cdot\|_2)$. By assumption, we have $\|T(x)\|_2 = \|x\|_2 \leq M \|x\|_1$ for all $x \in E$ and hence T is a continuous linear bijection. By the Banach isomorphism theorem, T is a linear homeomorphism. In

particular, T^{-1} is continuous and so, is bounded. This implies that there exists $m > 0$ with

$$\|x\|_1 = \|T^{-1}(x)\|_1 \leq m \|x\|_2$$

which shows that $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent. $\square$

The last consequence of the open mapping theorem we shall see is the first isomorphism theorem for Banach spaces which says that the quotient of a Banach space by the kernel of a surjective continuous linear map is an isomorphism to its image.

Theorem 37 (The First Isomorphism Theorem). *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be Banach spaces and $T : E \rightarrow F$ be a continuous linear surjection. Then the Banach spaces $(E/\ker(T), \|\cdot\|)$ and $(F, \|\cdot\|_F)$ are isomorphic.*

Proof. Observe that, as T is continuous, $\ker(T) = T^{-1}(\mathbf{0}_F)$ is closed and hence $E/\ker(T)$ indeed has a norm space structure with the quotient norm $\|\cdot\|$. Moreover, since E is Banach, the quotient space $(E/\ker(T), \|\cdot\|)$ is Banach by Theorem 13. Consider the map $S : E/\ker(T) \rightarrow F$ given by

$$S(x + \ker(T)) = T(x)$$

for all $x \in E$. Since we defined S in terms of coset representatives, we first need to check that S is well-defined. Let $x, y \in E$. Then we have

$$\begin{aligned} x + \ker(T) = y + \ker(T) & \text{ if and only if } x - y \in \ker(T) \\ & \text{ if and only if } T(x - y) = \mathbf{0}_F \\ & \text{ if and only if } T(x) = T(y) \\ & \text{ if and only if } S(x + \ker(T)) = S(y + \ker(T)) \end{aligned}$$

This not only shows that S is well-defined but also shows that S is injective. Having seen that the definition of S is independent of the coset representative, it immediately follows that S is linear because T is linear. Let $y \in F$. Since T is surjective, there exists $x \in E$ with $T(x) = y$ which means that $S(x + \ker(T)) = y$. Thus S is surjective. We now check that S is bounded and hence is continuous.

Let $x + \ker(T) \in E/\ker(T)$. Then, for any $y \in \ker(T)$, we have that

$$\begin{aligned} \|S(x + \ker(T))\|_F &= \|T(x)\|_F \\ &= \|T(x) - T(y)\|_F \\ &= \|T(x - y)\|_F \\ &\leq \|T\| \|x - y\|_E \end{aligned}$$

Taking infimum over all $y \in \ker(T)$, we now obtain

$$\|S(x + \ker(T))\|_F \leq \inf_{y \in \ker(T)} \|T\| \|x - y\|_E = \|T\| \|x + \ker(T)\|$$

which means that S is bounded. By the Banach isomorphism theorem, S being a continuous linear bijective map between two Banach spaces, it is an isomorphism. $\square$

An interesting consequence of Theorem 37 is that separable Banach spaces are precisely the quotients of $(\ell_1, \|\cdot\|_1)$ by closed subspaces.

Theorem 38. *Every separable Banach space is isomorphic to a quotient of $(\ell_1, \|\cdot\|_1)$ by a closed subspace.*

Proof. Let $(E, \|\cdot\|_E)$ be a separable Banach space. We shall construct a continuous linear surjection from ℓ_1 to E . Since E is separable and metrizable, its topological subspace $B_{\|\cdot\|_E}(\mathbf{0}_E, 1)$ is separable. Thus we can pick a countable dense

$$D = \{d_n : n \in \mathbb{N}\} \subseteq B_{\|\cdot\|_E}(\mathbf{0}_E, 1)$$

Consider the map $T : \ell_1 \rightarrow E$ given by

$$T(\mathbf{a}) = \sum_{n=0}^{\infty} a_n d_n$$

for all $\mathbf{a} \in \ell_1$. It is not immediately clear that the definition of T makes sense, that is, the series on the right-hand side actually converges for all $\mathbf{a} \in \ell_1$. Let us now check that this is the case. Let $\mathbf{a} \in \ell_1$. Then we have that

$$\sum_{n=0}^{\infty} \|a_n d_n\|_E = \sum_{n=0}^{\infty} |a_n| \|d_n\|_E \leq \sum_{n=0}^{\infty} |a_n| < \infty$$

Thus $T(\mathbf{a})$ is indeed defined for all $\mathbf{a} \in \ell_1$. It is clear that T is a linear map. Moreover, by the same argument that we have seen above, we have

$$\|T(\mathbf{a})\|_E = \left\| \sum_{n=0}^{\infty} a_n d_n \right\|_E \leq \sum_{n=0}^{\infty} \|a_n d_n\|_E \leq \sum_{n=0}^{\infty} |a_n| = \|\mathbf{a}\|_1$$

Thus T is bounded and so is continuous. We will now show that T is surjective. Since T is linear, in order to show that T is surjective, it suffices to show that $B_{\|\cdot\|_E}(\mathbf{0}_E, 1) \subseteq \text{Im}(T)$. Let $z \in B_{\|\cdot\|_E}(\mathbf{0}_E, 1)$. We will inductively construct a sequence of points $(d_{n_k})_{k \in \mathbb{N}}$ of D such that

$$\left\| z - \sum_{i=0}^k \frac{d_{n_i}}{2^i} \right\|_E \leq \frac{1}{2^{k+2}}$$

for all $k \in \mathbb{N}$.

- For $k = 0$, we can choose some $d_{n_0} \in D$ with $\|z - d_{n_0}\|_E \leq \frac{1}{2^2}$ since D is dense in $B_{\|\cdot\|_E}(\mathbf{0}_E, 1)$.

- Let $k \in \mathbb{N}$ and suppose that such $d_{n_0}, \dots, d_{n_k} \in D$ have been constructed. We will show that we can choose $d_{n_{k+1}} \in D$ so that

$$\left\| z - \sum_{i=0}^{k+1} \frac{d_{n_i}}{2^i} \right\|_E \leq \frac{1}{2^{(k+1)+2}}$$

By the inductive assumption, we know that

$$\left\| z - \sum_{i=0}^k \frac{d_{n_i}}{2^i} \right\|_E \leq \frac{1}{2^{k+2}}$$

Multiplying both sides by 2^{k+1} , we obtain

$$\left\| 2^{k+1} \left(z - \sum_{i=0}^k \frac{d_{n_i}}{2^i} \right) \right\|_E \leq \frac{1}{2}$$

Thus

$$2^{k+1} \left(z - \sum_{i=0}^k \frac{d_{n_i}}{2^i} \right) \in B_{\|\cdot\|_E}(\mathbf{0}_E, 1)$$

By the density of D in $B_{\|\cdot\|_E}(\mathbf{0}_E, 1)$, we can now choose some $d_{n_{k+1}} \in D$ such that

$$\left\| \left(2^{k+1} \left(z - \sum_{i=0}^k \frac{d_{n_i}}{2^i} \right) \right) - d_{n_{k+1}} \right\|_E \leq \frac{1}{2^2}$$

Dividing both sides by 2^{k+1} , we now obtain

$$\left\| z - \sum_{i=0}^{k+1} \frac{d_{n_i}}{2^i} \right\|_E \leq \frac{1}{2^{k+3}}$$

Thus, by induction, there exists such a sequence $(d_{n_k})_{k \in \mathbb{N}}$. Observe that this sequence of points of D may contain repetitions. Set

$$\lambda_n = \sum_{k \in I_n} \frac{1}{2^k} \text{ where } I_n = \{k \in \mathbb{N} : n_k = n\}$$

Then we have $(\lambda_n)_{n \in \mathbb{N}} \in \ell_1$. It is an exercise² to the reader to check that the partial sums of $\sum_{n=0}^{\infty} \lambda_n d_n$ converge to z and hence, we have

$$T((\lambda_n)_{n \in \mathbb{N}}) = \sum_{n=0}^{\infty} \lambda_n d_n = z$$

This completes the proof that T is surjective. Having a continuous linear surjection from ℓ_1 to E , we finally apply the first isomorphism theorem for Banach spaces and get that $(\ell_1/\ker(T), \|\cdot\|)$ is isomorphic to $(E, \|\cdot\|_E)$. $\square$

²This is relatively easy but subtler than it seems at first glance because we are summing up the corresponding coefficients in λ_n 's which means that you have to do two approximation arguments.

3.4. The closed graph theorem. In this subsection, we shall prove the last of the three fundamental results regarding Banach spaces that we needed to cover in course, namely, the closed graph theorem.

It is perhaps more precise to say *a* closed graph theorem because there are many closed graph theorems in different contexts. What they have in common is that they all state that a “certain” function between “certain” topological spaces is continuous provided that its graph is a closed subset of the product space. For example, an abstract topological space version states that a map from a topological space to a compact Hausdorff space is continuous if and only if its graph is closed. The version of the closed graph theorem that we shall state is for Banach spaces.

Theorem 39 (The closed graph theorem). *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be Banach spaces and let $T : E \rightarrow F$ be a linear map. If the graph*

$$\text{graph}(T) = \{(x, y) \in E \times F : y = T(x)\}$$

is a closed subset of the product topological space $E \times F$, then T is continuous.

Proof. Suppose that $\text{graph}(T)$ is closed. Consider the map $\|\cdot\|' : E \rightarrow \mathbb{R}$ given by

$$\|x\|' = \|x\|_E + \|T(x)\|_F$$

for all $x \in E$. It is not difficult to check that $\|\cdot\|'$ is a norm on E . We will next show that $(E, \|\cdot\|')$ is a Banach space. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of points of E which is Cauchy with respect to the norm $\|\cdot\|'$. Then $(x_n)_{n \in \mathbb{N}}$ is Cauchy with respect to $\|\cdot\|_E$ and $(T(x_n))_{n \in \mathbb{N}}$ is Cauchy with respect to $\|\cdot\|_F$ simply because $\|x_n - x_m\|_E, \|T(x_n) - T(x_m)\|_F \leq \|x_n - x_m\|'$. Since E and F are Banach, there exists $x \in E$ and $y \in F$ such that $\lim_{n \rightarrow \infty} x_n = x$ in $(E, \|\cdot\|_E)$ and $\lim_{n \rightarrow \infty} T(x_n) = y$ in $(F, \|\cdot\|_F)$. But this means that the sequence $((x_n, T(x_n)))_{n \in \mathbb{N}}$ of points of $E \times F$ converges to (x, y) in the product topological space (whose topology can be given, for example, by the product metric arising from the individual metrics associated to norms.) Since (x, y) is a limit point of $\text{graph}(T)$ and $\text{graph}(T)$ is closed, we have that $(x, y) \in \text{graph}(T)$ and so $T(x) = y$. But then

$$\begin{aligned} \|x_n - x\|' &= \|x_n - x\|_E + \|T(x_n - x)\|_F \\ &= \|x_n - x\|_E + \|T(x_n) - T(x)\|_F \\ &= \|x_n - x\|_E + \|T(x_n) - y\|_F \longrightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. This means that $(x_n)_{n \in \mathbb{N}}$ is convergent with respect to $\|\cdot\|'$. Thus $(E, \|\cdot\|')$ is Banach. Since $\|\cdot\|_E \leq \|\cdot\|'$, applying Corollary 36, we see that $\|\cdot\|_E$ and $\|\cdot\|'$ are equivalent norms on E . This implies that there exists $M > 0$ such that

$$\|T(x)\|_F \leq \|x\|_E + \|T(x)\|_F = \|x\|' \leq M \|x\|_E$$

Thus T is bounded and hence is continuous. □

The assumptions that E and F are Banach in the statement of the closed graph theorem cannot be dropped. To see that we cannot drop the Banachness assumption on E , consider the subspace

$$C^1([0, 1]) = \{f : [0, 1] \rightarrow \mathbb{R} : f \text{ is continuously differentiable}\}$$

of the normed space $(C([0, 1]), \|\cdot\|_\infty)$. Recall that $(C^1([0, 1]), \|\cdot\|_\infty)$ is not a Banach space. We have seen before that the differentiation map $D : C^1([0, 1]) \rightarrow C([0, 1])$ given by $D(f) = f'$ is linear but not continuous.

We will now show that the graph of D is closed. Let $((f_n, D(f_n)))_{n \in \mathbb{N}}$ be a sequence of points of the graph of D which converge to $(f, g) \in C^1([0, 1]) \times C([0, 1])$ in the product space. Then $D(f_n) = f'_n \rightarrow_{n \rightarrow \infty} g$ uniformly and so

$$f_n(x) - f_n(0) = \int_0^x f'_n(t) dt \rightarrow_{n \rightarrow \infty} \int_0^x g(t) dt$$

for all $x \in [0, 1]$. On the other hand, since $f_n \rightarrow_{n \rightarrow \infty} f$ uniformly, we have that $f_n(x) - f_n(0) \rightarrow_{n \rightarrow \infty} f(x) - f(0)$. Thus

$$f(x) = f(0) + \int_0^x g(t) dt$$

which now gives $D(f) = g$. So (f, g) is in the graph of D which means that D is a closed subset of $C^1([0, 1]) \times C([0, 1])$. Thus the Banachness assumption on E is necessary for the closed graph theorem. To see that the Banachness assumption on F is necessary, we refer the reader to [Con90, 12.6] from which we also have taken the previous example.

Unlike the uniform boundedness principle and the open mapping theorem, we shall not learn many concrete consequences of the closed graph theorem. The closed graph theorem is often invoked in the middle of a bigger argument to establish the continuity of a linear map for which checking the continuity from the definition seems more difficult than dealing with its graph.

4. TOPOLOGICALLY COMPLEMENTARY SUBSPACES

In this short section, we shall introduce the notion of topologically complemented subspaces of a normed space and prove some basic results. This section is intended to be a brief overview of these concepts and by no means is it intended to be a comprehensive treatment.

Before we proceed, let us recall the definition of a projection map on a vector space. Let V be a vector space. A map $P : V \rightarrow V$ is called a *projection map* if P is linear and $P^2 = P$. The proof of the following lemma is left to the reader as an exercise.

Lemma 6. *Let $P : V \rightarrow V$ be a projection map. Then*

- *For all $x \in V$, $x \in \text{Im}(P)$ iff $P(x) = x$,*
- *$I - P : V \rightarrow V$ is a projection map on V , and*
- *$\ker(P) = \text{Im}(I - P)$ and $\text{Im}(P) = \ker(I - P)$.*

Recall that two subspaces $W, K \subseteq V$ are said to be *algebraically complementary subspaces* if we have $W + K = V$ and $W \cap K = \{\mathbf{0}_V\}$, in which case we simply write $V = W \oplus K$.

It is not difficult to check that $V = W \oplus K$ if and only if every element $v \in V$ can be uniquely written as $v = v_W + v_K$ for some $v_W \in W$ and $v_K \in K$. In this case, the uniqueness of the decomposition allows us to define two canonical maps $\pi_W : V \rightarrow W$ and $\pi_K : V \rightarrow K$ given by $\pi_W(v) = v_W$ and $\pi_K(v) = v_K$ that give the corresponding elements. The reader may verify that if $V = W \oplus K$, then π_W and π_K are projection maps with $\pi_W + \pi_K = I$. Thus every pair of algebraically complementary subspaces give us a pair of projections that add up to the identity map. Conversely, if $P : V \rightarrow V$ is a projection map, then it easily follows from Lemma 6 that $V = \text{Im}(P) \oplus \text{Im}(I - P) = \text{Im}(P) \oplus \ker(P)$.

Let $(E, \|\cdot\|)$ be a normed space and $F, G \subseteq E$ be subspaces. We say that F and G are *topologically complementary subspaces* if $E = F \oplus G$ and the projection maps π_F and π_G are continuous (equivalently, bounded). In this case, we shall write $E = F \oplus_{\text{top}} G$.

The reader may recall from their linear algebra course that every subspace is *algebraically complemented*, i.e. there exists another subspace with which they form a pair of algebraically complementary subspaces. Given $W \subseteq V$, we can choose a Hamel basis $B \subseteq W$ for the subspace W and extend it to some Hamel basis $B' \supseteq B$ for the space V . Setting $K = \langle B' \setminus B \rangle$, we then obtain $V = W \oplus K$.

The main question that we are interested in is the following: Given a normed space $(E, \|\cdot\|)$ and a subspace $F \subseteq E$, when is it the case that F is *topologically complemented*, i.e. there exists a subspace G such that $E = F \oplus_{\text{top}} G$? The

following proposition lets us rephrase this question in terms of the existence of suitable projection maps.

Proposition 13. *Let $(E, \|\cdot\|)$ be a normed space and $F \subseteq E$ be a subspace. Then F is topologically complemented if and only if there exists a continuous projection map $P : E \rightarrow E$ with $\text{Im}(P) = F$.*

Proof. Suppose that F is topologically complemented, say, $E = F \oplus_{\text{top}} G$ for some subspace $G \subseteq E$. Then, by definition, we know that the projection map $\pi_F : E \rightarrow E$ is continuous and $\text{Im}(\pi_F) = F$.

Conversely, suppose that there exists a continuous projection map $P : E \rightarrow E$ with $\text{Im}(P) = F$. Set $G = \text{Im}(I - P)$. Then, by our previous observations, we know that $E = \text{Im}(P) \oplus \text{Im}(I - P) = F \oplus G$. Since I and P are continuous, so is $I - P$ and hence both $\pi_F = P$ and $\pi_G = I - P$ are continuous. Thus $E = F \oplus_{\text{top}} G$. $\square$

Observe that this proposition immediately implies that a topologically complemented subspace F must be closed since $F = \text{Im}(P) = \ker(I - P) = (I - P)^{-1}(\mathbf{0}_E)$ and $I - P$ is a continuous map. Let us now rephrase our question in light of this new realizations: Given a normed space $(E, \|\cdot\|)$ and a closed subspace $F \subseteq E$, when is there a continuous projection $P : E \rightarrow E$ with $\text{Im}(P) = F$?

In general, the answer to this question is negative, even when E is a Banach space. For example, it is well-known that the closed subspace $c_0 \subseteq \ell_\infty$ is not topologically complemented in $(\ell_\infty, \|\cdot\|_\infty)$. The reader may check [Whi66] for a short proof of this fact. That said, the answer turns out to be positive for some special cases. For example, all closed subspaces are topologically complemented in Hilbert spaces. In an arbitrary normed space, finite-dimensional subspaces (consequently, finite-codimensional subspaces) are also topologically complemented, as shown by the next theorem.

Theorem 40. *Let $(E, \|\cdot\|)$ be a normed space and $F \subseteq E$ be a finite-dimensional subspace. Then F is topologically complemented.*

Proof. Let $B = \{b_i : 1 \leq i \leq n\} \subseteq F$ be a Hamel basis for the subspace F . Using the Hahn-Banach theorem, we can construct continuous linear functionals $f_i \in E'$ such that $f_i(b_j) = \delta_{ij}$ for all $1 \leq i, j \leq n$.¹ Consider the map $f : E \rightarrow E$ given by

$$f(x) = \sum_{i=1}^n f_i(x)b_i$$

¹To construct f_i , start with a functional g_i that is defined to be 0 on the subspace generated by $B \setminus \{b_i\}$. Extend g_i to F so that $g_i(b_i) = 1$, and apply the Hahn-Banach theorem to extend this further to all of E .

for all $x \in E$. It is not difficult to check that f is a linear map. Moreover, $f(b_i) = b_i$ for all $1 \leq i \leq n$. Consequently, we have

$$f(f(x)) = f\left(\sum_{i=1}^n f_i(x)b_i\right) = \sum_{i=1}^n f(f_i(x)b_i) = \sum_{i=1}^n f_i(x)f(b_i) = \sum_{i=1}^n f_i(x)b_i = f(x)$$

for all $x \in E$. Thus f is a projection. It is also easily seen that $\text{Im}(f) = F$. Thus there exists a continuous projection on E whose image is F . Therefore, by Proposition 13, F is topologically complemented. $\square$

Let us now ask a slightly different² question: Given a normed space $(E, \|\cdot\|)$ and two algebraically complementary subspaces $F, G \subseteq E$, when are F and G topologically complementary? Clearly both F and G must be closed as they would be the kernels of the relevant continuous projection maps. It turns out that this is a sufficient condition when E is a Banach space.

Theorem 41. *Let $(E, \|\cdot\|)$ be a Banach space and $F, G \subseteq E$ be closed subspaces. If $E = F \oplus G$, then $E = F \oplus_{\text{top}} G$.*

Proof. Assume that $E = F \oplus G$. Let π_F and π_G be the corresponding projection maps. We wish to show that π_F is continuous, which will automatically imply that $\pi_G = I - \pi_F$ is also continuous.

By the closed graph theorem, it suffices to show that $\text{graph}(\pi_F)$ is closed. Let $((x_n, \pi_F(x_n)))_{n \in \mathbb{N}}$ be a sequence of points of $\text{graph}(\pi_F)$ such that

$$(x_n, \pi_F(x_n)) \longrightarrow_{n \rightarrow \infty} (x, y)$$

in the product space $E \times E$. Recall that this implies that $x_n \longrightarrow_{n \rightarrow \infty} x$ and $\pi_F(x_n) \longrightarrow_{n \rightarrow \infty} y$ in the normed space $(E, \|\cdot\|)$.

We will show that $(x, y) \in \text{graph}(\pi_F)$, that is, $\pi_F(x) = y$. Since F is closed, $\pi_F(x_n) \in F$ for all $n \in \mathbb{N}$ and $\pi_F(x_n) \longrightarrow_{n \rightarrow \infty} y$, we know that $y \in F$. Similarly, since G is closed, $(I - \pi_F)(x_n) \in G$ for all $n \in \mathbb{N}$ and

$$(I - \pi_F)(x_n) = x_n - \pi_F(x_n) \longrightarrow_{n \rightarrow \infty} x - y$$

we know that $x - y \in G$. But then

$$0 = \pi_F(x - y) = \pi_F(x) - \pi_F(y) = \pi_F(x) - y$$

which means $\pi_F(x) = y$. Thus $\text{graph}(\pi_F)$ contains all its limit points and hence is closed. $\square$

²If you think about it, we are *not* really asking a different question here. Given an arbitrary closed subspace F , we already know that there exists an algebraic complement G . The question is whether this algebraic complement is a topological complement or not.

Therefore, in Banach spaces, closed subspaces whose algebraic complements are also closed are topologically complemented. Finally, before we conclude, let us show that the Banachness condition in Theorem 41 cannot be dropped. We shall work in $(\ell_1, \|\cdot\|_1)$ to construct E, F, G such that $F, G \subseteq E$ are closed subspaces with $E = F \oplus G$ but F and G are not topologically complemented in E . Consider the closed subspaces

$$F = \overline{\langle \mathbf{e}_{2n} : n \in \mathbb{N} \rangle}$$

and

$$G = \overline{\left\langle \mathbf{e}_{2n} + \frac{1}{2n+1} \mathbf{e}_{2n+1} : n \in \mathbb{N} \right\rangle}$$

where the closure is taken in the normed space $(\ell_1, \|\cdot\|_1)$.

Set $E = F + G$. We claim that $F \cap G = \{\mathbf{0}\}$. Let $\mathbf{x} \in F \cap G$. We will first show that $x_{2k+1} = 0$ for all $k \in \mathbb{N}$. Suppose that $x_{2k+1} \neq 0$ for some $k \in \mathbb{N}$. Observe that, for all $\mathbf{y} \in \langle \mathbf{e}_{2n} : n \in \mathbb{N} \rangle$, we have $y_{2k+1} = 0$. It follows that

$$\|\mathbf{x} - \mathbf{y}\|_1 \geq |x_{2k+1}| > 0$$

This means that $\mathbf{x}$ cannot be in F which is the closure of $\langle \mathbf{e}_{2n} : n \in \mathbb{N} \rangle$. This is a contradiction. Thus $x_{2k+1} = 0$ for all $k \in \mathbb{N}$.

Suppose that $\mathbf{x} \neq \mathbf{0}$. Then, by our previous observation, we must have that $x_{2k} \neq 0$ for some $k \in \mathbb{N}$. Observe that, since $\mathbf{x} \in G$, there exists a finite linear combination

$$\sum_{i=0}^N \mu_{n_i} \left(\mathbf{e}_{2n_i} + \frac{1}{2n_i+1} \mathbf{e}_{2n_i+1} \right) \in \left\langle \mathbf{e}_{2n} + \frac{1}{2n+1} \mathbf{e}_{2n+1} : n \in \mathbb{N} \right\rangle$$

with distinct n_i 's such that

$$\left\| \mathbf{x} - \sum_{i=0}^N \mu_{n_i} \left(\mathbf{e}_{2n_i} + \frac{1}{2n_i+1} \mathbf{e}_{2n_i+1} \right) \right\|_1 < \frac{|x_{2k}|}{2k+3}$$

Thus the index $2k$ must appear among n_i 's and moreover, we must have

$$|x_{2k} - \mu_{2k}| < \frac{|x_{2k}|}{2k+3}$$

This immediately implies $|\mu_{2k}| \geq \frac{|x_{2k}|(2k+2)}{2k+3}$ by the triangle inequality. But then we have

$$\left| x_{2k+1} - \frac{\mu_{2k}}{2k+1} \right| \leq \left\| \mathbf{x} - \sum_{i=0}^N \mu_{n_i} \left(\mathbf{e}_{2n_i} + \frac{1}{n_i+1} \mathbf{e}_{2n_i+1} \right) \right\|_1 < \frac{|x_{2k}|}{2k+3}$$

and so

$$\frac{|x_{2k}|}{2k+3} < \frac{|x_{2k}|(2k+2)}{(2k+3)(2k+1)} \leq \left| \frac{\mu_{2k}}{2k+1} \right| = \left| x_{2k+1} - \frac{\mu_{2k}}{2k+1} \right| < \frac{|x_{2k}|}{2k+3}$$

which is a contradiction. Thus $\mathbf{x} = \mathbf{0}$.

Having shown that $F \cap G = \{\mathbf{0}\}$, we obtain $E = F \oplus G$. Since F and G are closed in ℓ_1 , they are closed subspaces of the normed space $(E, \|\cdot\|_1)$ whose topology coming from the restriction of the norm to E coincides with its subspace topology coming from ℓ_1 . On the other hand, the projection map $\pi_F : E \rightarrow F$ is not continuous since

$$\frac{-1}{2n+1} \mathbf{e}_{2n+1} \xrightarrow{n \rightarrow \infty} \mathbf{0}$$

in the normed space $(E, \|\cdot\|_1)$, however,

$$\pi_F \left(\frac{-1}{2n+1} \mathbf{e}_{2n+1} \right) = \pi_F \left(\mathbf{e}_{2n} - \left(\mathbf{e}_{2n} + \frac{1}{2n+1} \mathbf{e}_{2n+1} \right) \right) = \mathbf{e}_{2n} \not\xrightarrow{n \rightarrow \infty} \mathbf{0}$$

Therefore the assumption that E is Banach in Theorem 41 cannot be dropped.

5. WEAK AND WEAK-* TOPOLOGIES

In this section, we shall define the weak and weak-* topologies on a normed space and its dual space respectively. Our aim is to prove the Bourbaki-Alaoglu theorem which states that the closed unit ball is compact with respect to weak-* topology.

In order to understand this section, the reader needs to know some basic preliminary material from general topology. Although we have covered this material in class during live lectures, it will not be included in these lecture notes. As before, the reader who wishes to learn this background material may consult to MATH535 lecture notes of the author [from this link](#).

Up to this point, we only had the topology induced by the metric associated to a norm $\|\cdot\|$ in hand and never explicitly mentioned the topology while talking about continuity, convergence, openness or closedness. From now on, this will be important since there will be different topologies on the same norm space. For the remainder of this section, let us denote the topology of a normed space coming from its norm by τ_{norm} and call it the norm-topology.

5.1. Weak topology. Let $(E, \|\cdot\|)$ a normed space. Recall that the dual space of E is the vector space

$$E' = L(E, \mathbb{K}) = \{f : E \rightarrow \mathbb{K} : f \text{ is linear and continuous with respect to } \tau_{norm}\}$$

together with the operator norm defined before. By definition, all elements of E' are continuous with respect to τ_{norm} . Suppose that, for a certain purpose, it suffices for all elements of E' to be continuous with respect to some topology. These functionals are also continuous if one puts, say, the discrete topology on E . So why are we not working with that topology? Because discrete topology has too many open sets which disrupt the normed space structure and is useless. So why not consider the weakest topology which makes all functionals in E' continuous?

Let $\mathcal{C} \subseteq E'$. The weakest topology on E with respect to which every $f \in E'$ is continuous is called *the weak topology on E determined by $\mathcal{C}$* . In the case $\mathcal{C} = E'$, this topology is referred to as *the weak topology on E* and denote it by τ_w .

Unpacking the definition, we see that τ_w is the intersection of all topologies τ with respect to which each $f \in E'$ is continuous, i.e. we have $f^{-1}(O) \in \tau$ for all $f \in E'$ and all open $O \subseteq \mathbb{K}$. Using this observation, we can find a basis for the weak topology. Set

$$\mathcal{B} = \left\{ \bigcap_{i=1}^n f_i^{-1}(B_i) : f_1, \dots, f_n \in E', B_1, \dots, B_n \subseteq \mathbb{K} \text{ are open balls} \right\}$$

We claim that $\mathcal{B}$ is a basis for τ_w , i.e. every element of τ_w is the union of elements of $\mathcal{B}$. First note that our previous observation implies that each element of $\mathcal{B}$ must be open in the weak topology. Therefore $\mathcal{B} \subseteq \tau_w$. Moreover, if we consider the

collection $\hat{\tau}$ consisting of the unions of elements of $\mathcal{B}$, which the reader can check is indeed a topology, then we see that each $f \in E'$ is continuous with respect to $\hat{\tau}$. This is because, given any open set $O \subseteq \mathbb{K}$, we can write $O = \bigcup_{i \in I} B_i$ for some open balls $\{B_i\}_{i \in I}$ in $\mathbb{K}$ and we make sure that $f^{-1}(O) = \bigcup_{i \in I} f^{-1}(B_i) \in \hat{\tau}$ by the construction of $\hat{\tau}$. Therefore, $\hat{\tau}$ is a topology with respect to which each $f \in E'$ is continuous. This means $\hat{\tau} \supseteq \tau_w$. Since $\mathcal{B} \subseteq \tau_w$ and τ_w is closed under taking unions by the definition of a topology, we also have that $\hat{\tau} \subseteq \tau_w$. Thus $\hat{\tau} = \tau_w$ and so $\mathcal{B}$ is a basis for the weak topology τ_w .

Now let us introduce some notation for certain basis elements which will be of importance. Let $e \in E$, $f_1, \dots, f_n \in E'$ and $\epsilon > 0$. We set

$$\begin{aligned} U_{f_1, \dots, f_n, \epsilon}^e &= \{x \in E : |f_i(x) - f_i(e)| < \epsilon \text{ for all } i = 1, \dots, n\} \\ &= \bigcap_{i=1}^n \{x \in E : |f_i(x) - f_i(e)| < \epsilon\} \\ &= \bigcap_{i=1}^n f_i^{-1}(B_{\mathbb{K}}(f_i(e), \epsilon)) \end{aligned}$$

Clearly we have $U_{f_1, \dots, f_n, \epsilon}^e \in \mathcal{B}$. Next will be shown that such sets form a *neighborhood basis* around the point e , i.e. for every open neighborhood O of e in the weak topology, we can find some $e \in U_{f_1, \dots, f_n, \epsilon}^e \subseteq O$. Let $e \in E$ and let $O \subseteq E$ be an open neighborhood of e in the weak topology. Then, $\mathcal{B}$ being a basis for τ_w , there exists a basis element

$$e \in \bigcap_{i=1}^n f_i^{-1}(B_i) \subseteq O$$

where $f_1, \dots, f_n \in E'$ and $B_1, \dots, B_n \subseteq \mathbb{K}$ are open balls, say, $B_i = B_{\mathbb{K}}(\alpha_i, \epsilon_i)$ for some $\alpha_i \in \mathbb{K}$ and $\epsilon_i > 0$. Then we have

$$|f_i(e) - \alpha_i| < \epsilon_i$$

for all $1 \leq i \leq n$. We claim that

$$e \in U_{f_1, \dots, f_n, \min\{\hat{\epsilon}_1, \dots, \hat{\epsilon}_n\}}^e \subseteq \bigcap_{i=1}^n f_i^{-1}(B_i) \subseteq O$$

where $\hat{\epsilon}_i = \epsilon_i - |f_i(e) - \alpha_i|$. Let $x \in U_{f_1, \dots, f_n, \min\{\hat{\epsilon}_1, \dots, \hat{\epsilon}_n\}}^e$. Then, for each $1 \leq i \leq n$, we have

$$|f_i(x) - f_i(e)| < \min\{\hat{\epsilon}_1, \dots, \hat{\epsilon}_n\} \leq \hat{\epsilon}_i = \epsilon_i - |f_i(e) - \alpha_i|$$

and hence

$$|f_i(x) - \alpha_i| \leq |f_i(x) - f_i(e)| + |f_i(e) - \alpha_i| < \epsilon_i$$

which means $f_i(x) \in B_i$ for all $1 \leq i \leq n$. Thus $x \in \bigcap_{i=1}^n f_i^{-1}(B_i)$ which completes the proof that, given any $e \in E$, the sets of the form

$$U_{f_1, \dots, f_n, \epsilon}^e$$

form a neighborhood basis of e . Let us now record some basic important facts.

Proposition 14. $\tau_w \subseteq \tau_{norm}$

Proof. τ_{norm} is a topology with respect to which all $f \in E'$ are continuous and τ_w is, by definition, the smallest such topology. $\square$

Proposition 15. (E, τ_w) is a Hausdorff space.

Proof. Let $x, y \in E$ be with $x \neq y$. Then, by an application of the Hahn-Banach theorem, we can find some $f \in E'$ such that $f(x) = \alpha$ and $f(y) = \beta$ and $\alpha \neq \beta$. Set $\epsilon = \frac{|\alpha - \beta|}{2}$. We then have

$$x \in U_{f, \epsilon}^x \text{ and } y \in U_{f, \epsilon}^y \text{ and } U_{f, \epsilon}^x \cap U_{f, \epsilon}^y = \emptyset$$

Thus we can separate distinct points in (E, τ_w) by disjoint open neighborhoods. $\square$

It follows from Proposition 14 that, given a sequence $(x_n)_{n \in \mathbb{N}}$ of points of E and $x \in E$, that $x_n \rightarrow x$ in the norm topology implies that $x_n \rightarrow x$ in the weak topology. The converse is, as we shall see later, false in general.

From now on, we shall write $x_n \rightharpoonup x$ and say $(x_n)_{n \in \mathbb{N}}$ *weakly converges to* x if $x_n \rightarrow x$ with respect to the weak topology. We will now provide a useful criterion for weak convergence.

Theorem 42. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of elements of E and $x \in E$. Then $x_n \rightharpoonup x$ if and only if $f(x_n) \rightarrow f(x)$ for all $f \in E'$.

Proof. Suppose that $x_n \rightharpoonup x$. Let $f \in E'$. By definition, f is continuous with respect to τ_w . It is true for arbitrary topological spaces that the continuity of a function implies its sequential continuity.¹ Thus we have $f(x_n) \rightarrow f(x)$ by the sequential continuity of f .

For the converse direction, suppose that $f(x_n) \rightarrow f(x)$ for all $f \in E'$. We wish to show that $x_n \rightharpoonup x$, that is, $x_n \rightarrow x$ with respect to τ_w . Let $x \in U \subseteq E$ be an open neighborhood of x . We can then find a basis element

$$x \in \bigcap_{i=1}^n f_i^{-1}(B_i) \subseteq U$$

where $f_1, \dots, f_n \in E'$ and $B_1, \dots, B_n \subseteq \mathbb{K}$ are open balls. By assumption, we have $f_i(x_n) \rightarrow_{n \rightarrow \infty} f_i(x)$ for each $i = 1, \dots, n$. Therefore, by the definition of convergence of a sequence used with the neighborhood B_i of $f_i(x)$, for each $1 \leq i \leq n$, there exists $N_i \in \mathbb{N}$ such that for all $n \geq N_i$ we have

$$f_i(x_n) \in B_i$$

¹The converse requires the underlying space to be first-countable which is not the case here unless the normed space is finite-dimensional.

Set $N = \max_{1 \leq i \leq n} N_i$. Then, for all $n \geq N$, we have that

$$x_n \in \bigcap_{i=1}^n f_i^{-1}(B_i) \subseteq U$$

Therefore $x_n \rightarrow x$. □

Before we proceed, let us see some examples of normed spaces where the weak topology is strictly weaker than the norm topology. Consider the Banach space $(\ell_2, \|\cdot\|_2)$. We clearly have $\mathbf{e}_n \rightarrow \mathbf{0}$ in the norm topology. However, we do have $\mathbf{e}_n \rightharpoonup \mathbf{0}$. In order to see this, let $\varphi \in \ell'_2$. Recall from Section 2.4.2 that ℓ'_2 is isomorphic to $(\ell_2, \|\cdot\|_2)$ via the isomorphism $T : \ell_2 \rightarrow \ell'_2$ given by $T(\mathbf{a}) = f_{\mathbf{a}}$ where $f_{\mathbf{a}}(\mathbf{x}) = \sum_{i=1}^{\infty} a_i x_i$. Thus $\varphi = f_{\mathbf{a}}$ for some $\mathbf{a} \in \ell_2$. Then

$$\varphi(\mathbf{e}_n) = f_{\mathbf{a}}(\mathbf{e}_n) = \sum_{i=1}^{\infty} a_i \delta_{in} = a_n \rightarrow 0$$

Thus $\varphi(\mathbf{e}_n) \rightarrow \varphi(\mathbf{0}) = 0$ for all $\varphi \in \ell'_2$ which means that $\mathbf{e}_n \rightharpoonup \mathbf{0}$. This example is no coincidence. In infinite-dimensional spaces, the weak topology is strictly weaker than the norm topology.²

Theorem 43. *Let $(E, \|\cdot\|)$ be an infinite-dimensional normed space and τ_w be the associated weak topology. Then τ_w is not equal to any norm topology on E .*

Proof. Suppose towards a contradiction that τ_w is equal to a norm topology $\tau_{norm'}$ induced by a norm $\|\cdot\|'$ on E . Since

$$\mathbf{0} \in \{x \in E : \|x\|' < 1\}$$

is open with respect to $\tau_{norm'} = \tau_w$, there exists a neighborhood basis element

$$U_{f_1, \dots, f_n, \epsilon}^{\mathbf{0}} \subseteq \{x \in E : \|x\|' < 1\}$$

for some $f_1, \dots, f_n \in E'$ and $\epsilon > 0$. Observe that the linear map $F : E \rightarrow \mathbb{R}^n$ given by

$$F(x) = (f_1(x), \dots, f_n(x))$$

has to have non-trivial kernel since, otherwise, we need to have that $\dim_{\mathbb{K}}(E) \leq n$. Therefore, there exists $0 \neq e \in E$ such that

$$e \in \ker(F) = \bigcap_{i=1}^n \ker(f_i)$$

²While the existence of a sequence which is convergent with respect to a topology and not convergent with respect to another implies that these topologies are different, it is not true that distinct topologies must have distinct sets of convergent sequences. For example, the sets of weakly convergent sequences and norm convergent sequences coincide for $(\ell_1, \|\cdot\|_1)$.

But then, for all $\alpha \in \mathbb{K}$, we must have

$$\alpha e \in \ker(F) = \bigcap_{i=1}^n \ker(f_i) \subseteq U_{f_1, \dots, f_n, \epsilon}^0 \subseteq \{x \in E : \|x\|' < 1\}$$

This means that $|\alpha| \|e\| = \|\alpha e\| < 1$ for all $\alpha \in \mathbb{K}$, which is a contradiction. $\square$

Thus the weak topology on an infinite-dimensional space is not well-behaved enough to be induced by a norm. Indeed, the weak topology on an infinite-dimensional space is not a first-countable topology³ which automatically rules out the possibility that it is a metrizable topology. We shall not prove this fact here, which the reader can find in various functional analysis books.

In the finite-dimensional setting, as expected, the weak topology coincides with the norm topology.

Theorem 44. *Let $(E, \|\cdot\|)$ be a finite-dimensional normed space and τ_w be the associated weak topology. Then $\tau_w = \tau_{norm}$.*

Proof. By Proposition 14, it suffices to prove that $\tau_{norm} \subseteq \tau_w$. This amounts to proving the following: For every $x \in E$ and for every $O \in \tau_{norm}$ with $x \in O$, there exists $U \in \tau_w$ such that $x \in U \subseteq O$.

Let $x \in E$ and $O \in \tau_{norm}$ be with $x \in O$. Since O is open with respect to the norm topology, there exists $\epsilon > 0$ such that $B_{\|\cdot\|}(x, \epsilon) \subseteq O$. Since $\dim_{\mathbb{K}}(E) < \aleph_0$, there exists a Hamel basis $\{v_1, \dots, v_n\} \subseteq E$ for E . By normalizing if necessary, we may assume without loss of generality that $\|v_i\| = 1$ for all $1 \leq i \leq n$. For each $1 \leq i \leq n$, let $f_i \in E'$ denote the corresponding i -th coefficient functional as defined in Section 3.2. We claim that

$$U = U_{f_1, \dots, f_n, \epsilon/(n+1)}^x \subseteq B_{\|\cdot\|}(x, \epsilon) \subseteq O$$

Let $y \in U$. Then

$$\begin{aligned} \|x - y\| &= \left\| \sum_{i=1}^n f_i(x - y) v_i \right\| \leq \sum_{i=1}^n \|f_i(x - y) v_i\| \\ &\leq \sum_{i=1}^n |f_i(x - y)| \|v_i\| \\ &\leq \sum_{i=1}^n |f_i(x) - f_i(y)| \\ &\leq \sum_{i=1}^n \frac{\epsilon}{n+1} < \epsilon \end{aligned}$$

Thus $y \in B_{\|\cdot\|}(x, \epsilon)$ which completes the proof. $\square$

³Recall that a topology is said to be first-countable if every point has a *countable* neighborhood basis.

From now on, in order to increase readability and save time and space, we shall use the terminology *norm-open, weakly-open, norm-closed, weakly-closed* etc.

Since $\tau_w \subseteq \tau_{norm}$, every weakly-closed set is norm-closed. The converse of this statement fails in general. As we have seen before, $\{\mathbf{e}_n : n \in \mathbb{N}\}$ is not a weakly-closed set in ℓ_2 but is a norm-closed set. It turns out that, if one demands a norm-closed set to be convex, then it automatically becomes weakly-closed. Let us prove this for normed spaces over $\mathbb{R}$.

Theorem 45. *Let $(E, \|\cdot\|)$ be a normed space over $\mathbb{R}$ and $A \subseteq E$ be a convex set. If A is norm-closed, then A is weakly-closed.*

Proof. Suppose that A is norm-closed. We will show that $E \setminus A$ is weakly-open. Let $x \notin A$. Then, using Theorem 26, we can find $\gamma \in \mathbb{R}$ and $f \in E'$ such that

$$f(x) < \gamma < f(a) \text{ for all } a \in A$$

Since f is continuous with respect to the weak topology, $U = f^{-1}((-\infty, \gamma))$ must be weakly-open. Observe that $x \in U$ and $U \subseteq E \setminus A$. Thus $E \setminus A$ is weakly-open and so A is weakly-closed. $\square$

In order to prove this result for normed spaces over $\mathbb{C}$, we need analogues of hyperplane separation theorems for normed spaces over $\mathbb{C}$. The reader may check [Con90, Theorem 3.13] for such a statement. Using these, one can show that convex norm-closed sets are weakly-closed in general.

5.2. Weak-* topology. Let $(E, \|\cdot\|)$ a normed space. By applying the methods that we have learned in the previous subsection, we can put various weak topologies on the dual space E' by considering various subsets of the double dual E'' . Of all these weak topologies on E' , one seems to stand out: The weak topology on E' determined by $i(E) \subseteq E''$ where $i : E \rightarrow E''$ is the natural isometry given by

$$i(e) = \varphi_e \text{ where } \varphi_e : E' \rightarrow \mathbb{K} \text{ is given by } \varphi_e(f) = f(e) \text{ for all } f \in E'$$

for all $e \in E$. This special weak topology on E' is called *the weak-* topology on E'* .

We will now explore some basic properties of the weak-* topology. A moment's thought reveals that many propositions that we prove in the previous chapter actually holds for not just the full weak topology on a normed space but for all weak topologies determined by various subsets, when modified appropriately. For example, the reader may verify that

$$\mathcal{B} = \left\{ \bigcap_{i=1}^n f_i^{-1}(B_i) : f_1, \dots, f_n \in i(E), B_1, \dots, B_n \subseteq \mathbb{K} \text{ are open balls} \right\}$$

is a basis for the weak-* topology. Moreover, the collection

$$\left\{ U_{\varphi_{x_1}, \dots, \varphi_{x_n}, \epsilon}^f : x_1, \dots, x_n \in E, \epsilon > 0 \right\}$$

is a neighborhood basis for the point $f \in E'$ in the weak-* topology. Here

$$\begin{aligned} U_{\varphi_{x_1}, \dots, \varphi_{x_n}, \epsilon}^f &= \{g \in E' : |\varphi_{x_i}(g) - \varphi_{x_i}(f)| < \epsilon \text{ for all } i = 1, \dots, n\} \\ &= \{g \in E' : |g(x_i) - f(x_i)| < \epsilon \text{ for all } i = 1, \dots, n\} \end{aligned}$$

For notational simplicity, we shall drop φ 's and write

$$U_{x_1, \dots, x_n, \epsilon}^f$$

instead of $U_{\varphi_{x_1}, \dots, \varphi_{x_n}, \epsilon}^f$. The weak-* convergence of a sequence of functions in E' , which we shall use $f_n \rightharpoonup^* f$ to denote, can be characterized in a similar fashion to Theorem 46 as follows.

Theorem 46. *Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of elements of E' and $f \in E$. Then $f_n \rightharpoonup^* f$ if and only if $f_n(x) \rightarrow f(x)$ for all $x \in E$.*

The proof is exactly the same as that of Theorem 46 except while carrying out that proof with E replaced by E' , one needs to replace E'' by $i(E)$.

Since the weak-* topology on E' is the weakest topology on E' that make *fewer* functionals in E'' continuous than the weak topology on E' , we have that

$$\tau_{w^*} \subseteq \tau_w \subseteq \tau_{norm}$$

on E' . For finite-dimensional E , all three topologies clearly coincide. More generally, for a reflexive space E i.e. a normed space for which the canonical evaluation map $i : E \rightarrow E''$ is surjective, the weak and weak-* topologies on E' coincide for trivial reasons. Indeed, the converse of this statement also holds. We will now prove the last important theorem of this course.

Theorem 47 (Banach-Alaoglu-Bourbaki theorem). *Let $(E, \|\cdot\|)$ be a normed space. Then the closed unit ball*

$$S = \{f \in E' : \|f\| \leq 1\}$$

is weak- compact.*

Proof. Consider the product space

$$P = \prod_{x \in E} I_x \quad \text{where } I_x = [-\|x\|, \|x\|]$$

with the product topology where each component $I_x \subseteq \mathbb{R}$ is endowed with its usual Euclidean subspace topology. By Tychonoff's theorem, we have that P is a compact topological space.

We claim that

$$S = \{f \in E' : \|f\| \leq 1\} = \{f \in P : f \text{ is linear}\}$$

To see the left-to-right direction, let $f \in S$. Then we have that f is linear and that $|f(x)| \leq \|f\| \|x\| \leq \|x\|$ and so $f(x) \in I_x$ for all $x \in E$ which means $f \in P$. To see

the right-to-left direction, let $f \in P$ be linear. Then we have that $f(x) \in I_x$ for all $x \in E$ and so $|f(x)| \leq \|x\|$ for all $x \in E$. This implies that f is bounded and so $f \in E'$. Moreover, by taking the supremum over $x \in E$ with $\|x\| \leq 1$, we obtain $\|f\| \leq 1$. Thus $f \in S$.

We will next prove that S is a closed subset of P with respect to the product topology. Before we proceed to prove this, let us make an important observation. Let $f \in E'$, $x, y, z \in E$ and $\epsilon > 0$. Observe that

$$V_{x,y,z,\epsilon}^f = \{g \in P : |g(x) - f(x)|, |g(y) - f(y)|, |g(z) - f(z)| < \epsilon\} = \prod_{i \in E} U_i$$

where $U_i = I_i$ for all $i \in E$ with $i \neq x, y, z$ and

$$U_i = B_{\mathbb{K}}(f(i), \epsilon) \text{ for } i = x, y, z$$

Thus $V_{x,y,z,\epsilon}^f$ is a basis element for the product topology on P .

Let us now show that S is closed in P with respect to the product topology. Let $f \in \overline{S}$ where the closure is taken in P with respect to the product topology. We wish to show that $f \in S$. Since $S = \{f \in P : f \text{ is linear}\}$, it suffices to show that f is linear. Let $x, y \in E$ and $\alpha, \beta \in \mathbb{K}$. Let $\epsilon > 0$. Then

$$f \in V_{x,y,\alpha x + \beta y, \frac{\epsilon}{|\alpha| + |\beta| + 1}}^f$$

is an open neighborhood of f and hence there exists

$$u \in V_{x,y,\alpha x + \beta y, \frac{\epsilon}{|\alpha| + |\beta| + 1}}^f \cap S$$

Observe that

$$\begin{aligned} & |\alpha f(x) + \beta f(y) - f(\alpha x + \beta y)| \\ &= |\alpha f(x) + \alpha u(x) - \alpha u(x) + \beta f(y) + \\ & \quad \beta u(y) - \beta u(y) + u(\alpha x + \beta y) - u(\alpha x + \beta y) - f(\alpha x + \beta y)| \\ &= |\alpha f(x) - \alpha u(x) + \beta f(y) - \beta u(y) + u(\alpha x + \beta y) - f(\alpha x + \beta y)| \\ &\leq |\alpha| |f(x) - u(x)| + |\beta| |f(y) - u(y)| + |u(\alpha x + \beta y) - f(\alpha x + \beta y)| \\ &\leq |\alpha| \frac{\epsilon}{|\alpha| + |\beta| + 1} + |\beta| \frac{\epsilon}{|\alpha| + |\beta| + 1} + \frac{\epsilon}{|\alpha| + |\beta| + 1} = \epsilon \end{aligned}$$

Since this inequality is true for all $\epsilon > 0$, we must have that

$$|\alpha f(x) + \beta f(y) - f(\alpha x + \beta y)| = 0$$

which implies that f is linear. Thus $f \in S$ which completes the proof that S is closed. Being a closed subset of a compact space, S is compact with respect to its subspace topology coming from the product topology of P .

We shall now transport this result to the weak-* topology. In order to do this, consider a neighborhood basis element

$$U_{x_1, \dots, x_n, \epsilon}^f = \{g \in E' : |g(x_i) - f(x_i)| < \epsilon \text{ for all } i = 1, \dots, n\}$$

in the weak-* topology. Observe that the intersection

$$U_{x_1, \dots, x_n, \epsilon}^f \cap S = \left(\prod_{x=x_1, \dots, x_n} B_{\mathbb{K}}(f(x), \epsilon) \times \prod_{x \neq x_1, \dots, x_n} I_x \right) \cap S$$

is a basis element for the subspace topology of S coming from the product topology of P . Since any open set of E' in the weak-* topology can be written as a union of these neighborhood basis elements, we obtain that any open set in the subspace topology of S coming from the weak-* topology of E' is open in the subspace topology of S coming from the product topology of P .

Let $S \subseteq \bigcup_{j \in J} O_j$ be a weak-* open covering of S with respect to its subspace topology coming from the weak-* topology of E . We know by the previous observation that, for all $j \in J$, the set O_j is open in the subspace topology of S coming from the product topology of P . Since we already have that S is compact with respect to this topology, there exists finitely many $j_1, \dots, j_n \in J$ with

$$S \subseteq \bigcup_{i=1}^n O_{j_i}$$

Thus S is compact with respect to the weak-* topology.⁴ □

We will next see some consequences of the Alaoglu-Bourbaki theorem.

Corollary 48. *Let $(E, \|\cdot\|)$ be a normed space and $A \subseteq E'$ be weak-* closed and norm-bounded. Then A is weak-* compact.*

Proof. Since A is norm-bounded, there exists $\lambda \in K$ such that

$$\lambda A \subseteq S = \{f \in E' : \|f\| \leq 1\}$$

Scaling by a constant is a homeomorphism of E' with respect to the weak-* topology and hence λA stays weak-* closed. By the Banach-Alaoglu-Bourbaki theorem, being a weak-* closed subset of a weak-* compact set, the set λA is weak-* compact. Hence A is weak-* compact. □

It turns out that the converse of this proposition holds for Banach spaces as well.

Corollary 49. *Let $(E, \|\cdot\|)$ be a Banach space and $A \subseteq E'$ be weak-* compact. Then A is norm-bounded.*

Proof. Let $x \in E$. Then we clearly have

$$A \subseteq \bigcup_{n \in \mathbb{N}} \{f \in E' : |f(x)| \leq n\} = \bigcup_{n \in \mathbb{N}} U_{x,n}^0$$

⁴While we did not have to use it, the reader may prove that the subspace topology of S coming from the weak-* topology coincides with the subspace topology of S coming from the product topology of P .

Since A is weak-* compact, there exists a finite subcover of this open covering and hence there exists $N \in \mathbb{N}$ such that

$$A \subseteq U_{x,N}^{\mathbf{0}}$$

But this precisely means that

$$\sup_{f \in A} |f(x)| \leq N < \infty$$

Since E is Banach, we can now apply the uniform boundedness principle and obtain that

$$\sup_{f \in A} \|f\| < \infty$$

which says that A is norm-bounded in E' . □

6. CODA

“Mathematics, rightly viewed, possesses not only truth, but supreme beauty - a beauty cold and austere, like that of sculpture, without appeal to any part of our weaker nature, without the gorgeous trappings of painting or music, yet sublimely pure, and capable of a stern perfection such as only the greatest art can show. The true spirit of delight, the exaltation, the sense of being more than Man, which is the touchstone of the highest excellence, is to be found in mathematics as surely as poetry.”

Bertrand Russell.

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